



Dolby Voice Room 5.0

Administrator's Guide

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Notices

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Introduction to this guide

Dolby Voice Room is an integrated software and hardware conferencing solution for huddle spaces and small conference rooms.

- [About this documentation](#)
- [New in this version](#)
- [Related documentation](#)
- [Configuration guides](#)
- [Problem reports](#)
- [Documentation feedback](#)

1.1 About this documentation

This documentation is for IT personnel who intend to deploy and administer one or more Dolby Voice Room systems.

Users of this documentation should be familiar with:

- Basics of computer networking and Linux administration
- IP private branch exchange (PBX) call control platform used by your organization
- Conferencing service provider functionality used by your organization

This documentation provides information about:

- Dolby Voice Room system architecture
- Preparation and requirements for setting up your Dolby Voice Room systems
- Provisioning methods and procedures
- Configuring the systems using the Dolby Conference Phone user interface and Dolby Voice Room web interface
- Handling errors

1.2 New in this version

This release of the Dolby Voice Room includes new and updated features.

- Initial support for installed rooms with Universal Serial Bus (USB) as both audio input and output devices
- Support enabling HDMI input detection logs
- Support PTZ presets
- Support Session Initiation Protocol (SIP) PTZ presets
- Option to disable content share option during SIP video calls
- Removed expiring CA certificates

Related information

[Configuring Dolby Voice Room to use USB audio devices](#) on page 73

[Enabling HDMI logging](#) on page 84

[Creating and editing presets using PTZ control](#) on page 53

[Creating and editing presets using PTZ control during a SIP video call](#) on page 55

[Setting and viewing configuration parameters](#) on page 41

[Trusted certificates and issuers](#) on page 45

1.3 Related documentation

Review all of the Dolby Voice Room documents to ensure that the system is set up correctly and that you have the best conferencing experience possible.

These documents are available:

- *Dolby Voice Room Quick Start Guide*

This guide describes the contents of the Dolby Voice Room package, how to assemble the system, and how to connect the system to the network. The quick start guide is included in the package. It is also available from the Dolby Voice Room support pages.

- *Dolby Voice Room Third-Party Software Guide*

This guide describes the open source software used in the Dolby Voice Room.

- *Dolby Voice Room Peripherals Compatibility Guide*

This document describes the compatibility relationships between Dolby Voice Room and third-party USB cameras.

- This guide describes using Dolby Conferencing Console as a provisioning server for Dolby Voice Room.
- *Dolby Satellite Microphone Quick Start Guide*
This guide describes the contents of the microphone package and how to connect them to the Dolby Conference Phone. The quick start guide is included in the microphone package. It is also available from the Dolby Conference Phone support pages.
- *Dolby Voice Product Compatibility Guide*
This guide describes the compatibility relationships between the various Dolby Voice products.

1.4 Configuration guides

For information about integrating the Dolby Voice Room system with your conferencing service provider and a supported IP PBX, see the configuration guide for your call control platform.

The following configuration guides are available:

- *Dolby Voice Room Configuration Guide for Avaya Aura Platform 6*
- *BlueJeans Rooms with Dolby Voice Room Advanced Configuration Guide*
- *Dolby Voice Room Configuration Guide for Cisco Unified Communications Manager*
- *Dolby Voice Room Configuration Guide for Microsoft Skype for Business*
- *Dolby Voice Room Configuration Guide for RingCentral*
- *Dolby Voice Room Configuration Guide for Unify OpenScape*

1.5 Problem reports

When you report problems to your conferencing service provider, be prepared to provide basic system information, log files, and a full description of the issue.

- Basic system information:
 - Serial number
 - Software version
- Any relevant log files showing the problem occurring, or anything that might be relevant in the log. Did anything unusual occur? Did the system generate any high-severity log messages? If so, attach an extract. If you have not enabled logging, enable it for the system experiencing the issue, reproduce the issue, and provide the logs.
- Description of the issue:
 - When did the problem occur? How often does it occur? Is there any pattern or trend to the occurrence?
 - What was the scope of the problem? How many users did it affect? Was there any pattern or trend to the affected users?
 - Any symptoms observed.
 - Any error messages displayed on the phone user interface or web interface.
 - Any obvious triggers. (What was the user doing before the error message displayed?)
 - If the issue was reproducible. (Does it happen regularly or after certain tasks are performed?) If you can reproduce the problem, detail how.
 - How widespread the issue appears to be. (Does it affect only a single user or the whole organization?)
 - Any other information you have that might assist in understanding the issue.

- What operating system and version are being used by the client? What browser and version are being used by the client?
- What kind of Internet connection is the client using (ADSL/cable/wireless/mobile)?
- What other observations have you made? Is there anything else you think might assist us in identifying the root cause of the problem?

Related information

[Finding the system serial number and software version](#) on page 96

[Viewing and managing event logs](#) on page 81

1.6 Documentation feedback

If you have comments or feedback about this documentation, send us an email at dolbyvoicedocs@dolby.com.

Dolby Voice Room Overview

Dolby Voice Room is an integrated hardware and software conferencing system for huddle spaces and small conference rooms.

- [Dolby Voice Room Features and Benefits](#)
- [Dolby Voice Room hardware components](#)
- [Dolby Voice Room software](#)
- [Network configurations](#)
- [Operation modes of Dolby Voice Room](#)
- [Supported IP PBX call control platforms](#)

2.1 Dolby Voice Room Features and Benefits

Dolby Voice Room offers these capabilities:

- Advanced video and audio conferencing when integrated with conferencing service providers that support Dolby Voice technologies.
- Traditional public switched telephone network (PSTN) calling features when paired with IP PBX call control platforms.

Benefits include:

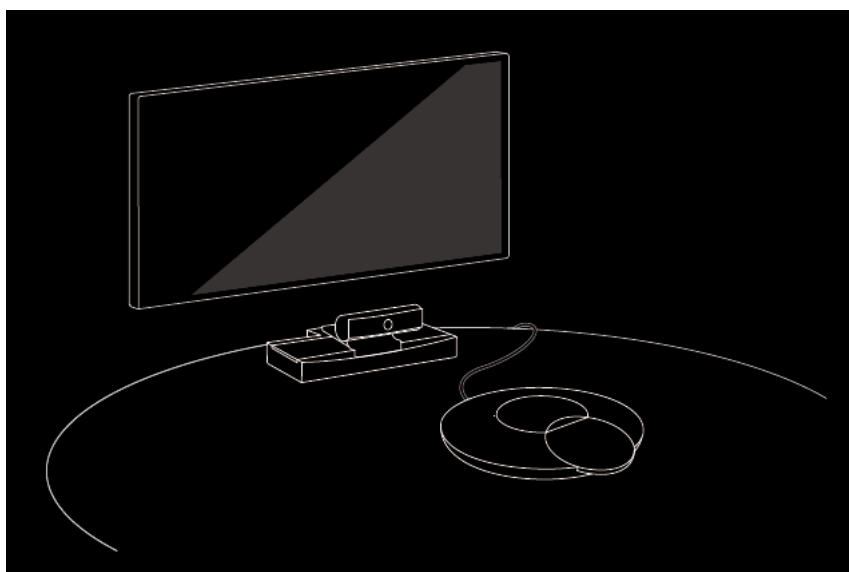
- 4K smart camera or 12X optical zoom camera that enables intelligent scene framing, which uses face recognition to automatically frame changing regions of interest, such as people moving in the room.
- HDR video that automatically adjusts contrast to help overcome lighting challenges.
- Whiteboard framing feature that enables you to share whiteboard content with meeting participants.
- Premium audio experience enabled by Dolby Voice and Dolby Conference Phone. Dynamic leveling continuously cancels out distracting noises, elevates the sound of speech, and enables full-room voice capture, even of quiet voices in environments with challenging acoustics. Spatial audio permits conference participants to discern where in the remote conference room voices are located, making it easier to listen and understand.
- Wired and wireless content sharing.
- Fully integrated and immersive service-provider experience built with Dolby Voice Room's application ecosystem and software development kit.
- Simplified setup and automated deployment.
- Easy conference joining.
- Remote device management for monitoring and troubleshooting.

2.2 Dolby Voice Room hardware components

The Dolby Voice Room system includes these hardware components:

- Dolby Voice Hub
- Dolby Voice Camera or Dolby Voice Camera Pro
- Dolby Conference Phone

To complete the conferencing system, you connect one or two high-definition televisions or monitors (displays).



2.2.1 Dolby Voice Hub

Dolby Voice Hub is the processing unit of Dolby Voice Room.

It includes a flexible mounting system that enables users to deploy the solution in one of three possible ways: display mount, camera and hub separated, and tabletop. See the *Dolby Voice Room Quick Start Guide* for illustrations of these configurations.

Dolby Voice Hub includes a power adapter and provides the following ports:

- Gigabit Ethernet
- Power over Ethernet (PoE) port for connection to the Dolby Conference Phone
- Three USB 3.0 ports for connection to Dolby Voice Camera and to a keyboard for diagnostics
- Two high-definition video ports and an input port for laptop for content sharing
- A USB OTG 3.0 port for future use

2.2.2 Dolby Voice Camera

Dolby Voice Camera is a smart camera with 4K resolution.

It includes these features:

- 95-degree wide-angle field of view
- Digital pan-tilt-zoom (PTZ) capability
- High dynamic range (HDR) capability

The camera includes a mounting system that permits mounting on a display (TV or monitor) or attaching directly to Dolby Voice Hub. See the *Dolby Voice Room Quick Start Guide* for more information.

2.2.3 Dolby Voice Camera Pro

Dolby Voice Camera Pro is a dual camera system with 1080p resolution.

It includes these features:

- The main camera supports 12X optical zoom and motorized pan/tilt control (MPTZ).
- The panoramic camera has 122 degrees diagonal, 98 degrees horizontal, and 72.5 degrees vertical field of view.
- The main camera has 72.5 degrees horizontal field of view.

The camera includes a wall mount kit that permits mounting on the wall.

2.2.4 Dolby Conference Phone

Dolby Conference Phone supports Dolby Voice codecs and spatial audio. It includes these components:

- Touch-screen display for interacting with the Dolby Voice Room software and the conferencing service provider's application
- Ethernet port for connection to Dolby Voice Hub
- Ports for connecting up to two satellite microphones

2.3 Dolby Voice Room software

The Dolby Voice Room system software includes the Dolby Voice Hub software, the Dolby Voice Camera firmware, and the Dolby Conference Phone firmware.

Dolby Voice Hub manages software updates automatically for the phone and the camera.

The software handles system functions, manages multiple user interfaces, and provides a platform for running conferencing service provider applications.

System functions include:

- Management of the protocol stack, the DSP, the user interface (UI), and the network interaction
- Voice over Internet Protocol (VoIP) signaling for a range of voice telephony functions using SIP signaling for call setup and control
- Advanced video features, including intelligent scene framing and whiteboard configuration and viewing
- Advanced audio signal processing for communications using a group of audio codecs
- Industry-standard security techniques for ensuring that all provisioning, signaling, and media transactions are authenticated and encrypted
- Flexible provisioning methods to support various deployment scenarios

User interfaces for administration include:

- Touch-screen interface on the Dolby Conference Phone
- Web interface
- Command-line interface with the diagnostic shell

2.4 Network configurations

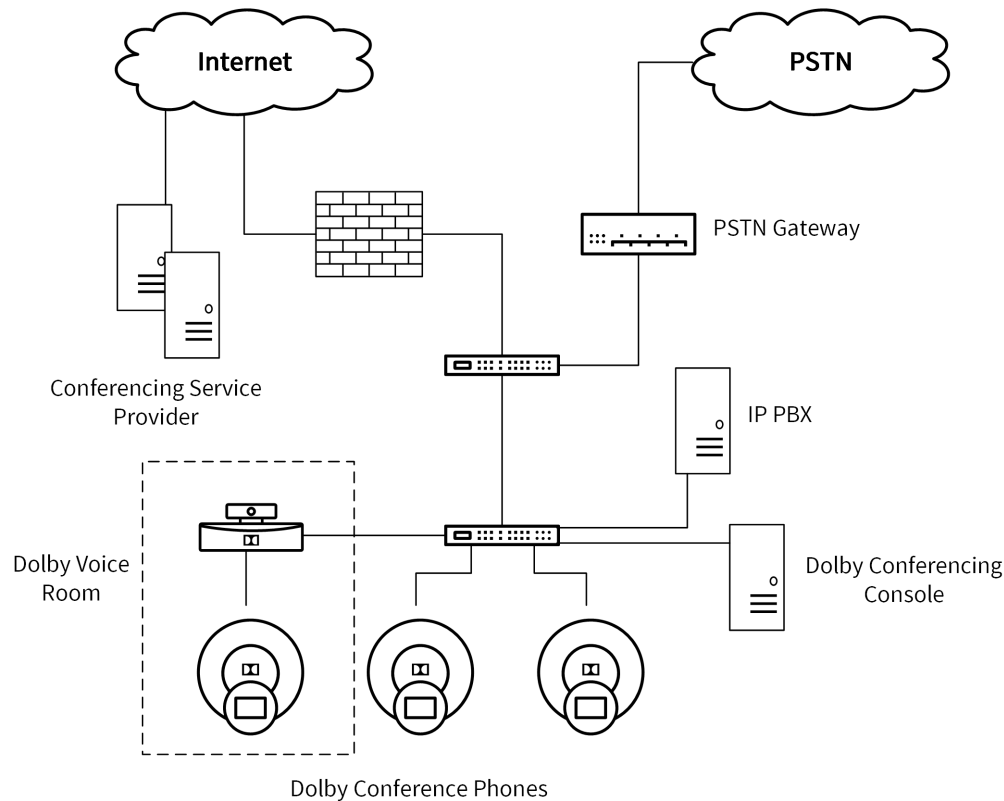
From an administrator's perspective, Dolby Voice Room is a conferencing endpoint and SIP endpoint in a network that interoperates with other compatible components, such as application servers, media servers, gateways, switches, and other endpoints.

For Dolby Voice Room to operate, you must have:

- A working IP network
- Network routers and switches configured for VoIP and WebRTC
- VoIP gateways configured for SIP
- Adequate firewall security
- The latest Dolby Voice Room software image
- An active, configured call server to receive and send SIP messages
- If not using a provisioning server provided by the conferencing server provider, a Dolby Conferencing Console on the IP network

You can set up a network using Dolby Voice Room in a number of ways. This figure shows one example of a network setup.

Figure 1: Dolby Voice Room in a network

**Related information**

[Provisioning Overview](#) on page 24

2.5 Operation modes of Dolby Voice Room

The Dolby Voice Room system can integrate with the conferencing service provider as a conference endpoint, register to an IP PBX call control platform as a SIP audio-only endpoint, or do both simultaneously.

The operation mode determines which features are supported. For example, if the operation mode is set to **Conferencing Only Mode**, it cannot be configured to connect to an IP PBX, and features such as dialing a local extension, call forwarding, and call transferring are unavailable.

If the operation mode is set to **Dual Mode**, the system can connect to both an IP PBX and a conferencing service enabled for Dolby Voice.

The default mode is **Dual Mode**.

For more information, see the configuration guide for your call control platform.

Related information

[Setting the Dolby Voice Room operational mode](#) on page 81

[Configuration guides](#) on page 9

2.6 Supported IP PBX call control platforms

An IP PBX provides audio communications through a TCP/IP protocol stack for endpoints on internal networks and interconnects with a PSTN for telephony connections for external endpoints.

Dolby Voice Room supports these IP PBX call control platforms:

- Avaya Aura Platform
- BroadSoft BroadWorks
- Cisco Unified Communications Manager
- Unify OpenScape Enterprise Express

Standard SIP call control platform

Dolby Voice Room is a SIP-compliant endpoint and interoperates with call control platforms designed to comply with RFC 3261. For Transport Layer Security (TLS) transport, the system is limited by the existing security configuration. Keep in mind:

- If the system transport is configured as AUTO, TLS transport is attempted only if the system fails to connect to Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) transports.
- The CA certificate of the SIP server must be downloaded to the system through provisioning.
- TLS 1.0, 1.1, and 1.2 are supported. TLS 1.2 is the preferred protocol, but version 1.0 or 1.1 can be used, depending on peer support.
- Because the system acts as a client, the server determines whether or not mutual authentication is required. (Normally, SIP servers are configured only with client authentication.)
- Certificate revocation lists (CRLs) are not supported.
- Simple Certificate Enrollment Protocol (SCEP) and Online Certificate Status Protocol (OCSP) are not supported.
- The system device certificate cannot be replaced at this time.
- If you try to set up IP PBX with explicit TLS transport instead of using the **Auto** transport feature, you must set the transport port explicitly. If you change your configuration from TLS to **Auto**, you must set the port back to the correct default port.

Contact your network administrator to find out which port is used for the TLS connection in your environment.

When TLS is operational on the system, Secure Real-time Transport Protocol (SRTP) is enabled; when TLS is not operational on the system, SRTP is disabled. By default, if TLS is enabled, SRTP is set to Mandatory. Dolby Voice Room offers a predefined list of cipher suites.

Related information

[TLS cipher suites](#) on page 46

Preparation and requirements

To successfully deploy Dolby Voice Room, you must first complete some tasks to satisfy both room requirements and network requirements.

Before you begin, read the *Dolby Voice Room Quick Start Guide* .

- [Conference Room requirements](#)
- [Network requirements](#)

3.1 Conference Room requirements

There are specific requirements for the conference room, including size, acoustics, lighting conditions, and necessary equipment.

3.1.1 Room size and acoustics

Before you install Dolby Voice Room, make sure that the room does not exceed the maximum recommended room size.

From an acoustics point of view, choosing a Dolby Voice Room solution based on the size of the room is important because it ensures natural and effective conversational flow for the usual seating positions in the room. Participant behavior and environmental factors influence the overall voice quality in the room and impact your experience; these factors include:

- Participant seating position
- Participant voice volume
- Ambient noise level
- Reverberation level
- Floor, wall, and ceiling materials
- Ceiling fans
- Projector placement

When Dolby Voice Room is deployed in acoustically challenging rooms, we recommend testing the setup to determine whether the actual results meet your requirements.

Related information

[Optimizing Dolby Voice Room for small rooms](#) on page 74

3.1.2 Power

The Dolby Voice Room system requires a nearby power outlet, 110–240 V, 50–60 Hz.

Dolby Voice Camera Pro requires its own power outlet, 100-240V, 50-60Hz, 0.8A.

Dolby Voice Camera Pro does not have a power control button on the camera.

 **Note:** Be aware that the system powers on as soon as it is plugged in, without the need to press the power button. Therefore, avoid connecting the system to power until you have connected all hardware components.

3.1.3 Display

Dolby Voice Room supports single-display and dual-display configurations, where the display is either a TV or monitor. The dual-display configuration, in which the two displays are arranged side by side, acts as a single wide display. Check with your conferencing service provider to see if they support the dual-display configuration.

This table summarizes display requirements.

Table 1: Display Requirements

Rear clearance	If you mount the Dolby Voice Camera on the display, allow 3 inches (7.6 cm) clearance from the front of the display bezel.
Positioning for camera field of view	If you intend to mount the Dolby Voice Camera on the display, keep in mind that the camera has a 95°-field of view. Consider also the position of the display relative to any whiteboard. For changing to whiteboard mode, we suggest that you be prepared to slide the camera along the top of the display to achieve the best whiteboard image quality.
Resolution	Supported resolutions: • 1080p • 4K (ultra-high definition, 3840 × 2160, 16:9) For a dual-display configuration, we recommend that the displays have the same native resolution.

Related information

[Whiteboard framing performance](#) on page 20

3.1.4 Cabling

An unshielded twisted-pair CAT6 cable is required to connect the Dolby Conference Phone to the Dolby Voice Hub .

If you choose to perform screen sharing with a wired connection between a computer and the system, you must also run a CAT2 (high-speed) high-definition video cable.



Note: You might also be able to perform screen sharing with a wireless connection to the system. Check with your conferencing service provider.

When the Dolby Voice Hub is mounted on the display, we recommend that you enclose the CAT6 and high-definition video cables in a cable-management system.

3.1.5 Lighting conditions and image quality

We recommend that you optimize lighting conditions for best image quality.

For example, ensure that the room is well lit. If it is not practical to adjust lighting conditions, you can adjust these image settings with the phone UI:

- Brightness
- Color intensity
- Contrast
- HDR (High Dynamic Range) Video Mapping



Note: Dolby Voice Camera Pro does not support HDR

Related information

[Adjusting the camera for lighting conditions](#) on page 50

3.1.6 Intelligent scene framing performance

Intelligent scene framing performance is the automatic zooming of the local conference room view, using face recognition to limit the view to the people in the room, eliminating unoccupied conference room space.

For optimal intelligent scene framing performance, the maximum recommended room sizes are:

- Small to medium conference rooms up to 15 ft × 15 ft (4.5 m × 4.5 m). Use Dolby Voice Camera
- Large conference rooms up to 20 × 40 ft (6 × 12 m). Use Dolby Voice Camera Pro.

If intelligent scene framing performance degrades in large rooms, activate room mode.

Intelligent scene framing is not supported for third-party cameras.

When using intelligent scene framing, the maximum distance from the person to the camera cannot exceed 20 ft. If the room length is beyond 20 ft, Dolby recommends disabling intelligent scene framing on the system.

3.1.7 Whiteboard framing performance

The whiteboard framing feature enables you to share whiteboard content with meeting participants. This requires a one-time setup procedure to configure the position of the whiteboard relative to Dolby Voice Camera or Dolby Voice Camera Pro. When you then switch the Dolby Voice Room system to whiteboard mode, the system fills the frame with the whiteboard.

For optimum whiteboard framing performance, the camera must be positioned at a minimum distance and minimum angle from the whiteboard.

The optimal distance from the Dolby Voice Camera to the whiteboard is less than 10 feet (3 meters), with the camera directly facing the whiteboard.

The maximum distance from the Dolby Voice Camera Pro to the whiteboard is 25 feet (6.5 meters).

Whiteboard framing can also work when the camera is at an angle to the whiteboard. When the camera is not directly facing the whiteboard, the output quality is determined by several factors:

- Size of the whiteboard
- Mounting position of the camera
- Distance from the camera to the whiteboard
- Angle between the camera and the whiteboard

 **Note:** The whiteboard framing feature works with any non-reflective surface, not just a whiteboard.

Related information

[Configuring a whiteboard](#) on page 51

3.2 Network requirements

Network connectivity must meet certain minimum requirements, and must be available in all locations where you intend to deploy Dolby Voice Room.

3.2.1 DHCP and DNS servers

Both Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) servers must be available to support Dolby Voice Room.

If a DHCP server is not available, you can set a static IP address. However, DHCP addressing is preferred, because it provides a simpler setup experience.

3.2.2 Firewall

Dolby Voice Room requires TCP port 443 for the domain *dolbyvoice.com*. Check with your conferencing service provider to determine other firewall requirements, including WebRTC port requirements.

3.2.3 VLAN

The Dolby Voice Room system requires internet access to connect to the conferencing service. In some environments, voice and data traffic are handled by different virtual LAN (VLAN)s.

3.2.4 Network ports used

The network port used by the Dolby Voice Room depends on protocol, data type, and whether it is sent by UDP or TCP.

Table 2: Network ports used

Port number	Protocol	Data	UDP or TCP
22	SSH	Diagnostics	TCP
80 ^[a]	Hypertext Transfer Protocol (HTTP)	Not applicable	TCP
443	Hypertext Transfer Protocol Secure (HTTPS)	Not applicable	TCP
5060	SIP	SIP signaling	UDP or TCP
5061	SIP over TLS	Secure signaling	TCP
16400–16431 ^[b]	RTP/RTCP	Media, audio	UDP
16400–16431 ^[c]	RTP/BFCP	Video	UDP

[a] Port 80 is always redirected to port 443.

[b] Real-time Transport Protocol (RTP) ports are always even numbers; RTP Control Protocol (RTCP) ports are always odd numbers. The numbers always start from 16,400/16,401 and progress to 16,402/16,403 and so forth until they reach the end of the range. At this point, they revert to 16,400/16,401 and start over again.

[c] RTP ports are always even numbers; RTCP ports are always odd numbers. The numbers always start from 16,400/16,401 and progress to 16,402/16,403 and so forth until they reach the end of the range. At this point, they revert to 16,400/16,401 and start over again.

3.2.5 802.1X

If your network uses 802.1X authentication, then you must understand the restrictions placed on devices that do not have 802.1X credentials. You must obtain 802.1X credentials for every Dolby Voice Room system.

You can customize these Dolby Voice Room supported authentication methods:

- EAP-PEAP: Requires the device to validate the CA certificate, and the user to enter a user name and/or password, for network authentication.
- EAP-TLS: Provides for certificate-based and mutual authentication of the client and the network. It relies on client-side and server-side certificates to perform authentication.
- EAP-MD5: Requires the creation of an MD5 hash of a random value from the network, and the user to enter a password, to authenticate.
- EAP-TTLS: The only supported secondary authentication method is MSCHAPv2.
- EAP-FAST with automatic proxy automatic configuration (PAC) provisioning: The system supports anonymous or authenticated in-band PAC provisioning. Manual PAC provisioning and the "do not use PAC" provisioning modes are not supported.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 31

[Configuring 802.1X after out-of-the-box setup](#) on page 32

3.2.6 HTTP proxy

Dolby Voice Room can use an HTTP proxy server to connect to services on the Internet.

Dolby Voice Room supports Web Proxy Automatic Discovery (WPAD) and PAC.

Static proxy server configuration is not supported.

3.2.7 NTP server

Dolby Voice Room can use a network time server, so your DHCP server should supply Network Time Protocol (NTP) server information.

If your DHCP server does not supply this information, you can configure NTP server information manually. If the desired NTP server is not on your network, ensure that your firewall rules allow connection to *time.nist.gov* and *pool.ntp.org*.

Related information

[Configuring the time server](#) on page 73

3.2.8 Bandwidth planning

Dolby Voice Room supports the H.264 video codec and the Opus 8 kHz (20ms) audio codec. Check with your conferencing service provider for audio and video bandwidth requirements.

Normally a minimum of 4 Mbps, for both upload and download, is recommended.

3.2.9 QoS limitation

Dolby Voice Room does not support parameter configurations for Quality of Service (QoS).

Deploying Dolby Voice Room

As the Dolby Voice Room system administrator, you work with IT to deploy the system in your organization.

- [Deployment overview](#)
- [Provisioning Overview](#)
- [Deploying the plug and play provisioning method](#)
- [Deploying local setup](#)
- [Handling network issues during provisioning](#)
- [Setting up individual Dolby Voice Room systems](#)
- [Setting up Dolby Voice Room with detached configuration](#)

4.1 Deployment overview

Deploying Dolby Voice Room in your organization consists of these tasks.

1. Select a method by which to provision your Dolby Voice Room systems.
2. Complete some deployment tasks to support the selected method.
3. Install and provision a Dolby Voice Room system to test the deployment. Handle network and other issues as needed.
4. After a successful test, set up Dolby Voice Room in your conference rooms and huddle spaces.

4.2 Provisioning Overview

As a Dolby Voice Room administrator, you must understand the Dolby Voice Room provisioning process. Provisioning refers to performing out-of-the-box (OOB) setup of the Dolby Voice Room system.

During provisioning, the system downloads configuration files, updates software if needed, and downloads the conferencing service provider application with which users interact. Provisioning can also be initiated on demand and configured to recur at regular intervals.

There are multiple ways to provision Dolby Voice Room:

- Plug and play provisioning method is a fast and easy way to set up Dolby Voice Room using your conferencing service provider's provisioning server. It makes trials and small-scale deployments of Dolby Voice Room easier for administrators. This is the preferred method.
- Local setup method is for users who want to use an on-site provisioning server, such as Dolby Conferencing Console. There are two local setup methods:
 - Automatic provisioning method
 - Manual provisioning method

When the Dolby Voice Room system starts up for the first time out of the box, it follows this provisioning procedure:

1. Contact the DHCP server to obtain an IP address and discover network resources, including automatic discovery of a proxy server.
2. Query the DNS server for the host name of a possible on-premises Dolby Conferencing Console.
3. Contact Dolby Voice Console in the cloud, and if Dolby Conferencing Console information was found, pass along that information.

Dolby Voice Console then determines which provisioning server the system should use: the on-premises Dolby Conferencing Console, if any (automatic provisioning method), or the conferencing service provider's provisioning server (plug and play provisioning method).

If the system is registered with Dolby Voice Console for conferencing service provider provisioning, and an on-premises Dolby Conferencing Console is present, the latter takes precedence. If the system is not registered with Dolby Voice Console, the system displays a screen asking the user to select a conferencing service provider.

Typically, your conferencing service provider or distributor registers your systems with Dolby Voice Console.

4. Download the provisioning information from the selected provisioning server, guide the user through the setup experience defined by the conferencing service provider, and reboot.

Errors might prevent the system from discovering a provisioning server. The most common errors are network issues, including:

- Physical connectivity issues
- HTTP/HTTPS proxy server issues
- 802.1X authentication issues

- VLAN issues
- NTP issues

The system displays manual setup screens to enable you to correct errors, for example, by entering network parameters or missing credentials. After you correct the errors, the system then attempts to complete the provisioning process. At certain points in this process, you can choose to skip the remainder of the steps and start the manual provisioning method.

For more information on Dolby Conferencing Console, see *Dolby Conferencing Console Administrator's Guide*.

4.3 Deploying the plug and play provisioning method

The plug and play provisioning method requires coordination with your conferencing service provider.

Prerequisites

Ensure that you have completed all preparations and requirements, keeping in mind that all Dolby Voice Room systems must connect to a provisioning server outside the firewall.

Procedure

1. Confirm with your conferencing service provider that you have adequate service accounts to deploy Dolby Voice Room.
2. If your network uses 802.1X authentication, obtain credentials for every Dolby Voice Room system.

Related information

[Preparation and requirements](#) on page 17

4.3.1 Testing a plug and play provisioning deployment

After you deploy the plug and play provisioning method, test it by provisioning a single Dolby Voice Room system.

Procedure

1. Unbox, install, and power up a new Dolby Voice Room system.
 - a) Mount Dolby Voice Hub. Connect one or two displays to the high-definition video ports labeled 1 and 2 on the Dolby Voice Hub. Do not connect power yet.
 - b) Connect Dolby Voice Camera to one of the USB ports.
 - c) Connect Dolby Conference Phone, taking care to connect the CAT6 cable to the PoE port on Dolby Voice Hub. This port is labeled with a Dolby Conference Phone icon.
 - d) Connect the network cable to the Dolby Voice Hub Ethernet port labeled with the network icon.
 - e) Connect power.

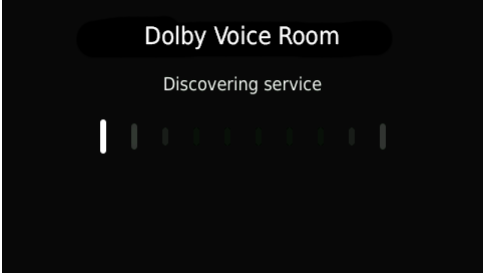
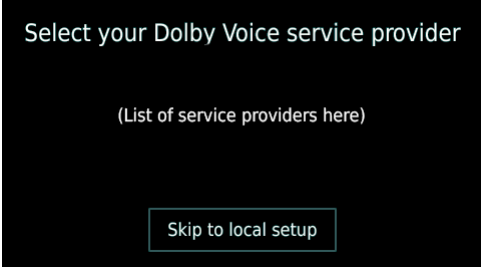
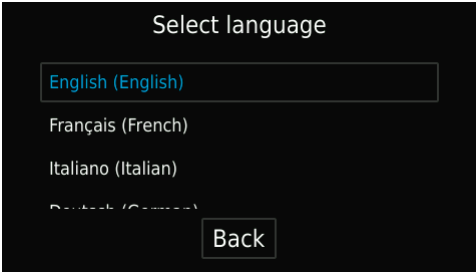
Refer to the *Dolby Voice Room Quick Start Guide* for more information.



Note: Be aware that the system powers on as soon as it is plugged in, without the need to press the power button. Therefore, avoid connecting the system to power until you have connected all hardware components.

2. Observe the Dolby Conference Phone display.

The screen progression starts with the Dolby Voice logo, and then **Getting ready, please wait**. The next screen may take a minute or more to appear. The following table shows the possible screens that may appear next.

Screen	Meaning/action
Discovering service, followed by Completing setup, and then Rebooting. 	Provisioning successful. After reboot, the conferencing service provider application launches. Refer to your service provider's documentation for information on any additional setup steps.
Select your Dolby Voice service provider 	Connection to Dolby Voice Console succeeded, but this Dolby Voice Room system is not registered. Do one of the following: - Select a service provider. The plug and play provisioning method continues, and if successful, the system reboots and launches the service provider application. - Tap Skip to local setup to manually specify the URL of an on-premises Dolby Conferencing Console for provisioning.
Select language 	Dolby Voice Room was unable to discover a provisioning server. Proceed as follows to determine the reason: - Select a language. Scroll with your finger if needed. - Select a time zone. The Setup unsuccessful screen displays, with a message explaining the issue. For example, the message might state that the system was unable to contact the DHCP server or that 802.1X credentials are needed. - Click Next. - On the Network setup screen, work with the menu choices to address the issue. After correcting the issue and returning to the Network setup screen, click Next to continue provisioning.

Related information

[Handling network issues during provisioning](#) on page 30

[Troubleshooting](#) on page 99

[Error Messages](#) on page 119

4.4 Deploying local setup

You can configure Dolby Voice Room for local setup if you require a configuration that is more advanced than the plug and play provisioning method.

You can deploy Dolby Voice Room using local setup for these scenarios:

- You want to control the firmware and application version that runs on Dolby Voice Room with Dolby Conference Phone.
- You have more than six Dolby Voice Rooms and want to control the Dolby Voice Room configuration, such as IP PBX configuration.
- You want an aggregated report for both Dolby Voice Room and IP PBX usage in your rooms.

There are two local setup methods:

- Automatic provisioning method that simplifies deployment and minimizes configuration errors for large or multiple-site deployment scenarios. Uses an on-premises Dolby Conferencing Console, which you manage, to provision each Dolby Voice Room system.
- Manual provisioning method used if Dolby Voice Room did not successfully discover a provisioning server, or if you want manual control.

4.4.1 Deploying the automatic provisioning method

We recommend using the automatic provisioning method. Using this method, the out-of-the-box setup wizard can discover the Dolby Conferencing Console or the provisioning server on your network. The automatic provisioning method is an on-premises, bulk method that simplifies deployment and minimizes configuration errors for multiple-site deployment scenarios.

Prerequisites

- Verify that you have administrative access to the DNS server.
- Ensure that your Dolby Conferencing Console is reachable on the network.
- Ensure that you have completed all preparations and requirements.

Procedure

1. Prepare the configuration files on Dolby Conferencing Console. Create profiles and device pools. See the *Dolby Conferencing Console Administrator's Guide* for details.
2. Create a Domain Name System service record (DNS SRV) for your on-premises Dolby Conferencing Console.
3. If your network uses 802.1X authentication, obtain credentials for every Dolby Voice Room system.

Related information

[Preparation and requirements](#) on page 17

[Setting and viewing configuration parameters](#) on page 41

Testing an automatic provisioning deployment

After you deploy the automatic provisioning method, test it by provisioning a single Dolby Voice Room system.

Procedure

1. Unbox, install, and power up a new Dolby Voice Room system.
 - a) Mount Dolby Voice Hub. Connect one or two displays to the high-definition video ports labeled 1 and 2 on the Dolby Voice Hub. Do not connect power yet.
 - b) Connect Dolby Voice Camera to one of the USB ports.
 - c) Connect Dolby Conference Phone, taking care to connect the CAT6 cable to the PoE port on Dolby Voice Hub. This port is labeled with a Dolby Conference Phone icon.
 - d) Connect the network cable to the Dolby Voice Hub Ethernet port labeled with the network icon.
 - e) Connect power.

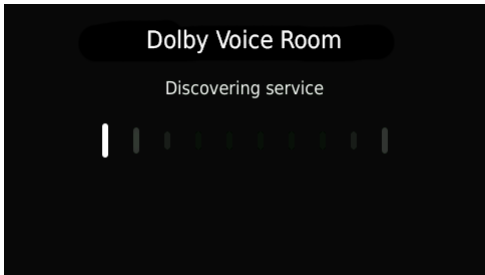
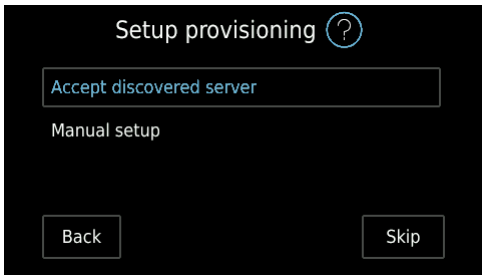
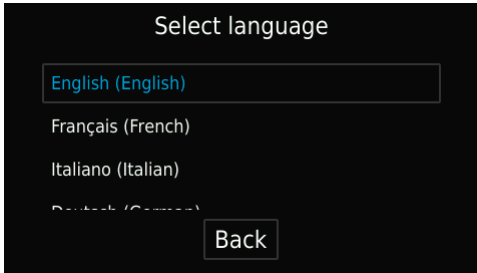
Refer to the *Dolby Voice Room Quick Start Guide* for more information.



Note: Be aware that the system powers on as soon as it is plugged in, without the need to press the power button. Therefore, avoid connecting the system to power until you have connected all hardware components.

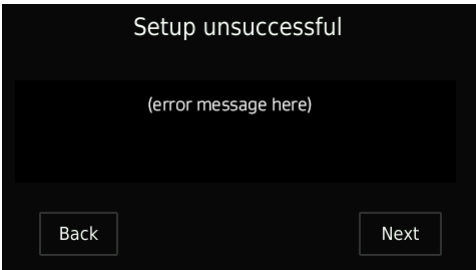
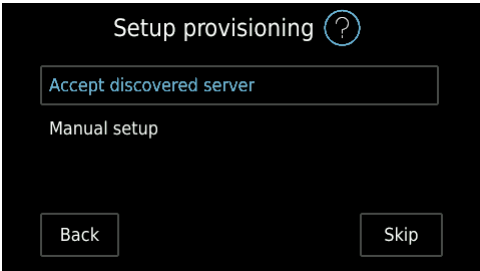
2. Observe the Dolby Conference Phone display.

The screen progression starts with the Dolby Voice logo, and then **Getting ready, please wait**. The next screen may take a minute or more to appear. The following table shows the possible screens that may appear next.

Screen	Meaning/action
Discovering service, followed by Completing setup, and then Rebooting. 	Provisioning was successful. After reboot, the conferencing service provider application launches. Refer to your service provider's documentation for information on any additional setup steps. Skip the rest of this procedure.
Setup provisioning 	Dolby Voice Room successfully contacted the on-premises Dolby Conferencing Console for provisioning information. Do one of the following: • Tap Accept discovered server to complete automatic provisioning. (Tap the question mark icon to view the host address of the discovered server.) Skip the rest of this procedure. • Tap Skip to stop the provisioning process. The system loads a default application that includes a Settings icon, so that you can complete setup later.
Select language 	One of these occurred: • Dolby Voice Room was unable to discover a provisioning server. • Dolby Voice Room could not contact Dolby Voice Console, but then successfully contacted the on-premises Dolby Conferencing Console for provisioning information.

3. If the **Select Language** screen displayed, select a language (scroll with your finger if needed), and then select a time zone.

The system displays one of the following screens.

Screen	Meaning/action
Setup unsuccessful  Includes a message describing an issue that prevented provisioning. For example, the message might state that the system was unable to contact the DHCP server or that 802.1X credentials are needed.	1. Click Next. 2. On the Network setup screen, work with the menu choices to correct the issue. After correcting the issue and returning to the Network setup screen, click Next to continue provisioning.
Setup provisioning 	Dolby Voice Room successfully contacted the on-premises Dolby Conferencing Console for provisioning information. Do one of the following: • Tap Accept discovered server to complete automatic provisioning. (Tap the question mark icon to view the host address of the discovered server.) • Tap Manual setup to manually specify a provisioning server. After entering the server protocol (HTTP or HTTPS) and address, tap Save to continue provisioning with the specified server. • Tap Back to go to the Network setup screen. • Tap Skip to stop the provisioning process. The system loads a default application that includes a Settings icon, so that you can complete setup later.

Related information

[Resetting to factory defaults](#) on page 93

[Handling network issues during provisioning](#) on page 30

[Troubleshooting](#) on page 99

[Error Messages](#) on page 119

DNS SRV record syntax for the automatic provisioning method

When this DNS SRV record is present, Dolby Voice Room can discover the on-premises Dolby Conferencing Console for provisioning.

Syntax

The DNS SRV record is an entry in the forward zone file and has this syntax:

```
_service._protocolName IN SRV priority weight portNO target subdns.domain.com
```

_service._protocolName

_dccc._tcp if you want the system to connect to Dolby Conferencing Console with HTTPS, or *_dcc._tcp* if you want the system to connect to Dolby Conferencing Console with HTTP. If records for both of these protocols are present, HTTPS takes precedence.

priority

Numeric. Lower gets first priority.

weight

Numeric. Higher gets used more often.

portNO

The port used.

443: The typical port used by the HTTPS provisioning server.

80: The typical port used by the HTTP provisioning server.

subdns.domain.com

The fully qualified domain name (FQDN) of the Dolby Conferencing Console on your network.

4.4.2 Deploying the manual provisioning method

Use the manual provisioning method if the system did not discover the Dolby Conferencing Console or a provisioning server, or if you want manual control. The manual provisioning method begins when you skip the selection of a service provider during system setup.

About this task

The first part of out-of-the-box setup involves network setup. You can use the setup wizard to connect the system to the network manually, if desired. After you complete this portion and have connected to the network, the next step is connecting to the provisioning server.

Procedure

1. Connect the Dolby Voice Room system to an a PoE ethernet port on the network using an ethernet cable.
The setup wizard displays on the phone UI.
2. If a provisioning server is not found, tap **Manual setup**.
3. From the **Select language** screen, choose a language. If needed, scroll up or down to find the language.
4. From the **Select time zone** screen, choose a time zone. If needed, scroll up or down to find the time zone.
5. **Optional:** From the **Network setup** screen, you can change your virtual LAN settings and proxy settings, if needed.
6. If challenged for 802.1X authentication, tap **Confirm certificate**, provide the valid credentials, and then follow the instructions in [Deploying with 802.1X authentication](#).
You will be presented with a certificate, depending on the protocol method you selected.
7. Wait for the phone to connect to the network.

What to do next

After you have connected to the network, continue through the setup wizard to connect to the provisioning server.

To reenable Link Layer Discovery Protocol, you must do it manually in **Settings** either on the phone UI or with the Dolby Conferencing Console.

4.5 Handling network issues during provisioning

Network issues might prevent provisioning from completing. Here are the common network setup tasks that you might need to perform during provisioning.

About this task

If the provisioning process halts at the **Select language** screen, this indicates that Dolby Voice Room was unable to connect to a provisioning server. From the **Network setup** screen, you can correct network issues and then complete the provisioning process.

Procedure

- Do one of the following:
 - Correct any network issues, and then tap **Refresh** to query the network again and update this screen.
 - Tap one of the menu choices on this screen, depending on the error message received, and address the issue, after which this screen redisplay. Tap **Next** to continue provisioning.

4.5.1 Configuring 802.1X during out-of-the-box setup

If your network requires 802.1X during the network setup portion of the setup wizard, you are required to complete 802.1X setup. By default, 802.1X authentication is disabled during out-of-the-box setup.

About this task

You can use configuration parameters, the phone UI, or the web interface to enable 802.1X authentication.

Once you have enabled 802.1X authentication for Dolby Voice Room, you can customize the protocol methods you want to use.

 **Note:** Dolby recommends using attached mode in an 802.1X environment. Attached mode is when Dolby Voice Room is connected to Dolby Conference Phone, using an Ethernet cable on a Dolby Conference Phone port.

Procedure

- Set the configuration parameter as described in the following table.

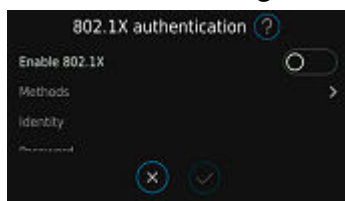
Configuration parameter name	Configuration parameter description
Network.Eap01.Enabled	Enables/disables 802.1X authentication. Set to false.

- To enable 802.1X authentication from the web interface:

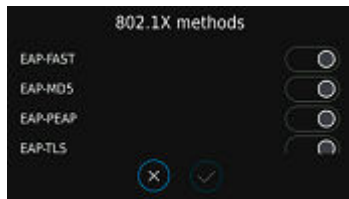
 **Note:** If you need a Wi-Fi connection to access your device, see *Web administration portal access over Wi-Fi* in the *Troubleshooting* chapter.

- Access the Dolby Voice Room web interface.
 - Navigate to **Settings > Network > IEEE 802.1X**.
 - Toggle **Enable IEEE 802.1X Authentication** to turn 802.1X authentication on.
- To enable 802.1X authentication from the phone UI:
 - Access the Dolby Voice Room **Settings** screen.
 - Tap this sequence:

Administrative settings > Network configuration > 802.1X

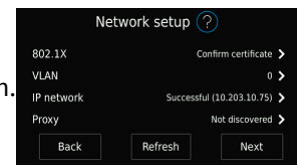


- Toggle **Enable 802.1X** to turn 802.1X authentication on.
After enabling 802.1X, you can customize protocol methods, by selecting **Methods** from the **802.1X authentication** screen.
- Enable the protocol methods you want to use, and tap the confirm button.



Note: Depending on the authentication server configuration, you may be prompted to enter authentication credentials before you accept the server certificate.

The **Network setup** screen returns, showing successful authentication.



e) Tap **Next** to continue provisioning.

What to do next

After you complete these steps, return to the provisioning task that you were working on before you were prompted to confirm this certificate.

Related information

[Configuring 802.1X after out-of-the-box setup](#) on page 32

[Primary and secondary VLANs](#) on page 33

[Changing the VLAN setting](#) on page 36

[Manually specifying a provisioning server](#) on page 37

[802.1X](#) on page 21

4.5.2 Configuring 802.1X after out-of-the-box setup

You can configure 802.1X authentication after you have completed the out-of-the-box setup.

About this task

If a guest VLAN is not configured, you must use the embedded UI to enable 802.1X authentication.

If a guest VLAN is permitted, you can use the embeddedUI and web UI to enable 802.1X authentication.

Note: Dolby recommends using attached mode in an 802.1X environment. Attached mode is when Dolby Voice Room is connected to Dolby Conference Phone, using an Ethernet cable on a Dolby Conference Phone port.

Procedure

- Set the configuration parameters as described in the following table.

Configuration parameter description	Configuration parameter name
Enables or disables 802.1X authentication	Network.Eap01.Enabled
Sets the Institute of Electrical and Electronics Engineers (IEEE) 802.1X user identity	Network.Eap01.Identity

Configuration parameter description	Configuration parameter name
Sets the IEEE 802.1X password	Network.Eapol.Password
Sets the 802.1X authentication protocol method	Network.Eapol.EnableEapFast
	Network.Eapol.EnableEapMd5
	Network.Eapol.EnableEapPeap
	Network.Eapol.EnableEapTls
	Network.Eapol.EnableEapTtls

- Use the web interface:
 - a) Access the Dolby Voice Room web interface.
 - b) Navigate to **Settings > Network > IEEE 802.1X**.
 - c) Toggle **Enable IEEE 802.1X Authentication** to turn 802.1X authentication on.

Go to **Settings > Network > IEEE 802.1X**, and enter the parameters or toggle the buttons to enable the protocol methods.

- Use the phone UI:
 - a) Access the Dolby Voice Room **Settings** screen.
 - b) Tap this sequence: **Administrative settings > Network configuration > 802.1X**
 - c) Toggle **Enable 802.1X** to turn 802.1X authentication on.
 - d) Select **Methods**
 - e) Enable the protocol methods you want to use, and tap the confirm.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 31

[Primary and secondary VLANs](#) on page 33

[Changing the VLAN setting](#) on page 36

[Manually specifying a provisioning server](#) on page 37

[802.1X](#) on page 21

4.5.3 Primary and secondary VLANs

The Dolby Voice Room system requires internet access to connect to the conferencing service. In some environments, voice and data traffic are handled by different VLANs.

Note:

If you are converting your system to a dual VLAN configuration, we recommend doing it attached mode.

If the voice VLAN lacks Internet connectivity, modify firewall policies to allow the system to connect to your conferencing service. Alternatively, you can set up the system on the data VLAN. If neither of these solutions works for you, then you need a dual-VLAN setup. After you enable dual VLANs, communication over the Internet with servers for your conferencing service provider passes through the secondary VLAN. With separate VLANs, the system has two IP addresses, one for each VLAN, but the same media access control (MAC) address.

When the secondary VLAN is enabled, the primary VLAN is designated as the voice VLAN, and the secondary VLAN is designated as the data VLAN. When the secondary VLAN is disabled, the system can be placed on either the data VLAN or the voice VLAN. If your organization has a strict policy on segregating voice and data VLANs, and you want to configure the system for both IP PBX and conferencing service, enable the secondary VLAN.

Check the following to verify VLAN status:

DNS Server Status

Displays status of DNS for the voice and data VLAN.

The **Status** screens with voice and data VLAN settings are shown in these figures.

Figure 2: Status screen displaying Voice VLAN settings

The screenshot displays the 'Network' status page. It includes sections for Ethernet settings, DHCP configuration, Virtual LAN settings, and a comprehensive 'Alarms and Status' section with various system indicators.

Network	
Host Name	VCS8000-E0001178
Ethernet	
Ethernet Duplex	full
Ethernet Speed (Mbps)	1000
Total Packets Received	13122
Total Packets Transmitted	6648
DHCP	
DHCP Server Address	10.0.0.1
Virtual LAN	
VLAN ID In Use	0
Alarms and Status	
DHCP Status	✓
DNS Server Status	✓
Default Gateway Status	✓
Hostname Resolvable	!
IEEE 802.1X Authentication Method	Unknown
IEEE 802.1X Authentication Phase 2 Method	Unknown
IEEE 802.1X Authentication Status	●
IEEE 802.1X Client Certificate	Dolby Factory Installed Certificate
IP Address Conflict	
Link Layer State	✓
NTP Server Discovery	✓
PAC Status	●
PDS Connection Status	●
Voice VLAN Status	●
WPA Status	●

Figure 3: Status screen displaying Data VLAN settings

+
Network

-
Network - Secondary VLAN

Alternative DNS Server Address
DNS Server Address
Default Gateway
Domain Name
IP Address
Subnet Mask

+
DHCP

-
Status

DHCP Status

DNS Server Status

Default Gateway Status

IP Address Conflict

PAC Status

Secondary VLAN Status

WPAD Status

Network traffic is allocated onto different VLANs based on the traffic type, as listed in this table.

Table 3: Traffic-type-to-network-interface mapping

Traffic type	Network interface	Comment
DHCP	Primary and secondary	Works independently on both
WPAD	Primary and secondary	If dual VLAN is enabled, the secondary is supported.
PAC	Primary and secondary	If dual VLAN is enabled, the secondary is supported.
DNS ^[a]	Primary and secondary	DNS queries made on the network interface where the DNS requestor application runs
Conferencing services	Secondary	
Lightweight Directory Access Protocol (LDAP)	Secondary	
Web server	Primary and secondary	
SSH server	Primary and secondary	
Syslog	Secondary	
NTP	Secondary	

Table 3: Traffic-type-to-network-interface mapping (continued)

Traffic type	Network interface	Comment
Provisioning	Secondary	
SIP call	Primary	

[a] If the primary VLAN shares the same DNS settings, then DNS traffic runs only on the primary virtual LAN.

Dual VLAN limitations

Dual VLAN has these limitations:

- In detached mode, users can directly plug into the network and use Bluetooth to automatically discover the IP address of the Dolby Voice Hub.
- In attached mode, if dual VLAN is enabled, a voice VLAN must be assigned to the Dolby Voice Hub.
- In detached mode, Dolby Conference Phone must be assigned to the untagged VLAN.
- The secondary VLAN must be untagged on the Dolby Voice Hub.
- On Dolby Voice Room, provisioning requests can only be performed on the secondary VLAN.
- If dual VLAN is enabled for Dolby Voice Room in a 802.1x environment and the Cisco ethernet switch, you must use multi-host mode on the Cisco switch.
- If dual VLAN is enabled, the primary VLAN proxy is disabled.
- Secondary VLAN IP assignment is only supported through DHCP, user cannot assign static IP to the secondary VLAN.

Voice VLAN checklist

Check the following to determine whether your network is ready for dual VLAN deployment.

- Manually configure the VLAN ID to place the device on the voice VLAN.
- Verify that the network configuration is working.

Data VLAN checklist

- Remove the VLAN ID manual configuration.
- Verify that the network configuration is working.

If both checklist tests work

If both tests work, you can proceed with the dual VLAN configuration.

 **Note:** When the dual-VLAN feature is used in conjunction with a Cisco switch port with 802.1X authentication enabled, the Ethernet port on the switch must be configured to Multiple-Hosts mode. If the port is set to Multidomain Authentication or Multiauthentication mode, dual VLAN does not work on the system.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 31

[Configuring 802.1X after out-of-the-box setup](#) on page 32

[Changing the VLAN setting](#) on page 36

[Manually specifying a provisioning server](#) on page 37

4.5.4 Changing the VLAN setting

Change the VLAN setting if your VLAN 0 (the default) has no internet access and thus prevents provisioning from proceeding. Or if provisioning succeeds, you can change the VLAN later if you choose not to use the default VLAN.

About this task

This procedure assumes that a VLAN issue occurred, preventing provisioning from completing, and that you have viewed the error message on the **Setup unsuccessful** screen and tapped **Next** to go to the **Network setup** screen.

Procedure

1. Access the Dolby Voice Room settings menu.
This may be different for each service provider.
2. Enter the admin password.
3. Tap **Network Configuration > Ethernet > Manual VLAN ID**.
4. Enter the **Manual VLAN ID**.
5. Tap **Enter**.
6. Tap the confirm button.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 31

[Configuring 802.1X after out-of-the-box setup](#) on page 32

[Primary and secondary VLANs](#) on page 33

[Manually specifying a provisioning server](#) on page 37

4.5.5 Manually specifying a provisioning server

During out-of-the-box setup, you can specify a provisioning server to use by manually entering the server address.

About this task

There are a few ways to manually specify a provisioning server:

- With the phone UI
- With the web interface

Using the phone UI

1. On the **Network setup** screen, tap **Next**.
2. On the **Warning** screen, tap **Continue**.
3. On the **Setup provisioning** screen, tap **Manual setup**.



Note: Tapping **Skip** enables you to skip the provisioning process, in which case a default application is loaded. This application includes a **Settings** button that enables you to complete the provisioning process later.

4. On the next screen, select the protocol and enter the server address.
5. Tap **Save**.

Using the web interface

1. Access the Dolby Voice Room web interface.
2. Go to **Settings > Provisioning > Server**.
3. Enable **Manual Server Configuration**.
4. Specify the **Provisioning Server Type**.
5. Enter the **Provisioning Server Address**.
6. Click **Save**.

If the server configuration is valid, provisioning proceeds with the specified server.

Related information

[Configuring 802.1X during out-of-the-box setup](#) on page 31

[Configuring 802.1X after out-of-the-box setup](#) on page 32

[Primary and secondary VLANs](#) on page 33

[Changing the VLAN setting](#) on page 36

4.6 Setting up individual Dolby Voice Room systems

After deploying and testing the desired provisioning method, you can install and set up individual Dolby Voice Room systems.

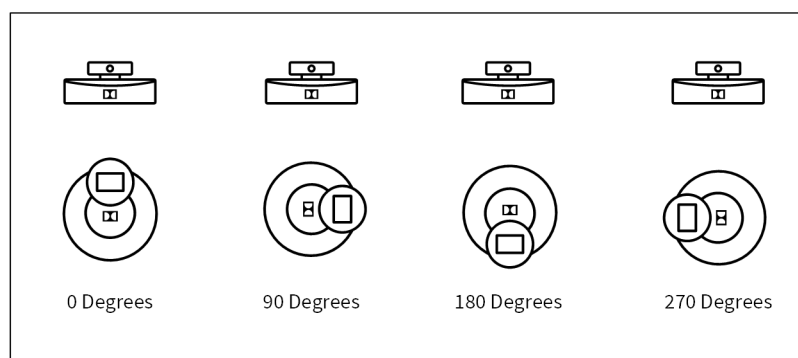
Procedure

1. Unbox, install, and power up a new Dolby Voice Room system.
 - a) Mount Dolby Voice Hub. Connect one or two displays to the high-definition video ports labeled 1 and 2 on the Dolby Voice Hub. Do not connect power yet.
 - b) Connect Dolby Voice Camera to one of the USB ports.
 - c) Connect Dolby Conference Phone, taking care to connect the CAT6 cable to the PoE port on Dolby Voice Hub. This port is labeled with a Dolby Conference Phone icon.
 - d) Connect the network cable to the Dolby Voice Hub Ethernet port labeled with the network icon.
 - e) Connect power.

Refer to the *Dolby Voice Room Quick Start Guide* for more information.

 **Note:** Be aware that the system powers on as soon as it is plugged in, without the need to press the power button. Therefore, avoid connecting the system to power until you have connected all hardware components.

2. Go through the setup steps. Refer to the documentation from your conferencing service provider and this guide for information on:
 - Testing a plug and play provisioning deployment
 - Testing an automatic provisioning deployment
 - Handling network issues during provisioning
3. **Optional:** For accurate spatial audio, align the Dolby Conference Phone in one of the orientations shown in the following diagram, and then configure the **Spatial capture origin** parameter to indicate the chosen orientation. The default is 0 degrees.



- a) Access the Dolby Voice Room web interface.
- b) Expand **Settings** > **Dolby Conference Phone** > **Audio** and set **Spatial capture origin** to the angle that corresponds to the chosen orientation.

Related information

[Testing a plug and play provisioning deployment](#) on page 25

[Testing an automatic provisioning deployment](#) on page 27

[Handling network issues during provisioning](#) on page 30

[Accessing the Dolby Voice Room web interface](#) on page 77

4.7 Setting up Dolby Voice Room with detached configuration

Dolby Voice Room supports detached configuration, meaning that the Dolby Conference Phone does not have to be plugged into the Dolby Voice Hub's PoE port for the system to function.

Prerequisites

- Dolby Conference Phone must be configured while connected to Dolby Voice Hub prior to setting up Dolby Voice Room with detached configuration.
- Ensure that the network port that the Dolby Conference Phone is plugged into and configures the Dolby Conference Phone to be able to connect to the Dolby Voice Hub's IP address.

About this task

A static IP address is not required for the Dolby Voice Hub to work in detached mode. Once Dolby Conference Phone has been paired with Dolby Voice Hub, Dolby Conference Phone will automatically follow the Dolby Voice Hub IP address changes.



Note: Ignore the error message on the Dolby Voice Hub requesting you to plug the Dolby Conference Phone into the Dolby Voice Hub.

Procedure

1. Set up Dolby Voice Room using the plug and play provisioning method.

You must plug the Dolby Conference Phone into the Dolby Voice Hub's PoE port and follow the instructions to set the system up and verify that the system works properly with your conferencing service provider.

2. Detach the Dolby Conference Phone from the Dolby Voice Hub, and plug the Dolby Conference Phone into a separate network port.



Note: Ignore the error message on the Dolby Voice Hub requesting you to plug the Dolby Conference Phone into the Dolby Voice Hub.

Related information

[Dolby Conference Phone pairing issues](#) on page 117

5

Configuring Dolby Voice Room features

As the Dolby Voice Room administrator, you can configure many features of the system, including network settings, corporate directory features, display and camera settings, calling features, and more. You configure these features by setting and changing configuration parameters.

- [Configuration parameters](#)
- [Setting and viewing configuration parameters](#)
- [Configuring certificates](#)
- [Changing display settings](#)
- [Changing audio input and output devices](#)
- [Configuring Dolby Voice Camera and Dolby Voice Camera Pro](#)
- [Configuring IP PBX](#)
- [Configuring preferences](#)
- [Configuring contact directories](#)
- [Configuring the time server](#)
- [Configuring Dolby Voice Room to use USB audio devices](#)
- [Optimizing Dolby Voice Room for small rooms](#)
- [Disabling the reboot prompt](#)
- [Disabling Bluetooth advertisement](#)

5.1 Configuration parameters

A large set of configuration parameters controls Dolby Voice Room features and settings. The system receives and stores the majority of configuration parameters from the provisioning server. You can manually set other configuration parameters using one of the interfaces that the system provides.

A detailed description of all configuration parameters, along with the default or permitted values for those parameters, is available in the **Help** section of the web interface.

 **Note:** If a configuration parameter is set by the provisioning server, you cannot change it manually.

When you set Dolby Voice Room configuration parameters, you are setting them for Dolby Voice Hub. Dolby Voice Hub then sets any required configuration parameters for Dolby Conference Phone.

Default values

If a configuration parameter has not been set manually and is not present in the configuration files on the provisioning server, a default value is used.

Configuration parameters requiring reboot

Some configuration parameters require a reboot when changed. If the system requires a reboot, a reboot prompt displays on the phone user interface and web interface. If you do not cancel the reboot within 20 seconds, the system reboots automatically. Active calls are not interrupted; the reboot is delayed until those calls have ended. You can turn off the reboot prompt if desired.

Device pools and profiles

With Dolby Conferencing Console, you can easily assign multiple configuration settings to multiple devices in a single operation. You can define device pools, which are collections of Dolby Voice Room systems, and you can create profiles, which are collections of configuration parameters and their values. You can then apply profiles to pools. See the *Dolby Conferencing Console Administrator's Guide* for details.

 **Note:** If you use the plug and play provisioning method, you cannot configure features for a pool of systems, because your systems are provisioned by your conferencing service provider. You can still configure features one system at a time (for example, you can configure IPPBX settings, which are not provisioned by your conferencing service provider's provisioning service).

Related information

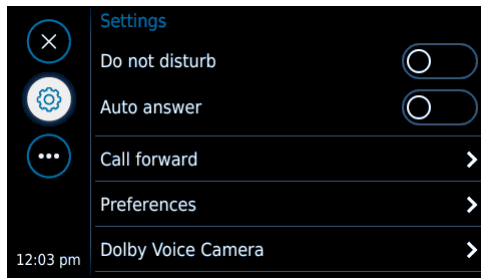
[Disabling the reboot prompt](#) on page 74

5.2 Setting and viewing configuration parameters

There are a few ways to set and view configuration parameters.

Procedure

- Use the phone UI.
 - a) Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

b) **Optional:** For advanced settings tap

Note: If a setting is dimmed, it means that this configuration parameter is set by the provisioning server and cannot be changed manually.

- Use the web interface.

Note: If a setting is dimmed, it means that this configuration parameter is set by the provisioning server and cannot be changed manually.

- Use the **configure** command in the diagnostic shell to set a parameter.
For example, to configure the system to automatically answer calls, enter this command:

```
configure Preferences.AutoAnswerEnabled true
```

To set the logging level to DEBUG for the SIP protocol, enter this command:

```
configure Logging.Level.Sip DEBUG
```

- Use the **show configuration** command in the diagnostic shell to view a parameter.
For example, to see how long a web interface or diagnostic shell session remains active before login is required again, enter this command:

```
show configuration Authentication.Local.SessionTimeout
```

To show all configuration parameters, enter this command:

```
show configuration all
```

To get a list of related configuration parameters, such as all parameters related to PBX, enter the first portion or first two portions of the parameter name, up to the period, and press Tab.

```
show configuration Sip.Pbx. (Tab)
```

```
Sip.Pbx.CallFailoverMode
Sip.Pbx.CallFailoverTimeout
Sip.Pbx.Codecs
Sip.Pbx.ContentShareEnabled
Sip.Pbx.Domain
Sip.Pbx.EnableP2PSIPVideo
Sip.Pbx.EnableURIDialingProxy
Sip.Pbx.InBandDTMF
Sip.Pbx.OnlyAcceptCallServerEvents
Sip.Pbx.OutboundProxy1
Sip.Pbx.OutboundProxy1Port
Sip.Pbx.OutboundProxy2
Sip.Pbx.OutboundProxy2Port
Sip.Pbx.RegistrationTimeout
Sip.Pbx.SipUriScheme
Sip.Pbx.TLSCertNameMatchingPolicy
Sip.Pbx.URIDialingEnabled
Sip.Pbx.URIDialingProxy
Sip.Pbx.URIDialingTransport
```

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

[Administering with the diagnostic shell](#) on page 78

5.3 Configuring certificates

The collection of certificate authority (CA) and device certificates installed on the Dolby Voice Room system is a security feature and is used for authentication with features such as HTTPS provisioning, LDAP TLS connection, TLS SIP signaling encryption, 802.1X, and conferencing services that support Dolby Voice.

The system can trust certificates issued by widely recognized certificate authorities when trying to establish a connection to a server. To establish new trust relationships, you can add new certificates to the system through Dolby Conferencing Console or the web interface.

5.3.1 Device certificate

A device certificate is installed on each Dolby Voice Room system at the factory.

The device certificate is an X.509 digital certificate signed by the Dolby root certificate authority. The digital certificate and associated private key are stored on the system in its nonvolatile memory as part of the manufacturing process. For more information on digital certificates, see www.ietf.org/html.charters/pkix-charter.html and IETF RFC 2459.

The Dolby Voice Room system has the following certificate installed:

Issuer name	Expiration date	CA Certificate download link
.Dolby.Comms.Devices.CA.1	Expires before 2050	https://certificates.dolbyvoice.com/dolby_comms_devices_ca1_chain.cer



Note: The connected Dolby Conference Phone may have two different sets of certificates installed. In order to configure some network security features that may require a device certificate, you need to identify the certificate installed on the Dolby Conference Phone. For more information about viewing the CA certificate issuer information, see the *Dolby Conference Phone Administrator's Guide*.

Related information

[Adding certificate authority certificates](#) on page 43

[Viewing certificate authority certificates status](#) on page 44

[Trusted certificates and issuers](#) on page 45

5.3.2 Adding certificate authority certificates

A certificate that is issued by a certificate authority establishes a point of trust between two entities, for example, the Dolby Voice Room system and the LDAP directory or the provisioning server.

If you want the system to trust any CAs, you must add their certificates to the system in a CA certificate directory. This file must be in Privacy-Enhanced Electronic Mail (PEM) encoded format.

You can add CA certificates from:

- Dolby Conferencing Console

For details, see the *Dolby Conferencing Console Administrator's Guide*.

- The web interface

1. Access the Dolby Voice Room web interface.

2. From the **Tools > Certificates > Import CA certificates**.

A standard file-selection dialog box displays.

3. Select the CA certificate file and click **Open**.

 **Note:** Certificates imported through the web interface are deleted after a factory reset. You must re-import the certificates.

Related information

[Device certificate](#) on page 43

[Viewing certificate authority certificates status](#) on page 44

[Trusted certificates and issuers](#) on page 45

[Accessing the Dolby Voice Room web interface](#) on page 77

5.3.3 Viewing certificate authority certificates status

With the CA certificates status screen, you can view the number of certificates that are expired, are about to expire, or are not yet valid.

About this task

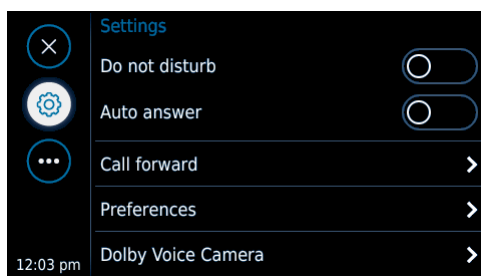
All certificates are validated every time the certificate store changes, as well as every 24 hours if scheduled provisioning is enabled. Warnings and error messages are available from the phone UI and the web interface. If a certificate expires, the **Expired** count increases on the CA certificates status screen. If a certificate is set to expire within seven days, the **Expiring soon** count increases on the CA certificates status screen. You must replace the certificate before expiry, or certain features will stop working on the system.

Figure 4: Errors and warnings on CA certificates status screens



Procedure

- To view CA certificates status from the phone user interface:
 - a) Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

- b) Tap this sequence:

 > **About > Status > CA Certificates**

- To view CA certificates status from the web interface:

- a) From the **Status** tab, click **Dolby Voice Hub**, then **Device**, then **CA Certificates**.
- b) **Optional:** To see a list of certificates, including a list of expired certificates, click the link next to **CA Certificates Details**.

i Tip: A list of expired certificates appears before the full list of certificates. You can select the hexadecimal number in the error message, copy it to the clipboard, and then use your browser's find command to find the matching certificate in the full list of certificates.

Related information

[Device certificate](#) on page 43

[Adding certificate authority certificates](#) on page 43

[Trusted certificates and issuers](#) on page 45

5.3.4 Trusted certificates and issuers

Dolby Voice Room trusts the certificates listed here.

Table 4: Trusted certificates

Certificate name	Issuer
*.lobby.pp.iconf.net	Entrust Certification Authority - L1K
Amazon Root CA 1	Starfield Services Root Certificate Authority - G2
Amazon Root CA 1	Self-signed
Amazon Root CA 2	Self-signed
Amazon Root CA 3	Self-signed
Amazon Root CA 4	Self-signed
Baltimore CyberTrust Root	Baltimore CyberTrust Root
Baltimore CyberTrust Root	Baltimore CyberTrust Root
C=US/O=Starfield Technologies, Inc./OU=Starfield Class 2 Certification Authority	http://www.valicert.com// emailAddress=info@valicert.com
C=US/O=The Go Daddy Group, Inc./OU=Go Daddy Class 2 Certification Authority	C=US/O=The Go Daddy Group, Inc./OU=Go Daddy Class 2 Certification Authority
C=US/O=VeriSign, Inc./OU=Class 2 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network	C=US/O=VeriSign, Inc./OU=Class 2 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network
C=US/O=VeriSign, Inc./OU=Class 3 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network	C=US/O=VeriSign, Inc./OU=Class 3 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network
Cybertrust Public SureServer SV CA	Baltimore CyberTrust Root
DigiCert Assured ID Root CA	DigiCert Assured ID Root CA
DigiCert Assured ID Root G2	DigiCert Assured ID Root G2
DigiCert Assured ID Root G3	DigiCert Assured ID Root G3
DigiCert Global Root CA	DigiCert Global Root CA
DigiCert Global Root G2	DigiCert Global Root G2
DigiCert Global Root G3	DigiCert Global Root G3
DigiCert High Assurance EV Root CA	DigiCert High Assurance EV Root CA

Table 4: Trusted certificates (continued)

Certificate name	Issuer
DigiCert SHA2 High Assurance Server CA	DigiCert High Assurance EV Root CA
DigiCert Trusted Root G4	DigiCert Trusted Root G4
DOLBY-PSM	DOLBY-PSM
Entrust Root Certification Authority	Entrust Root Certification Authority
Entrust Root Certification Authority - G2	Entrust Root Certification Authority - G2
Entrust.net Certification Authority (2048)	Entrust.net Certification Authority (2048)
GeoTrust Global CA	GeoTrust Global CA
GeoTrust Global CA 2	GeoTrust Global CA 2
GeoTrust Primary Certification Authority	GeoTrust Primary Certification Authority
GeoTrust Primary Certification Authority - G3	GeoTrust Primary Certification Authority - G3
GeoTrust Universal CA	GeoTrust Universal CA
GeoTrust Universal CA 2	GeoTrust Universal CA 2
GlobalSign	GlobalSign
GlobalSign Root CA	GlobalSign Root CA
Go Daddy Root Certificate Authority - G2	Go Daddy Root Certificate Authority - G2
Google Internet Authority G3	GlobalSign
Microsoft Root Certificate Authority 2011	Microsoft Root Certificate Authority 2011
Microsoft Secure Server CA 2011	Microsoft Root Certificate Authority 2011
O=RSA Security Inc/OU=RSA Security 2048 V3	O=RSA Security Inc/OU=RSA Security 2048 V3
Starfield Root Certificate Authority - G2	Starfield Root Certificate Authority - G2
Starfield Services Root Certificate Authority - G2	C=US/O=Starfield Technologies, Inc./OU=Starfield Class 2 Certification Authority
Symantec Class 3 Secure Server CA - G4	VeriSign Class 3 Public Primary Certification Authority - G5
TC TrustCenter Class 2 CA II	TC TrustCenter Class 2 CA II
TC TrustCenter Universal CA I	TC TrustCenter Universal CA I
thawte Primary Root CA	thawte Primary Root CA
thawte Primary Root CA - G3	thawte Primary Root CA - G3
Thawte Universal CA Root	Thawte Universal CA Root
VeriSign Class 3 Public Primary Certification Authority - G5	VeriSign Class 3 Public Primary Certification Authority - G5
VeriSign Universal Root Certification Authority	VeriSign Universal Root Certification Authority

Related information

[Device certificate](#) on page 43

[Adding certificate authority certificates](#) on page 43

[Viewing certificate authority certificates status](#) on page 44

TLS cipher suites

The Dolby Voice Room offers a predefined list of cipher suites. This list cannot be changed at this time.

 **Note:** The current SRTP implementation supports the AES_CM_128_HMAC_SHA1_80 and AES_CM_128_HMAC_SHA1_32 cipher suites. It does not support the F8_128_HMAC_SHA1_8 cipher suite.

Table 5: TLS cipher suites

Cipher	Cipher suites
0x2f	TLS_RSA_WITH_AES_128_CBC_SHA
0x30	TLS_DH_DSS_WITH_AES_128_CBC_SHA
0x31	TLS_DH_RSA_WITH_AES_128_CBC_SHA
0x35	TLS_RSA_WITH_AES_256_CBC_SHA
0x36	TLS_DH_DSS_WITH_AES_256_CBC_SHA
0x37	TLS_DH_RSA_WITH_AES_256_CBC_SHA
0x3c	TLS_RSA_WITH_AES_128_CBC_SHA256
0x3d	TLS_RSA_WITH_AES_256_CBC_SHA256
0x3e	TLS_DH_DSS_WITH_AES_128_CBC_SHA256
0x3f	TLS_DH_RSA_WITH_AES_128_CBC_SHA256
0x41	TLS_RSA_WITH_CAMELLIA_128_CBC_SHA
0x42	TLS_DH_DSS_WITH_CAMELLIA_128_CBC_SHA
0x43	TLS_DH_RSA_WITH_CAMELLIA_128_CBC_SHA
0x84	TLS_RSA_WITH_CAMELLIA_256_CBC_SHA
0x85	TLS_DH_DSS_WITH_CAMELLIA_256_CBC_SHA
0x86	TLS_DH_RSA_WITH_CAMELLIA_256_CBC_SHA
0x9c	TLS_RSA_WITH_AES_128_GCM_SHA256
0x9d	TLS_RSA_WITH_AES_256_GCM_SHA384
0xa0	TLS_DH_RSA_WITH_AES_128_GCM_SHA256
0xa1	TLS_DH_RSA_WITH_AES_256_GCM_SHA384
0xa4	TLS_DH_DSS_WITH_AES_128_GCM_SHA256
0xa5	TLS_DH_DSS_WITH_AES_256_GCM_SHA384
0xc004	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA
0xc005	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA
0xc009	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
0xc00a	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
0xc00e	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
0xc00f	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
0xc013	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
0xc014	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
0xc023	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
0xc024	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
0xc025	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256

Table 5: TLS cipher suites (continued)

Cipher	Cipher suites
0xc026	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
0xc027	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
0xc028	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
0xc029	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
0xc02a	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
0xc02b	TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
0xc02c	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
0xc02d	TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256
0xc02e	TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384
0xc02f	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
0xc030	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
0xc031	TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256
0xc032	TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384

5.4 Changing display settings

Using the web interface, you can set the preferred display resolution and, in a dual-display configuration, swap images between the displays.

Procedure

1. Access the Dolby Voice Room web interface.
2. Navigate to **Settings > Displays**
3. Change one of these settings:
 - **Swap monitor position**
Toggle the switch position to swap images between displays in a dual-display configuration.
 - **Preferred resolution**
Select the resolution recommended by your conferencing service provider. Dolby Voice Room uses that resolution if supported by the displays. The system falls back to a lower resolution if the displays do not support the selected resolution.
 - **Enable display mirroring**
Enable to mirror the image of one display to another in a dual-display configuration

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

5.5 Changing audio input and output devices

Using the web interface, you can manually change audio input and output devices. By default, the **Audio Device Selection** option is set to **Automatic** and audio input and output device selection is performed automatically at initialization. Audio input and output device manual selection options are dependent on paired or available devices.

Procedure

1. Access the Dolby Voice Room web interface.
2. Under the **Settings** tab, expand **Audio** and select **Manual** for **Audio Device Selection**.
3. Manually select your audio device by modifying these settings:

Option	Description
Input Audio Device	Select Dolby Conference Phone or USB Device .
Output Audio Device	Select Dolby Conference Phone or USB Device .

5.6 Configuring Dolby Voice Camera and Dolby Voice Camera Pro

Configure Dolby Voice Camera and Dolby Voice Camera Pro to take advantage of some of the advanced features of Dolby Voice Room.

 **Note:** Your conferencing service provider may provide their own methods for configuring Dolby Voice Camera and Dolby Voice Camera Pro. Refer to their documentation for details.

5.6.1 Camera modes

Dolby Voice Camera and Dolby Voice Camera Pro operate in three different modes.

Room mode

Captures a full view of the room, using the entire 95-degree field of view. Your conferencing service provider might provide manual pan, tilt, and zoom (PTZ) controls.

People mode

Enables intelligent scene framing, which uses face recognition to adjust the camera view automatically so that the people in the room fill the frame. In larger rooms with potentially more people, it is often desirable to turn off this feature.

Whiteboard mode

Fills the frame with a whiteboard, enabling you to share drawings with meeting participants. Requires a one-time configuration of the whiteboard location.

 **Note:** Your conferencing service provider's application might have different names for these modes.

Related information

[Changing the camera mode](#) on page 49

[Adjusting the camera for lighting conditions](#) on page 50

[Vertically flipping the camera position](#) on page 51

[Configuring a whiteboard](#) on page 51

[Creating and editing presets using PTZ control](#) on page 53

[Creating and editing presets using PTZ control during a SIP video call](#) on page 55

5.6.2 Changing the camera mode

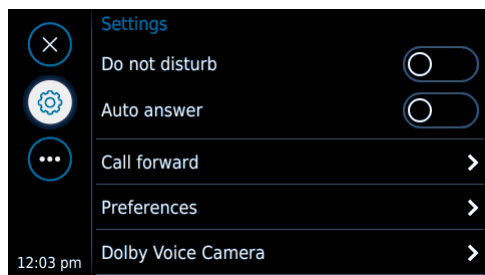
Use the phone user interface to select room mode, people mode, or whiteboard mode.

About this task

 **Note:** This procedure is available from the **Settings** screen. Your conferencing service provider's application might provide an alternate way to change the camera mode.

Procedure

1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap **Dolby Voice Camera** > **Change mode**.

3. Select a mode from the menu.

Related information

[Camera modes](#) on page 49

[Adjusting the camera for lighting conditions](#) on page 50

[Vertically flipping the camera position](#) on page 51

[Configuring a whiteboard](#) on page 51

[Creating and editing presets using PTZ control](#) on page 53

[Creating and editing presets using PTZ control during a SIP video call](#) on page 55

5.6.3 Adjusting the camera for lighting conditions

Use the phone UI to adjust the quality of the image captured by the camera.

About this task

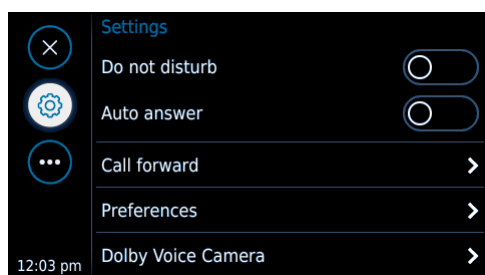
You can adjust the brightness, contrast, and color intensity (saturation) of the image. You can also turn on the HDR feature, which you can use to compensate for bright window light in the camera field of view. These adjustments are set to their default values when you clear user preferences.

- Tip:** For a room with even lighting and without an outside-facing window, we recommend turning HDR off.

Note: Dolby Voice Camera Pro does not support HDR.

Procedure

1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap **Dolby Voice Camera**.

An image preview is shown full screen on the display.

3. Tap **Adjust image quality**.

4. On the **Adjust image quality** screen, adjust the sliders and optionally turn HDR on or off.



Related information

- [Camera modes](#) on page 49
- [Changing the camera mode](#) on page 49
- [Vertically flipping the camera position](#) on page 51
- [Configuring a whiteboard](#) on page 51
- [Creating and editing presets using PTZ control](#) on page 53
- [Creating and editing presets using PTZ control during a SIP video call](#) on page 55

5.6.4 Vertically flipping the camera position

You can vertically flip the camera position using the web interface or through the command-line interface.

Procedure

- From the web interface:
 - a) Using the web interface, select **Settings > Dolby Voice Camera > Flip image vertically**.
 - b) Select **On**.
 - c) Click **Save**.
- From the command-line interface, enter:


```
DolbyCli>configure DVCamera.VerticalFlip <string>(false)
```

The parameter values are true and false.

Related information

- [Camera modes](#) on page 49
- [Changing the camera mode](#) on page 49
- [Adjusting the camera for lighting conditions](#) on page 50
- [Configuring a whiteboard](#) on page 51
- [Creating and editing presets using PTZ control](#) on page 53
- [Creating and editing presets using PTZ control during a SIP video call](#) on page 55

5.6.5 Configuring a whiteboard

Before you can use whiteboard mode, you must configure the location of the whiteboard in the conference room.

Prerequisites

We recommend that you draw something on the whiteboard and clearly mark all four corners so that you can evaluate the quality of the configuration.

About this task

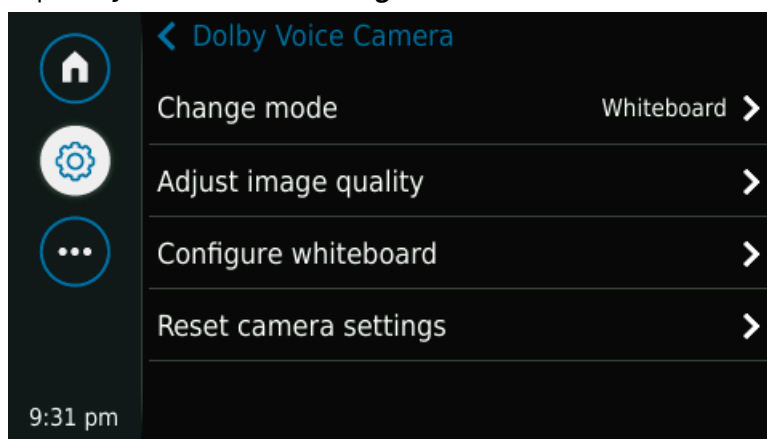
The following procedure is available from the **Settings** screen. Your conferencing service provider's application might provide an alternate way to configure a whiteboard. When configuring your whiteboard

for optimal whiteboard framing performance, review the relevant information about angle and distance limitations.

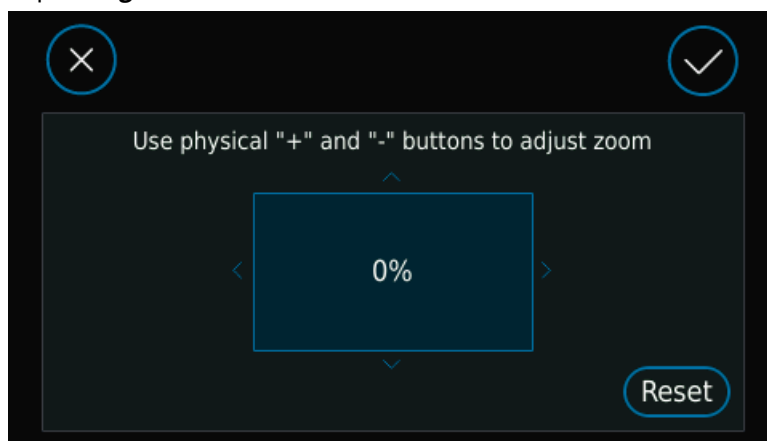
Note: The following procedure is available from the **Settings** screen. Your conferencing service provider's application might provide an alternate way to configure a whiteboard.

Procedure

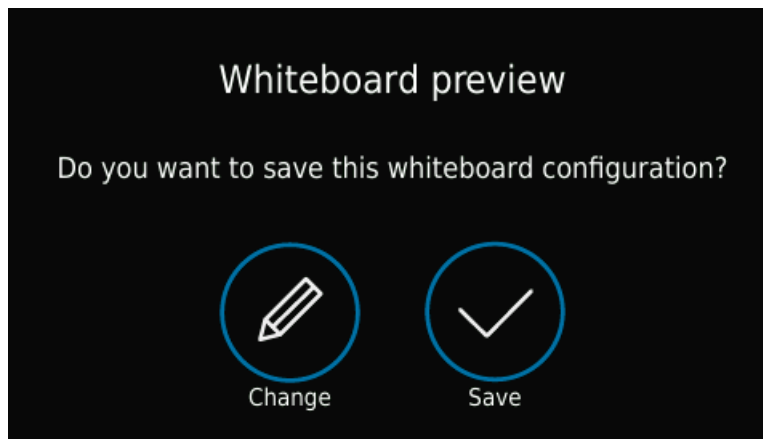
1. Access the Dolby Voice Room **Settings** screen.
2. Tap **Dolby Voice Camera** > **Change mode** > **Whiteboard**.



3. Tap **Configure whiteboard**.



4. Drag the rectangle to set the PTZ settings.
5. Use the physical +/- buttons to adjust the zoom.
This step is only needed for Dolby Voice Camera Pro.
6. Tap the check mark icon, once you have set the PTZ and zoom settings.
7. Drag the corners of the rectangle, so that the capture area rectangle on the display lines up with the corners of the whiteboard.
8. Tap the forward arrow when you have framed the whiteboard.
You can preview the whiteboard



9. Tap **Save** when you are satisfied with the alignment.

10. On the Whiteboard preview screen, you can:

- Tap **Save** to save the whiteboard configuration.
- Tap **Change** to revise the whiteboard configuration.

Related information

[Camera modes](#) on page 49

[Changing the camera mode](#) on page 49

[Adjusting the camera for lighting conditions](#) on page 50

[Vertically flipping the camera position](#) on page 51

[Creating and editing presets using PTZ control](#) on page 53

[Creating and editing presets using PTZ control during a SIP video call](#) on page 55

[Whiteboard framing performance](#) on page 20

5.6.6 Creating and editing presets using PTZ control

You can create and edit up to three presets and assign specific pan, tilt, and zoom settings for each one.

About this task

You can only create and edit presets in Room Mode. Otherwise, this option is not available.

Procedure

1. Access the Dolby Voice Room **Settings** screen.

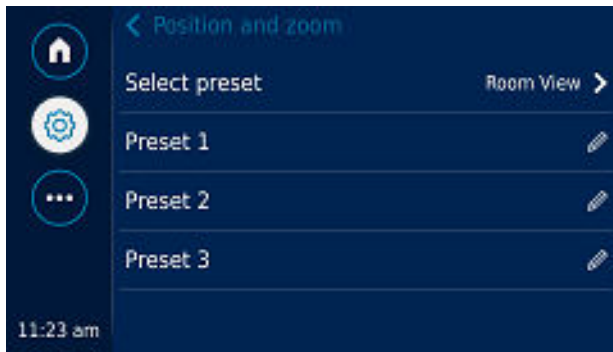
2. Tap **Dolby Voice Camera**.

The **Dolby Voice Camera** screen displays.

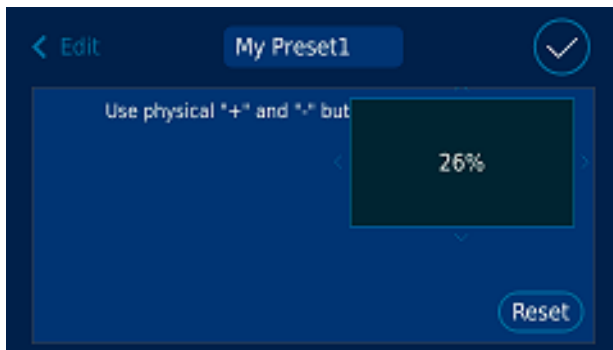


3. Tap **Position and zoom**.

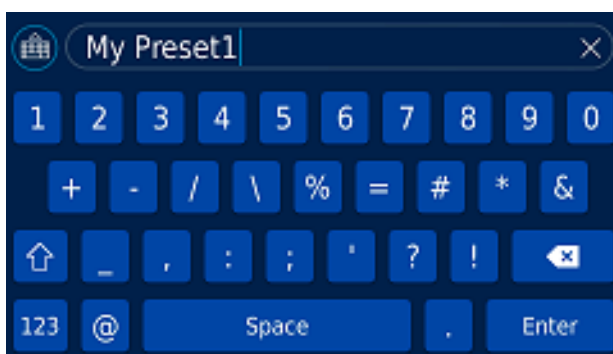
The **Position and zoom** screen displays.



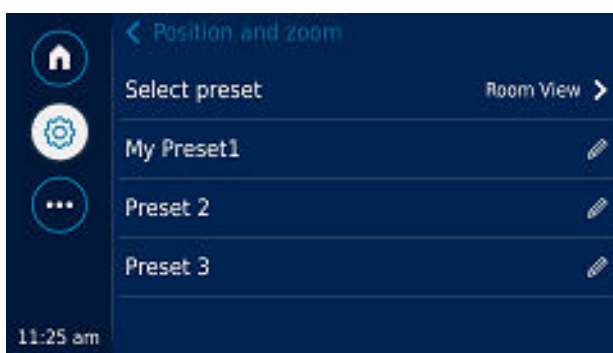
4. Select the preset you want to create or edit and click the edit button. The **Preset** screen displays.



5. Drag the rectangle to adjust the pan and tilt positions.
6. Use the +/- buttons or the sliding scale to adjust the zoom.
7. Click the confirm button to save your changes.
8. Click in the Preset box to rename the preset.



9. Click **Enter** to save the name.



Related information

[Camera modes](#) on page 49

[Changing the camera mode](#) on page 49

[Adjusting the camera for lighting conditions](#) on page 50

[Vertically flipping the camera position](#) on page 51

[Configuring a whiteboard](#) on page 51

[Creating and editing presets using PTZ control during a SIP video call](#) on page 55

5.6.7 Creating and editing presets using PTZ control during a SIP video call

During a SIP video call, you can create and edit up to three presets and assign specific pan, tilt, and zoom settings for each one.

About this task

You can only create and edit presets in Room Mode. Otherwise, this option is not available.

Procedure

1. While on a SIP video call, tap **Camera > Setup**.
2. Select the preset you want to create or edit.
The **Preset** screen displays.
3. Drag the rectangle to adjust the pan and tilt positions.
4. Use the -/+ buttons or the sliding scale to adjust the zoom.
5. Click the confirm button to save your changes.
6. Click in the Preset box to rename the preset.
7. Tap the confirm button.

Related information

[Camera modes](#) on page 49

[Changing the camera mode](#) on page 49

[Adjusting the camera for lighting conditions](#) on page 50

[Vertically flipping the camera position](#) on page 51

[Configuring a whiteboard](#) on page 51

[Creating and editing presets using PTZ control](#) on page 53

5.7 Configuring IP PBX

To use Dolby Conference Phone as an IP telephone to make IP audio calls, you must configure IP PBX settings.

About this task

Use the web interface or the Dolby Conferencing Console to configure the account settings for your IP PBX and register the device. This involves modifying the **Account**, **Server**, and **Credential** settings in the **IP PBX** section.

 **Note:** Check with your conferencing service provider for the recommended way to configure IPPBX. Some service providers might disable IPPBX functions.

Results

After you finish setting up the device for IP PBX, the phone icon on the home screen no longer appears dimmed.

5.7.1 Prerequisites for configuring IP PBX

Configuring IP PBX for the Dolby Voice Room system requires that you have information about the SIP domain name and the outbound proxy server address.

Your IP PBX service provider or IT department may provide the following information:

- One of the following SIP server/service connection information:
 - SIP domain name only.
 - SIP outbound proxy server address and/or SIP domain name. There may be two addresses for redundancy.
 - SIP domain name and SIP transport type.
 - SIP domain name, outbound proxy server address, and SIP transport type.
- SIP user account (required).
- SIP authentication user name; also known as the digest authentication user name (required).
- SIP authentication user password (required).

Related information

[Support for SIP high availability](#) on page 56

[Configuring IP PBX using the web interface](#) on page 57

[Configuring IP PBX using the Dolby Conferencing Console](#) on page 58

[Adjusting codec preferences](#) on page 61

[Enabling Opus on Dolby Voice Room](#) on page 61

[Configuring voice activation detection](#) on page 62

[Configuring IP PBX for SIP URI dialing](#) on page 63

[SIP video configuration](#) on page 64

5.7.2 Support for SIP high availability

You can configure IP PBX settings for the Dolby Voice Room to provide uninterrupted server availability and DNS-based redundancy.

The Dolby Voice Room supports the following for SIP high availability:

- Server failover/failback
- DNS redundancy
- DNS-based load balancing
- NAPTR record support

Server failover has two modes: `ACTIVE_SERVER` and `PREFERRED_ORDER`. The `ACTIVE_SERVER` mode switches/fails over to a secondary server after ending the call. The `PREFERRED_ORDER` mode attempts to fail over to the preferred list of IP addresses during the call. You can specify the server failover mode in the `Sip.Pbx.CallFailoverMode` field.

If your IP PBX service provider configures the Dolby Voice Room with only the SIP domain, the Dolby Voice Room performs the DNS SRV query on the domain host name. After four attempts to register to the IP address, the Dolby Voice Room attempts to register to the next IP in the DNS response.

If your IP PBX service provider configures the Dolby Voice Room with the SIP domain name and outbound proxy server address configured, the Dolby Voice Room performs the DNS SRV query on the outbound proxy server. After four attempts to register to the IP address, the Dolby Voice Room attempts to register to the next IP in the DNS response.

You can specify NAPTR record as the transport type in the `Sip.Account.TransportType` field.

The following table lists the various high-availability configuration options available:

Table 6: High availability deployment configuration options

IP PBX server configuration options	Outcome
Sip.Pbx.Domain as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.Domain field.
Sip.Pbx.Domain as <i>hostname</i> :5060	SIP requests are sent to the IP address resolving the host name configured in the Sip.Pbx.Domain field.
Sip.Pbx.Domain as <i>hostname</i> and Sip.Pbx.OutboundProxy1 as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.OutboundProxy1 field.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as IP address, and Sip.Pbx.OutboundProxy2 as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.OutboundProxy1 field or the IP address configured in the Sip.Pbx.OutboundProxy2 field.
Sip.Pbx.Domain as <i>hostname</i> or <i>hostname</i> :0 only and Sip.Account.TransportType set to NAPTR record	NAPTR record query (to discover transport) followed by SRV query (to discover port) followed by A query.
Sip.Pbx.Domain as <i>hostname</i> or <i>hostname</i> :0 only and Sip.Account.TransportType set to AUTOSRV	SRV query (to discover port), followed by A query.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as <i>hostname</i> , Sip.Pbx.OutboundProxy1Port as 0, and Sip.Account.TransportType as AUTO	SRV query (to discover port) followed by A query.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as <i>hostname</i> , Sip.Pbx.OutboundProxy1Port as 0, Sip.Pbx.OutboundProxy2 as <i>hostname</i> , Sip.Pbx.OutboundProxy2Port as 0, and Sip.Account.TransportType as AUTO.	SRV query (to discover port) followed by A query.

Related information

- [Prerequisites for configuring IP PBX](#) on page 56
- [Configuring IP PBX using the web interface](#) on page 57
- [Configuring IP PBX using the Dolby Conferencing Console](#) on page 58
- [Adjusting codec preferences](#) on page 61
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- [Configuring IP PBX for SIP URI dialing](#) on page 63
- [SIP video configuration](#) on page 64

5.7.3 Configuring IP PBX using the web interface

You can use the web interface to configure IP PBX for the Dolby Voice Room system.

Procedure

1. Access the Dolby Voice Room web interface.
2. Ensure that IP PBX calling is enabled. On the **Settings** tab, expand **Features** and choose **Dual Mode** for the **Operation Mode**.
3. On the **Settings** tab, navigate to **IP PBX Settings > Account** and enter this information:
 - In the **Display Name** field, enter the device name.
 - In the **Extension number/Address** field, enter the extension or SIP user account.
 - If the dial-in number is different from the extension number/address, enter the dial-in number in the **Display Number** field.

- If your IP PBX service provider has specified the transport type, choose the appropriate setting from **Transport Type**.
4. On the **Settings** tab, navigate to **IP PBX Settings > Server** and enter this information:
 - If a SIP domain is available, enter it in the **SIP Domain Name** field.
 - If an outbound proxy server is available, enter it in the **Primary Call Server/Outbound Proxy** field. To enable redundancy, enter an alternative outbound proxy server address in the **Secondary Call Server/Outbound Proxy** field.
 - If your IP PBX service provider requires a port other than 5060 or 5061, enter the port numbers in the **Primary Server/Outbound Proxy Port** and **Secondary Server/Outbound Proxy Port** fields.
 - To restrict or reorder the audio codecs used, modify the list in the **PBX Codecs List**. By default, the list value is *G722, G711U, G711A, G729AB*. For example, to only use only G.711, edit the list to include only *G711U, G711A*.

For specific instructions, see the configuration guide for your conferencing service provider or call control platform.

5. Expand the **Credential** section, and in the **User Credential Name** and **User Credential Password** fields, enter the SIP authentication user name and password, respectively.

If your IP PBX service provider requires a specific realm for authentication, enter it in the **Server Realm** field.

For specific instructions, see the configuration guide for your call control platform.

Related information

- [Prerequisites for configuring IP PBX](#) on page 56
- [Support for SIP high availability](#) on page 56
- [Configuring IP PBX using the Dolby Conferencing Console](#) on page 58
- [Adjusting codec preferences](#) on page 61
- [Enabling Opus on Dolby Voice Room](#) on page 61
- [Configuring voice activation detection](#) on page 62
- [Configuring IP PBX for SIP URI dialing](#) on page 63
- [SIP video configuration](#) on page 64
- [Operation modes of Dolby Voice Room](#) on page 15
- [Accessing the Dolby Voice Room web interface](#) on page 77

5.7.4 Configuring IP PBX using the Dolby Conferencing Console

You can use the Dolby Conferencing Console to configure IP PBX for the Dolby Voice Room system.

Procedure

1. Create a profile, and configure these parameters:

- In **Features > Operation Mode**:

Configuration parameter	Setting
Features.OperationMode	Dual Mode

- In **IP PBX Settings > Server**:

Configuration parameter	Setting
Sip.Pbx.Domain	The name of the SIP domain, if one is available.
Sip.Pbx.OutboundProxy1	The outbound proxy server address.

Configuration parameter	Setting
Sip.Pbx.OutboundProxyPort1	The outbound proxy server port number. Change the default value of 5060 if your IP PBX service provider requires a different port.
Sip.Pbx.OutboundProxy2	An alternative outbound proxy server address, if one is available.
Sip.Pbx.OutboundProxyPort2	The alternative outbound proxy server port number. Change the default value of 5060 if your IP PBX service provider requires a different port.
Sip.Pbx.SipUriScheme	Configures the URI scheme. Options are SIP and SIPS. The default is On = SIP.
Sip.Pbx.OnlyAcceptCallServerEvents	Configuration to allow SIP events from the call server only. If set to True, allows SIP events from the call server only. If set to False, allows SIP events from any address.
Sip.Pbx.CallFailoverMode	Configures call failover mode. Select one of these: <i>Active Server</i> - Call is always sent to the active server. <i>Preferred Server's Order</i> - Call is sent to the server in the order of preference.
Sip.Pbx.RegistrationTimeout	SIP registration time out. The range is from 4-59. The default is 32.
Sip.Pbx.CallFailoverTimeout	Call failover timeout. Range is 4-60. The default is 4.
Sip.Pbx.Codecs	Audio codecs list. Audio codecs used: G722, G711U, G711A, G729AB by default. Modify the list to specify allowable codecs or change order of preference. For example, to use only G.711, edit the list to include only G711U, G711A.
Sip.Pbx.InBandDTMF	Default is off. When set to Off out-of-band dual-tone multifrequency (DTMF) is enabled.
Sip.Pbx.EnableP2PSIPVideo	Enables Peer-to-peer SIP video calls through IP-PBX server. Default is off.
Sip.Pbx.ContentShareEnabled	Configures content share for SIP video calls is enabled or not. The default is On.
Sip.Pbx.URIDialingEnabled	Enables calls by URI dialing.
Sip.Pbx.URIDialingTransport	URI dialing transport. Select one of these: <i>None</i> <i>TLS</i> <i>TCP</i> <i>UDP</i>
Sip.Pbx.URIDialingProxy	URI dialing proxy server address.

Configuration parameter	Setting
Sip.Pbx.EnableURIDialingProxy	URI Dialing Proxy server to route all the SIP URI calls, ignoring URIProxy parameter.
Sip.Pbx.TLSCertNameMatchingPolicy	Matches the DNS name with the certificate name using a wildcard rule. Select: <i>Strict Wildcard Matching</i> - If the certificate name is *.abc.com and the DNS name is 1234.abc.com they are matched, but DNS name abc.com does not. <i>Lenient Wildcard Matching</i> - Includes <i>Strict Wildcard Matching</i> and allows DNS name abc.com to be matched with *.abc.com <i>Ignore name matching</i> - No name matching at all. If the certificate is valid it is accepted. The default is <i>Strict Wildcard Matching</i>.

2. Assign the profile you created to the device pool that contains the systems you are configuring.

3. For each system in the device pool, set these parameters:

- In **IP PBX Settings > Account**:

Configuration parameter	Setting
Sip.Account.DisplayName	The name of the device.
Sip.Account.UserName	The extension or SIP address.
Sip.Account.DisplayNumber	The dial-in number, if different than the extension number/address.
Sip.Account.TransportType	The transport type. Select <i>AUTO</i> <i>UDP</i> <i>TCP</i> <i>TLS</i> <i>NAPTR</i> The default value is <i>AUTO</i>.

- In **IP PBX Settings > Credential**:

Configuration parameter	Setting
Sip.Credential.Name	The SIP authentication user name
Sip.Credential.Password	The SIP authentication user password
Sip.Credential.Realm	The realm for authentication, if your IP PBX service provider requires a specific realm

For specific instructions, see the configuration guide for your call control platform and the *Dolby Conferencing Console Administrator's Guide*.

Related information

- [Prerequisites for configuring IP PBX](#) on page 56
- [Support for SIP high availability](#) on page 56
- [Configuring IP PBX using the web interface](#) on page 57
- [Adjusting codec preferences](#) on page 61
- [Enabling Opus on Dolby Voice Room](#) on page 61

[Configuring voice activation detection](#) on page 62

[Configuring IP PBX for SIP URI dialing](#) on page 63

[SIP video configuration](#) on page 64

5.7.5 Adjusting codec preferences

Dolby Voice Room allows you to adjust the IP PBX codec preferences using the `Sip.Pbx.Codecs` parameter.

About this task

When the device functions as a VoIP phone, the `Sip.Pbx.Codecs` parameter determines which audio codecs are used and their priority order. By default, this parameter is set to `G722, G711U, G711A, G729AB`. You can add or remove codecs and change the order of codecs in the list. For example, you might want to move `G729AB` to the beginning of the list to reduce VoIP traffic across the wide area network (WAN).

This parameter is not required when the system is used in **Conferencing Only Mode**.

Procedure

- From the web interface:
 - a) On the **Settings** tab, expand **IP PBX Settings**.
 - b) Expand the **Server** section.
 - c) In the **PBX Codec List** field, add, remove, or change the order of codecs as needed.
- From the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Expand **IP PBX Settings > Server**.
 - e) Drag the **PBX Codec List** to the device configuration.
The `Sip.Pbx.Codecs` parameter appears in the device configuration.
 - f) In the **Sip.Pbx.Codecs** field, add, remove, or change the order of codecs as needed.

Related information

[Prerequisites for configuring IP PBX](#) on page 56

[Support for SIP high availability](#) on page 56

[Configuring IP PBX using the web interface](#) on page 57

[Configuring IP PBX using the Dolby Conferencing Console](#) on page 58

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[SIP video configuration](#) on page 64

5.7.6 Enabling Opus on Dolby Voice Room

You can enable Opus on Dolby Voice Room for WebRTC calls. Consult your conferencing service provider confirm if Opus must be turned on.

You can enable Opus by setting parameters or through the web UI.

Set the configuration parameters:

To enable Opus for Dolby Voice Room, set the `Dvms.Audio.EnableOpus` parameter to true. By default, this parameter is set to false.

Use the web UI :

1. Navigate to **Settings > Dolby Voice Conferencing Server > Audio**
2. Toggle the **Enable Opus** button.

Related information

[Prerequisites for configuring IP PBX](#) on page 56
[Support for SIP high availability](#) on page 56
[Configuring IP PBX using the web interface](#) on page 57
[Configuring IP PBX using the Dolby Conferencing Console](#) on page 58
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5.7.7 Configuring voice activation detection

The Dolby Voice Room by default signals G.729AB (G.729A with Annex B) to its peer. If your IP PBX configuration requires the device to signal G.729A, you can set the `Sip.Vad.SignalAnnexB` configuration parameter to `false` to override the default behavior.

About this task

The Dolby Voice Room supports the G.729AB codec, but it also allows Annex B in its Session Description Protocol (SDP) to be turned off.

You can perform this procedure from the web interface or from Dolby Conferencing Console.

Procedure

1. From the web interface:
 - a) From the **Settings** tab, expand the **IP PBX Settings** section, and then expand the **VAD** section.
 - b) Turn off `Signal G729 Annex B`.
This option controls the `Sip.Vad.SignalAnnexB` parameter.
2. Alternatively, from Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Navigate to **IP PBX Settings > VAD**.
 - e) Drag `Signal G729 Annex B` to the device configuration in the upper right-hand side of the screen.
The `Sip.Vad.SignalAnnexB` parameter appears in the device configuration.
3. Turn the `SIP.Vad.SignalAnnexB` parameter off.

Results

The device now signals peers using G.729A.

Related information

[Prerequisites for configuring IP PBX](#) on page 56
[Support for SIP high availability](#) on page 56
[Configuring IP PBX using the web interface](#) on page 57
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5.7.8 Configuring IP PBX for SIP URI dialing

You can use the web interface or the Dolby Conferencing Console to configure IP PBX for dialing using a SIP URI.

About this task

By default, SIP URI dialing is enabled and the configuration parameter is set to **TRUE**.

Procedure

- To disable SIP URI dialing from the web interface:
 - a) On the **Settings** tab, expand **IP PBX Settings**.
 - b) Expand the **Server** section.
 - c) Turn off **Enable SIP URI Dialing**.
- To disable SIP URI dialing from the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Expand **IP PBX Settings > Server**.
 - e) Drag **Enable SIP URI Dialing** to the device configuration.
The `Sip.Pbx.URIDialingEnabled` parameter appears in the device configuration.
 - f) Set **Sip.Pbx.URIDialingEnabled** to **FALSE**.

Related information

[Prerequisites for configuring IP PBX](#) on page 56

[Support for SIP high availability](#) on page 56

[Configuring IP PBX using the web interface](#) on page 57

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Making a call using SIP URI dialing

You can make a call using the SIP URI dialer after configuration through the web interface or the Dolby Conferencing Console. SIP URI dialing works for unregistered and unauthenticated SIP lines.

About this task

By default, SIP URI dialing uses TLS as the default protocol. Dolby Voice Room automatically adds `transport=tls` to the SIP URI if a transport parameter is not specified. You can specify an alternative transport protocol such as UDP or TCP. The Dolby Voice Room keyboard field supports a maximum of 128 characters.

Procedure

1. On the Dolby Voice Room UI, tap the **URI** button.

The onscreen keyboard displays with **sip:.**

2. Enter the URI, and tap the **Call** button to join the call.

5.7.9 SIP video configuration

Dolby Voice Room allows you to dial into third-party video conferences using SIP URI.

On Dolby Voice Room, SIP video calling is enabled by default. Dialing into video conferences using SIP URI does not require SIP registration. The default video codec used for URI-based video calls is H.264.

You can make video calls using UDP, TCP or TLS transport. The default is TLS. To make a call using TCP transport, the URI format is `user@domain.com;transport=tcp`. To make a call using TLS transport, the URI format is `user@domain.com;transport=tls`.

You also can make a SIP video call using only the host name or IP address.

If a video call needs to pass through a proxy server, you have to set the `Sip.Pbx.URIDialingProxy` parameter to specify the proxy server and set `uriproxy=1` to the URI. You can set this parameter in the Dolby Voice Room web UI configuration page by navigating to **IP PBX Settings > Server**.

If the `uriproxy=1` parameter is set, but the `Sip.Pbx.URIDialingProxy` parameter is not set, Dolby Voice Room will use the IP PBX configuration to make calls.

On video calls, you can:

- Disable the camera
- Use Room mode to capture the entire room
- Use People mode to capture individuals who come into the frame

For whiteboard and screen sharing, Dolby Voice Room supports full 1080p high definition (HD) content share.

To share your screen, your laptop must be connected to the Dolby Voice Hub. If it is not connected, it is grayed out on the TV display. If there are two monitors, one is the local preview and the other is remote preview.

To share the whiteboard, it must be preconfigured. Otherwise, the default view displays.

Dolby Voice Room provides a lock icon to identify a secure video connection. If the lock icon is white and locked, the connection is secure, and audio and video are transmitted using SRTP. If it is red and unlocked, it is insecure, and audio and video are transmitted using RTP.

Related information

- [Prerequisites for configuring IP PBX](#) on page 56
- [Support for SIP high availability](#) on page 56
- [Configuring IP PBX using the web interface](#) on page 57
- [Configuring IP PBX using the Dolby Conferencing Console](#) on page 58
- [Adjusting codec preferences](#) on page 61
- [Enabling Opus on Dolby Voice Room](#) on page 61
- [Configuring voice activation detection](#) on page 62
- [Configuring IP PBX for SIP URI dialing](#) on page 63

Making a SIP video call

You can dial into third-party video conferences using SIP URI.

About this task

You can make video calls using UDP, TCP or TLS transport. The default is TLS. To make a call using TCP transport, the URI format is `user@domain.com;transport=tcp`. To make a call using TLS transport, the

URI format is `user@domain.com;transport=tls`. The Dolby Voice Room keyboard field supports a maximum of 128 characters.

You can use the call log to view what type of calls were made on Dolby Voice Room by tapping the call log button. If it was a video call, a video icon displays.

Procedure

1. On the Dolby Voice Room UI, tap the video call button.
2. On the keypad screen, tap the **URI** button.
The onscreen keyboard displays with **SIP:**.
3. Enter the URI, and tap the **Call** button to join the call.

`sip:6505551212.com`

5.8 Configuring preferences

You can set up these preferences for your users, or you can allow users to set them themselves.

5.8.1 Disabling the recent-calls list

The Dolby Voice Room system can gather and display a list of the 100 most recently missed, placed, and received calls. This list is enabled by default.

About this task

The recent-calls list contains call information such as remote-party identification, time and date of the call, and call duration. The list is uploaded from the system to the provisioning server after each call is complete.

The list can be used to redial previous outgoing calls, return missed calls, and save contact information to the local contact directory. The list can be cleared manually by individual users. The list is retained when the system reboots or when the system software is upgraded.

Procedure

- Disable the recent-calls list with the web interface:
 - a) Access the web interface.
 - b) In the **Settings** tab, expand **Features**, and turn off **Enable Recent Calls**.
- Disable the recent-calls list by setting a configuration parameter.
 - a) Open the diagnostic shell.
 - b) Enter this command:

```
configure Features.RecentCallsEnabled false
```

Related information

[Configuring call forwarding](#) on page 65

[Enabling automatic answer](#) on page 66

[Enabling do not disturb](#) on page 67

[Setting the language](#) on page 67

[Setting the time and date display](#) on page 68

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

5.8.2 Configuring call forwarding

The system provides a flexible call-forwarding feature that enables you to forward incoming calls to another party.

About this task

There are three ways to apply call forwarding:

- To all calls
- When the system is busy
- When a call is not answered after 40 seconds

There are several ways to configure call forwarding:

Procedure

- Use the web UI:
 - a) Navigate to **Settings > User Preferences**.
 - b) Enter the phone number in the appropriate forwarding options.
- Use the phone UI:
 - a) Access the **Settings** screen .
 - b) tap **Call Forward**.
 - c) Enter the phone number in the appropriate forwarding option.
- Set these configuration parameters:

Configuration parameter name	Configuration parameter description
Preferences.ForwardAllNumber	Enter the number where all calls are forwarded.
Preferences.ForwardBusyNumber	Enter the number where the call is forwarded if the system is busy.
Preferences.ForwardNoAnswerNumber	Enter the number where the call is forwarded if the user does not answer.

To disable any of these call-forwarding options, specify a null string:

```
configure Preferences.ForwardAllNumber ""
```

Related information

[Disabling the recent-calls list](#) on page 65

[Enabling automatic answer](#) on page 66

[Enabling do not disturb](#) on page 67

[Setting the language](#) on page 67

[Setting the time and date display](#) on page 68

5.8.3 Enabling automatic answer

In certain circumstances, you may be unable to physically answer an incoming call. This feature allows you to configure the system to answer incoming calls automatically after the first ring. The call is answered with the microphone muted.

About this task

There are several ways to configure automatic answer:

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.AutoAnswerEnabled	Enable or disable automatic answer. Set to true or false.

- Use the webUI:
 - Go to **Settings > User Preferences**.
 - Select the automatic answer **Off/On** option.
- Use the phone UI:
 - Access the **Settings** screen.
 - Select the automatic answer **Off/On** option.

Related information

[Disabling the recent-calls list](#) on page 65

[Configuring call forwarding](#) on page 65

[Enabling do not disturb](#) on page 67

[Setting the language](#) on page 67

[Setting the time and date display](#) on page 68

5.8.4 Enabling do not disturb

You use the do not disturb feature to temporarily stop incoming calls and turn off audio alerts.

About this task

While do not disturb is turned on, incoming calls are sent a busy signal and are logged as missed.

There are several ways to configure the do not disturb feature.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.DndEnabled	Enable or disable do not disturb behavior. Set to true or false.

- Use the web UI:
 - Go to **Settings > User Preferences**.
 - Select the Do Not Disturb **Off/On** option.
- Use the phone UI:
 - Access the **Settings** screen.
 - Select the do not disturb **Off/On** option.

Related information

[Disabling the recent-calls list](#) on page 65

[Configuring call forwarding](#) on page 65

[Enabling automatic answer](#) on page 66

[Setting the language](#) on page 67

[Setting the time and date display](#) on page 68

5.8.5 Setting the language

You can change the language displayed on the phone UI. By default, English is shown.

About this task

There are several ways to set the language.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.Localization.Language	Possible values include: • English • French • German • Italian • Spanish • Portugal Portuguese • Brazilian Portuguese • Simplified Chinese • Traditional Chinese • Japanese • Korean • Russian • Czech • Hebrew • Arabic

- Use the web interface:
 - a) Navigate to **Settings > User Preferences > Localization**.
 - b) Select the language from the **Language** list.
- Use the phone UI:
 - a) Access the **Settings** screen.
 - b) Tap **Preferences > Language**
 - c) Select the language from the **Language** list.

Related information

[Disabling the recent-calls list](#) on page 65

[Configuring call forwarding](#) on page 65

[Enabling automatic answer](#) on page 66

[Enabling do not disturb](#) on page 67

[Setting the time and date display](#) on page 68

5.8.6 Setting the time and date display

A clock and calendar are enabled by default. You can display the time and date for your time zone in several formats. The system will synchronize with the configured time server.

About this task

The available date formats may change, depending on the language selection.

There are several ways to set the time and date display.

Procedure

- Set the configuration parameters:

Configuration parameter name	Configuration parameter description
Preferences.Localization.Timezone	Time zone. Permitted values are: • Pacific/Kwajalein • Pacific/Pitcairn • Pacific/Samoa • Pacific/Honolulu • America/Anchorage • America/Los_Angeles • America/Denver • America/Mazatlan • America/Phoenix • America/Chicago • America/Guatemala • America/Mexico_City • America/Bogota • America/New_York • America/Asuncion • America/Caracas • America/Campo_Grande • America/Santiago • America/Sao_Paulo • America/St_Johns • America/Argentina/Buenos_Aires • Atlantic/South_Georgia • Atlantic/Azores • UTC • Europe/London • Africa/Lagos • Europe/Paris • Africa/Johannesburg • Asia/Jerusalem • Asia/Baghdad • Asia/Tehran • Asia/Muscat • Asia/Kabul • Asia/Karachi • Asia/Kolkata • Asia/Kathmandu • Asia/Dhaka • Asia/Rangoon • Asia/Bangkok • Asia/Singapore • Asia/Tokyo • Australia/Adelaide • Australia/Darwin • Australia/Lord_Howe • Australia/Sydney • Asia/Magadan • Pacific/Auckland • Pacific/Tongatapu
Preferences.Localization.TimeFormat	Time format. Permitted values are: • <i>h:mm ap</i> (For 12-hour format. Enclose in quotes when specifying this value in the configure command.) • <i>h:mm</i> (For 24-hour format)
Preferences.Localization.DateFormat	Date format. Permitted values are:

- Use the web interface:
Go to **Settings > User Preferences > Localization**, and adjust the time and date parameters.
- Use the phone UI:
 - To change the time zone, access the **Settings** screen, tap **Preferences > Time zone**, and select the desired time zone from the list.
 - To change the time format, access the **Settings** screen, tap **Preferences > Time format**, and select the desired format from the list.
 - To change the date format, access the **Settings** screen, tap **Preferences > Date format**, and select the desired format from the list.

Related information

[Disabling the recent-calls list](#) on page 65

[Configuring call forwarding](#) on page 65

[Enabling automatic answer](#) on page 66

[Enabling do not disturb](#) on page 67

[Setting the language](#) on page 67

5.9 Configuring contact directories

For making IP PBX calls, Dolby Voice Room enables users to access a corporate directory and to create a local directory.

5.9.1 Corporate directory

Dolby Voice Room can connect to a directory server that supports the LDAP version 3.

You can configure LDAP directory parameters to make your corporate directory available. After it is set up, users can search the corporate directory.

If you are required to access the LDAP server in a secure manner, you must download the appropriate CA certificate to the system.

Only Microsoft Active Directory 2003 SP2 or later is supported at this time. Microsoft Active Directory supports server-side sorting.

Tip:

Consult your LDAP administrator when changing any corporate directory configuration information. For more information on LDAP attributes, see IETF RFC 4510.

Related information

[Local directory](#) on page 70

[Configuring the corporate directory](#) on page 71

[Creating a local contact directory from the provisioning server](#) on page 72

[Clearing the local contact directory](#) on page 72

[Adding certificate authority certificates](#) on page 43

5.9.2 Local directory

A local directory on the phone can hold up to 1,000 contacts.

You can enter contacts to a local contact directory manually or save recent calls and corporate directory search results.

 **Note:** For saved entries to work, you must format them in a manner that is consistent with your call-control platform. For example, you may need to include the number 9 or a plus sign (+) at the beginning of the entry.

A local directory can also be downloaded from the provisioning server. Users can then manually add entries to the downloaded directory.

Related information

[Corporate directory](#) on page 70

[Configuring the corporate directory](#) on page 71

[Creating a local contact directory from the provisioning server](#) on page 72

[Clearing the local contact directory](#) on page 72

5.9.3 Configuring the corporate directory

Use the web interface or set configuration parameters to configure an LDAP corporate directory.

About this task

You need this information to connect the system to the LDAP directory:

- The IP address or host name of the LDAP server
- The log-in name and password for the LDAP server
- The base domain name for the search of the LDAP server

 **Note:** Certain parameters are set by default (for example, search filter, scope, time-out, limit, and attributes).

Procedure

1. Do one of the following :
 - a) Specify LDAP parameters with the web interface:
Navigate to **Settings > Corporate Directory > LDAP**.
 - b) Set LDAP configuration parameters, either with the diagnostic shell or from Dolby Conferencing Console.

Parameter	Description
Directory.Ldap.URIList	Server URI. Use a comma-separated list to add more than one server. For example: ldap://10.101.21.5,ldap://10.120.21.6
Directory.Ldap.LoginName	Log-in name
directory.Ldap.Password	Password
Directory.Ldap.BaseDN	Base domain name. For example, dc=example, dc=com
Directory.Ldap.SecureConnection	Set to <i>true</i> or <i>false</i> , depending on the security settings on the LDAP server.

2. Reboot the system.
3. Make sure that the configuration process completed correctly by accessing the **Settings** screen and tapping this sequence on the phone:

 > **About > Status > Directory Service**

 **Note:** The **Status** menu contains **Directory Service** only if the service is configured.

Results

After the reboot is complete, users can search for contacts in the LDAP server.

Related information

[Corporate directory](#) on page 70

[Local directory](#) on page 70

[Creating a local contact directory from the provisioning server](#) on page 72

[Clearing the local contact directory](#) on page 72

[Corporate directory](#)

[Rebooting Dolby Voice Room](#) on page 92

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

5.9.4 Creating a local contact directory from the provisioning server

On the provisioning server, you can create a local contact directory for users.

When the system reboots, the local contact directory is downloaded from the provisioning server. The user can then modify the downloaded directory by adding contacts, either manually or by saving recent calls or corporate-directory search results.

See the *Dolby Conferencing Console Administrator's Guide* for details.

 **Note:** Dolby Conferencing Console version 1.2 and later allows you to provision a local contact directory.

Related information

[Corporate directory](#) on page 70

[Local directory](#) on page 70

[Configuring the corporate directory](#) on page 71

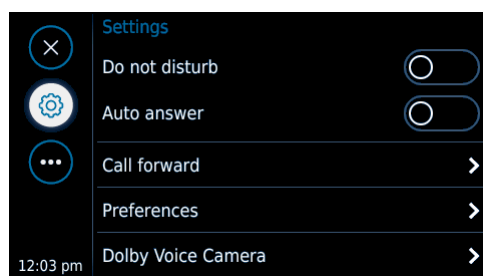
[Clearing the local contact directory](#) on page 72

5.9.5 Clearing the local contact directory

Use the phone user interface to clear the local contact directory.

Procedure

1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Select **Preferences** > **Clear personal data** > **Clear local contacts**.
3. Tap  to confirm.

Related information

[Corporate directory](#) on page 70

[Local directory](#) on page 70

[Configuring the corporate directory](#) on page 71

[Creating a local contact directory from the provisioning server](#) on page 72

5.10 Configuring the time server

The Dolby Voice Room system uses a time server to synchronize the clocks on all of the systems on the network. If your network has a local NTP server, we highly recommend that you configure it.

About this task

The Dolby Voice Room system uses DHCP option 42 to configure the time server. If the system cannot configure the time server using DHCP option 42, and the system is connected to an IP PBX, the IP PBX supplies the time, and the NTP status is shown as **Synchronized with SIP** on the Dolby Conference Phone UI and web interface.

If you perform a factory reset, and DHCP option 42 is not received and the NTP servers are not provisioned manually, the time servers are set by default to *time.nist.gov* as the primary NTP server and *pool.ntp.org* as the secondary NTP server.

 **Note:** The network must have a DNS server for the system to resolve the host names and connect to the default NTP servers. In addition, your firewall policy must permit access to these servers.

You can set and check the time server parameters in several ways:

Procedure

- Set the configuration parameters:

Configuration parameter name	Configuration parameter description
Network.Ntp.AutoDiscover	Enables NTP auto discovery (default true). When set to false, DHCP option 42 is ignored and you can manually set NTP server addresses with specified IP addresses or hostnames.
Network.Ntp.Server1	Sets the primary NTP server IP address or hostname
Network.Ntp.Server2	Sets the secondary NTP server IP address or hostname

- Use the web interface:
 - Go to **Settings > Network > NTP** and enter the parameters.
- Use the phone UI:
 - To review the time server status, access the **Settings** screen.
 - Tap  > **About > Status > Network time synchronization**.

5.11 Configuring Dolby Voice Room to use USB audio devices

You can configure Dolby Voice Room to use USB audio devices (microphones and speakers with an external DSP) in conference rooms.

Procedure

- Connect the DSP unit to Dolby Voice Room using a USB cable.
- Configure Dolby Voice Room to use USB as an input and output audio device.

- a) Log into Dolby Voice Room web UI.
Default username and password are admin/1739.
- b) Navigate to **Settings > Audio**.
- c) Change these settings:
 - **Audio Device Selection** to **Manual**
 - **Input Audio Device** to **USB Device**
 - **Output Audio Device** to **USB Device**
 - **Override Calibration Output Settings** to **On**
 - **Audio Device Latency** to **0**
 - **Audio Device AES/AEC Mode** to **NOLEAK_SLOW_CONVERGENCE**

 **Note:** Both **Input Audio Device** and **Output Audio Device** should be **USB Device**. Do not select a combination of **USB Device** and **Dolby Conference Phone**.

- d) Click **Save**.

5.12 Optimizing Dolby Voice Room for small rooms

In some cases, particularly in small or huddle conference rooms, participants stationed closer to the Dolby Conference Phone can sound much louder on the remote end than participants who are located further away from the phone.

About this task

If some conference participants in a small room sound much louder than others on the remote end, you can configure the room size to make voice volumes more uniform. To adjust microphone pickup for small or huddle conference rooms, follow these instructions:

Procedure

- From the web interface:
 - a) On the **Settings** tab, navigate to **Dolby Conference Phone > Audio**.
 - b) Set **Room size** to **Small**.
This option controls the Phone.Audio.RoomSize parameter.
- From the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters, and select the device to configure.
 - b) In the left navigation bar, click the **Edit** icon.
The **Device configuration** screen displays.
 - c) Navigate to **Dolby Conference Phone > Audio**.
 - d) Drag **Room size** to the device configuration.
 - e) Set the Phone.Audio.RoomSize parameter to **Small**.

5.13 Disabling the reboot prompt

The reboot prompt is disabled by default, prompting you to reboot due to configuration changes. If you take no action in 20 seconds—for example, cancel the reboot—the system reboots automatically.

About this task

This procedure disables both the reboot prompt.

You can use either configuration parameters or the web interface to disable the reboot prompt.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Provisioning.CfgChange. UserAcknowledgementEnabled	Enables/disables the reboot. Set to false.

- Use the web interface:
 - Go to **Settings > Provisioning > On Configuration Change**.
 - Set **User acknowledgment enabled** to **Off**.

5.14 Disabling Bluetooth advertisement

By default, the Dolby Voice Room system is configured to allow the phone to advertise a Bluetooth low energy (Bluetooth LE) signal that can be detected by other Bluetooth-enabled devices. When used in conjunction with other Bluetooth-enabled devices, such as mobile phones, this feature can allow the phone to be controlled remotely. You can turn it off for security reasons, if needed.

About this task

Check with your conferencing service provider before disabling Bluetooth advertisement.

Procedure

- From the web interface:
 - On the **Settings** tab, expand **Dolby Conference Phone**.
 - Turn **Allow bluetooth advertisement** off.
This option controls the Phone.EnableBluetoothAdvertisement parameter.
- Alternatively, from Dolby Conferencing Console:
 - From the home screen, select the device pool than contains the device for which you want to edit parameters.
 - Select the device you want to edit.
 - In the left navigation bar, click the **Edit** button.
The **Device configuration** screen displays.
 - Navigate to **Dolby Voice Conferencing Service > Service**.
 - Drag the **Allow bluetooth advertisement** to the device configuration.
The Phone.EnableBluetoothAdvertisement parameter appears in the device configuration.
 - Turn the parameter off.

6

Administering Dolby Voice Room

Administrative tasks include viewing system status, viewing and managing event logs, viewing call statistics, running diagnostics, and more.

- [Changing the administrative password](#)
- [Accessing the Dolby Voice Room web interface](#)
- [Administering with the diagnostic shell](#)
- [Viewing the status menu](#)
- [Setting the Dolby Voice Room operational mode](#)
- [Viewing and managing event logs](#)
- [Initiating Reprovisioning](#)
- [Enabling power saving mode](#)
- [Changing network settings](#)
- [Running Dolby Conference Phone diagnostics](#)
- [Bluetooth LE initialization and advertisement](#)
- [Confirming Bluetooth LE initialization and advertisement](#)
- [Enabling Privacy block](#)
- [Call and audio statistics](#)
- [Enabling WebRTC tracing](#)
- [Rebooting Dolby Voice Room](#)
- [Resetting to factory defaults](#)
- [Updating the Dolby Voice Room software](#)
- [Configuring HDMI input audio](#)
- [Finding the system serial number and software version](#)
- [Finding the Dolby Voice Camera serial number and software version](#)

6.1 Changing the administrative password

The Dolby Voice Room administrative settings are protected with an administrative password. The same password is used to log in to the web interface and diagnostic shell.

About this task

The default administrative user name is *admin* and the default password is *1739*. Change the administrative password from the default value as soon as possible. If you leave the administrative password unchanged or use weak passwords, the system is vulnerable to various network attacks.

 **Note:** If you changed and then subsequently forgot the administrative password, reset the system to factory defaults to restore the default password of *1739*. If you are not already logged in, you must use the fail-safe method to reset the system.

Use one of these methods to change the administrative password.

 **Note:** We recommend that you do not enter passwords in configuration files, as it poses a security risk.

Procedure

- Use the web interface.
 - a) Access the Dolby Voice Room web interface, and log in with the current administrative user name and password.
 - b) Go to **Settings > Authentication > Local Account**, enter a new administrative password, and click **Save**.
- Use the phone UI.
 - a) Tap  > **Administrative Settings**.
 - b) Enter the current administrative password.
 - c) Tap **Change password**.
 - d) Enter and confirm the new password.
- Use the diagnostic shell and the **configure** command.
 - a) Open the diagnostic shell, and log in with the current administrative user name and password.
 - b) Enter this command:

```
configure Authentication.Local.Admin newPassword
```

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

[Resetting to factory defaults](#) on page 93

6.2 Accessing the Dolby Voice Room web interface

Use the web interface to view and change Dolby Voice Room settings.

About this task

You can access the Dolby Voice Room web interface using any of these web browsers:

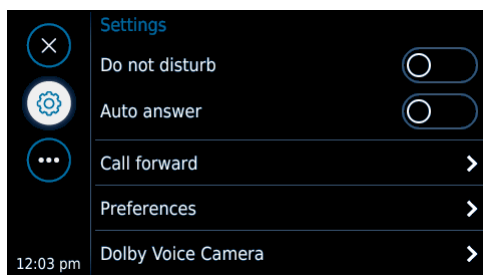
- Microsoft Internet Explorer 10.0.x or later on Windows-based computers (backward compatibility not supported)
- Google Chrome 31.0.x or later
- Apple Safari 7.0.2.x

Note: The Dolby Voice Room web interface has been optimized for modern browsers. The best experience is obtained using one of the supported browsers. Versions 8 and 9 of Internet Explorer work in a degraded fashion; support for these browsers is deprecated and will be removed in a future release.

By default, access to the web interface is enabled. You can disable access only through the configuration files by changing `Network.Services.WebServerMode`.

Procedure

1. Determine the system IP address:
 - a) Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

- b) Tap this sequence:



The IP address displays.

2. Open one of the supported web browsers.
3. Enter `http://your-system-IP-address` in the browser address bar.
The log-in page displays.

Note: If the browser displays a certificate-related error message, such as **Site is not secure** or **Your connection is not private**, then take the required steps to bypass the error page and proceed to the log-in page.

4. Log in as the administrator. (Default credentials are *admin* and *1739*.)
The web interface main page displays. Expand the headings and hover the mouse over a parameter name to see a parameter description, along with the name of the corresponding configuration parameter.

Note: We recommend that you change the default administrative password.

Related information

[Changing the administrative password](#) on page 77

6.3 Administering with the diagnostic shell

The diagnostic shell provides a command-line interface for administering Dolby Voice Room.

6.3.1 Diagnostic shell

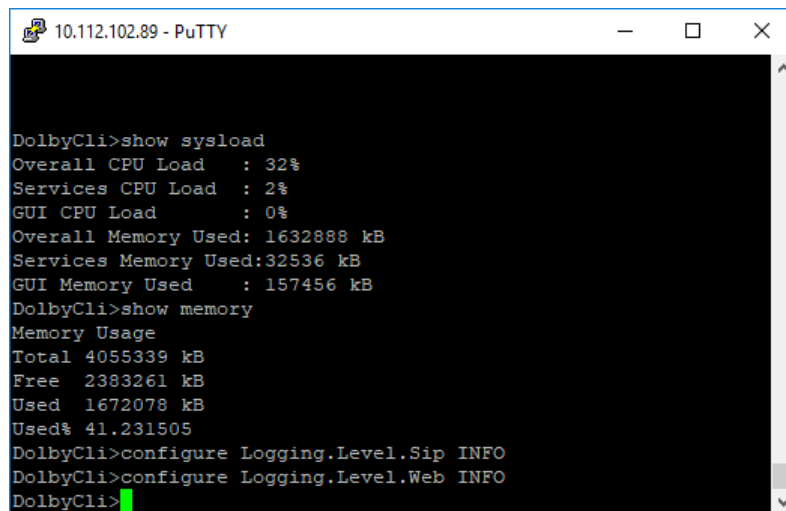
The diagnostic shell provides a command-line interface for you to query the system state, run diagnostics, and configure selected parameters.

You access the diagnostic shell remotely with the SSH protocol or locally with an external keyboard.

Only one instance of the shell service is supported. After a connection to the shell service is established, subsequent requests are denied until the current session is terminated. The session times out after a period of inactivity. The time-out period is set by the `Authentication.Local.SessionTimeout` configuration parameter. The shell service starts when the system boots up, but it is accessible remotely only if the configuration parameter `Network.Services.SSHEnabled` is enabled, which is the default. You must enter the administrative user name and password to use the diagnostic shell.

Shell commands include localized command help, function calls, diagnostic tests, commands to view and set configuration parameters, and more. A list of commands is available in the **Help** section of the web interface.

Figure 5: Diagnostic shell command sample



```

10.112.102.89 - PuTTY
DolbyCli>show sysload
Overall CPU Load   : 32%
Services CPU Load  : 2%
GUI CPU Load       : 0%
Overall Memory Used: 1632888 kB
Services Memory Used:32536 kB
GUI Memory Used    : 157456 kB
DolbyCli>show memory
Memory Usage
Total 4055339 kB
Free 2383261 kB
Used 1672078 kB
Used% 41.231505
DolbyCli>configure Logging.Level.Sip INFO
DolbyCli>configure Logging.Level.Web INFO
DolbyCli>

```

Tip: The shell provides help and shortcuts:

- Type `help` to see a list of commands.
- Press Up-arrow to view previous commands. Press Enter to submit.
- Press Tab to complete partially typed commands or to prompt for additional command information, such as command parameters. For example, if you enter `system`, and then press Tab, the possible system commands are listed and the command is repeated on the command line for you to finish command entry. If you enter `system traceroute` and press Tab, the system indicates that the command parameter is the IP address of the test host.

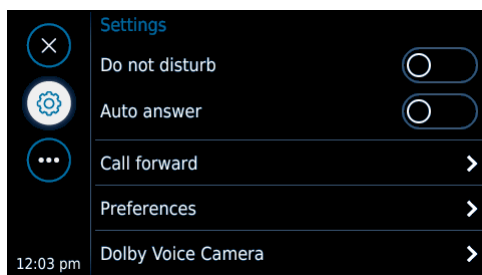
Related information

[Opening the diagnostic shell](#) on page 79

6.3.2 Opening the diagnostic shell

You access the diagnostic shell remotely with the SSH protocol or locally with an external keyboard.

- Open the diagnostic shell remotely:
 1. Determine the system IP address:
 - a. Access the Dolby Voice Room **Settings** screen.



This may be different for each service provider.

- b. Tap this sequence:



The IP address displays.

2. Open your preferred SSH client application.
 3. Connect to the system IP address using SSH (port 22).
 4. Log in with the administrative user name and password. (Default credentials are *admin* and *1739*.)
 5. Enter commands, and then type *exit* to exit the shell.
- Open the diagnostic shell locally:
 1. Connect a keyboard to one of the USB ports on Dolby Voice Hub.
 2. Press Ctrl+Alt+F3 on the keyboard.

The diagnostic shell opens on the display.

3. Log in with the administrative user name and password. (Default credentials are *admin* and *1739*.)
4. Enter commands, and then press Ctrl+Alt+F2 to dismiss the diagnostic shell and return to the default display.

Related information

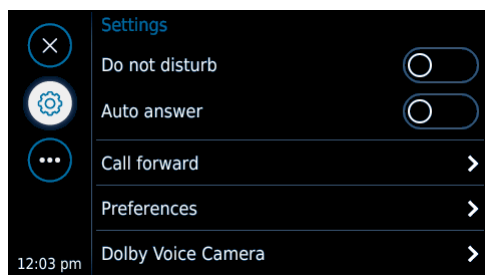
[Diagnostic shell](#) on page 78

6.4 Viewing the status menu

The status menu on the Dolby Conference Phone UI displays status information on the Dolby Voice Room components, the provisioning server, the network, and more.

Procedure

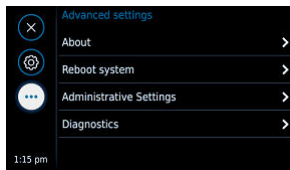
1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap .

The **Advanced settings** screen displays.



3. Tap **About > Status**.

Any statuses showing an error condition () are displayed at the top.

6.5 Setting the Dolby Voice Room operational mode

Set the operational mode to control whether Dolby Voice Room operates as a conferencing endpoint only, or both a conferencing endpoint and a SIP audio-only endpoint for IP PBX calls.

Procedure

- Use the web interface:
 - a) Access the Dolby Voice Room web interface.
 - b) Go to **Settings > Features** and, for **Operation Mode**, select **Conferencing Only Mode** or **Dual Mode**.
- Use the diagnostic shell:
 - a) Open the diagnostic shell.
 - b) Use the **configure** command to set the `Features.OperationMode` configuration parameter to one of the following values:
 - DUALMODE
 - CONFONLYMODE

Example: `configure Features.OperationMode DUALMODE`

Related information

[Operation modes of Dolby Voice Room](#) on page 15

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

6.6 Viewing and managing event logs

The Dolby Voice Room system has a sophisticated event-logging system for diagnostics and troubleshooting.

6.6.1 Event logging

The Dolby Voice Room system generates event-log entries on a continuous basis, both for ordinary operation events and for error conditions. The event logs provide an essential troubleshooting tool. Dolby Customer Support or your conferencing service provider might ask you to share logs during problem resolution.

Log files are stored in a flash file system on the Dolby Voice Room system. You can use the web interface to download the logs directly from the device.

If you are using automatic provisioning with Dolby Conferencing Console, Dolby Voice Room automatically uploads log entries to Dolby Conferencing Console, and you view the logs with the Dolby Conferencing Console web interface.

The maximum file size for the Dolby Voice Room `log/messages` and `log/messages.0` log files is 4 MB. The maximum file size for the Dolby Voice Room `uploads/SN-messages` and `uploads/SN-messages.0` log files is 2 MB.

Dolby Voice Room supports the syslog protocol. You can configure the system to automatically upload logs to a syslog receiver.

For information about viewing logs on Dolby Conferencing Console, see the .

Related information

- [Log levels](#) on page 82
- [Audio logging](#) on page 83
- [syslog logging](#) on page 83
- [Enabling HDMI logging](#) on page 84
- [Viewing log files](#) on page 84

6.6.2 Log levels

Each of the components of the Dolby Voice Room software is capable of logging events of different severity. This allows you to capture low-severity events in one component, and only high-severity events for other components.

Log-level configuration parameters enable you to indicate how much logging is done for each software component. The range of log levels, in order of detail and verbosity from lowest to highest, includes NONE, EMERGENCY, ALERT, CRITICAL, ERROR, WARNING, NOTICE, INFO, DEBUG, and DEBUG1. (DEBUG1 contains maximum verbose trace information.) By default, all log-level parameters are set to INFO.

 **Note:** You should not modify the default log levels unless directed to by Dolby Customer Support. Inappropriate logging levels can cause performance issues.

In the log files, messages are prefixed by timestamp and log level.

- `debug`: Indicates verbose traces generated only when log level is set to DEBUG
- `info`: Indicates normal phone behavior
- `notice`: Indicates minor issues or failures, but could be normal behavior
- `warn`: Indicates that an error may occur if action is not taken
- `err`: Indicates that a failure or loss of functionality occurred
- `crit`: Indicates that an unrecoverable failure of a semi-important component occurred
- `alert`: Indicates that an unrecoverable failure of an important component occurred
- `emerg`: Indicates that a complete unrecoverable system failure occurred

The `warn`, `err`, `crit`, `alert`, and `emerg` messages indicate problems of increasing severity.

A full description of all log-level parameters can be found under the **Help** section of the web interface.

To change the logging level of the SIP communication module to the DEBUG level, set `Logging.Level.Sip` to DEBUG. You can also make this change in the **Logging** section of the web interface.

 **Note:** If you set logging to a more verbose level for testing or troubleshooting, then when testing or troubleshooting is complete, remember to return logging levels to their default values.

Related information

- [Event logging](#) on page 81
- [Audio logging](#) on page 83
- [syslog logging](#) on page 83
- [Enabling HDMI logging](#) on page 84
- [Viewing log files](#) on page 84

6.6.3 Audio logging

For the audio module of the Dolby Voice Room software, you can enable logging through configuration files or the web interface.

 **Note:** Turn on audio logging only if requested by Dolby Customer Support.

Audio logging tracks these audio characteristics:

- Echo reference input
- Echo reference output
- Jitter buffer input
- Jitter buffer output

A full description of all audio logging configuration parameters can be found in the **Help** section of the web user interface.

By default, all audio logging is disabled. If you want the audio log files written to the provisioning server, you need to set the audio logging mode, `Logging.Audio.Mode`. The supported modes are `DISABLED`, `STATISTICS`, and `DIAGNOSTIC`.

If the audio logging mode is set to `STATISTICS` or `DIAGNOSTIC`:

- A call statistics log containing per-second statistical information (including jitter buffer and packet loss) is uploaded to the provisioning server. For example, the file `D0025454-call_stats_b502bb32-2bdd-49da-8ae7-0d89ff295a01.json` is uploaded from the device numbered D0025454.
- An audio dump of preprocessing statistics that contains metadata about the preprocessing algorithm is uploaded to the provisioning server. For example, the file `D0025454-vsv_stats_20140529163458.bin` is uploaded from the device numbered D0025454.

If the audio logging mode is set to `DIAGNOSTIC`, the recorded diagnostic data is uploaded to the provisioning server. For example, the file `D0025454-audiolog_20140529141935.gz` is uploaded from the device numbered D0025454.

 **Note:** The `STATISTICS` audio logging mode may impact the end of a call (when data is uploaded to the provisioning server). The `DIAGNOSTIC` audio logging mode impacts the entire call, adding approximately 10–15% overhead.

Related information

[Event logging](#) on page 81

[Log levels](#) on page 82

[syslog logging](#) on page 83

[Enabling HDMI logging](#) on page 84

[Viewing log files](#) on page 84

6.6.4 syslog logging

`syslog` is a standard for forwarding log messages in an IP network. You can configure the Dolby Voice Room system to ship log entries to a `syslog` receiver.

Because `syslog` is supported by a wide variety of devices and receivers, `syslog` can be used to integrate log data from many different types of systems into a central repository.

The `syslog` messages contain much of the same information as the Dolby Voice Room logs, with the exception of the events during the early stage of the system initialization (as the device is not yet configured to send `syslog` messages). `syslog` data is sent in clear text.

By default, syslog is disabled. (`Logging.RemoteSysLog.Enable` is `false`.) To use syslog, you must set `Logging.RemoteSysLog.Enable` to `true` and set the syslog server address (`Logging.RemoteSysLog.Address`). Messages relating to developer logs and intermodule communication are sent to the `DEBUG` level. Messages pertaining to user action (if logged) and module initialization (if any) are available in the `INFO` level. Error and warning messages are logged in `ERROR` and `WARNING`, respectively.

For more information on the syslog protocol, see RFC 3164.

Related information

[Event logging](#) on page 81

[Log levels](#) on page 82

[Audio logging](#) on page 83

[Enabling HDMI logging](#) on page 84

[Viewing log files](#) on page 84

6.6.5 Enabling HDMI logging

You can enable HDMI logging on the Dolby Voice Room HDMI-input drivers.

About this task

HDMI logging can be used to troubleshoot issues with HDMI sources. The following parameters are available:

- **Enable HDMI Input Detection logs:** Provides information on how the HDMI source connected to the system is detected and information about the incoming signal. The logs can be used for detecting HDMI source connection and disconnection problems.
- **Logging Interval:** Specifies the interval at the driver with log information.

Procedure

1. Access the Dolby Voice Room web interface.
2. In the **Settings** tab, expand **Logging > HDMI Input**.
3. Toggle **Enable HDMI Input Detection logs** and set the **Logging Interval**.

Related information

[Event logging](#) on page 81

[Log levels](#) on page 82

[Audio logging](#) on page 83

[syslog logging](#) on page 83

[Viewing log files](#) on page 84

6.6.6 Viewing log files

You can download and view log files stored in the Dolby Voice Room system or on Dolby Conferencing Console.

If logs are located here	Do this
Dolby Conferencing Console	Use the Dolby Conferencing Console web interface. See the .
Dolby Voice Room flash memory  Note: All logs originate from the device flash memory. The flash memory always has the most recent log entries, some of which may not have been uploaded yet.	1. Access the Dolby Voice Room web interface. 2. Select Tools, and then click Download Logs. 3. Save the <i>tar.gz</i> file, and unpack it with your preferred decompression utility. The TAR archive contains one or two log files, named <i>message</i> and <i>message.0</i>. The file named <i>message</i> contains the latest entries. 4. Open the log files with your preferred text editor.

Related information

[Event logging](#) on page 81

[Log levels](#) on page 82

[Audio logging](#) on page 83

[syslog logging](#) on page 83

[Enabling HDMI logging](#) on page 84

[Accessing the Dolby Voice Room web interface](#) on page 77

6.7 Initiating Reprovisioning

The most common reasons to initiate reprovisioning of Dolby Voice Room are to incorporate a configuration change (such as new IP PBX settings) and to update the system software.

About this task

There are several methods for initiating reprovisioning:

- With Dolby Conferencing Console
- With the web UI
- With a diagnostic shell command
- By rebooting
- By enabling scheduled provisioning

 **Note:** Check with your conferencing service provider to see if their application initiates reprovisioning by regularly rebooting the system.

Initiating reprovisioning with Dolby Conferencing Console

Initiating reprovisioning from Dolby Conferencing Console is how you push software updates to the devices. See the *Dolby Conferencing Console Administrator's Guide* for details.

Initiating reprovisioning with the web interface

1. Access the Dolby Voice Room web interface.
2. Go to **Tools** and click the **Provision** button.

Initiating reprovisioning with a diagnostic shell command

1. Access the diagnostic shell.
2. Enter this command:

```
system provision
```

Initiating reprovisioning by rebooting

You can initiate reprovisioning by rebooting the Dolby Voice Room.

Enabling scheduled provisioning

For scheduled provisioning, you define a daily time window by specifying an on-the-hour start time and a duration in minutes. Reprovisioning happens randomly within this window.

Enable scheduled provisioning using one of these methods:

- Enable with the web interface.
 1. Access the Dolby Voice Room web interface.
 2. Under the **Settings** tab, go to **Provisioning > Scheduled Provisioning**.
 3. Turn on **Enable Scheduled Provisioning**, and enter a start time and interval duration.
 4. Use the sliders to set a **Scheduled Reprovisioning Start Time (hour)** and a **Scheduled Reprovisioning Interval Duration (minutes)**.

For example, setting the start time to 23 and the interval duration to 90 defines a 90-minute window starting at 11:00 p.m. Reprovisioning takes place at a random time between 11:00 p.m. and 12:30 a.m.

- Enable by setting configuration parameters.
 1. Access the diagnostic shell.
 2. Set these parameters:
 - `Provisioning.Scheduled.Enable`
 - `Provisioning.Scheduled.StartTime`
 - `Provisioning.Scheduled.Duration`

See the **Help** section of the web interface for details on these parameters.

Related information

[Updating the Dolby Voice Room software](#) on page 94

[Rebooting Dolby Voice Room](#) on page 92

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

6.8 Enabling power saving mode

You can enable power saving mode and specify the number of minutes for which the system will idle before turning the TV off. By default, power saving is disabled.

About this task

There are two power saving settings: **Off Hours Idle Timeout** and **Work Hours Idle Timeout**. The working hours are defined by the start/end times and days of the work weeks. If power saving is not set within the specified work hours, then it is automatically set to **Off Hours Idle Timeout**.

Not all TVs support power saving mode. You can perform a diagnostic action through the Dolby Conference Phone web interface to determine whether your TV supports automatic power control. You can use the Dolby Conference Phone diagnostics menu to power the TV on and off.

Procedure

1. Access the Dolby Voice Room web interface.
2. Expand **Settings > User Preferences > Power Saving**.
3. Specify these settings:
 - **Off Hours Idle Timeout**

- **Work Hours Idle Timeout**
- **Work Week Start Day**
- **Work Week End Day**
- **Work Day Start Time**
- **Work Day End Time**

4. Click **Save**.

Related information

[Dolby Voice Room power saving issues and solutions](#) on page 103

6.9 Changing network settings

You might need to change network settings after provisioning is complete.

About this task

Using the web interface, the phone interface, or the diagnostic shell, you can change several network settings, including:

- 802.1X credentials
- NTP server address
- VLAN number
- Proxy settings
- Services

If DHCP network configuration is off, you can also set basic IP settings such as IP address, gateway, DNS server, and so on.

 **Note:** Many network setting changes cause a reboot to occur.

Procedure

- Change network settings with the web interface:
 - a) Access the Dolby Voice Room web interface.
 - b) Expand **Settings > Network**.
 - c) If **DHCP Network Configuration Method** is off, change basic IP settings. Or, expand one of these sections and enter your changes:
 - IEEE 802.1X
 - NTP
 - Virtual LAN
 - Proxy
 - Services
- Change network settings with the phone UI:
 - a) Access the Dolby Voice Room **Settings** screen.
 - b) Tap  > **Administrative Settings**.
 - c) Enter the administrative password.
The default administrative password is 1739. We recommend that you change it.
 - d) Tap **Network configuration**, and then tap one of these menu choices:
 - Ethernet
 - Time Server
 - 802.1X
- Change network settings with the diagnostic shell and the **configure** command:

- a) On the **Help** tab of the web interface, click the link to view the Dolby Voice Hub configuration parameters, and then click the **Network** link to scroll to the network-specific parameters.
- b) Locate the configuration parameter that you need to change.
- c) Open the diagnostic shell and log in.
The default credentials are *admin* and *1739*. We recommend that you change the administrative password.
- d) Enter the **configure** command to change the network setting.
For example, enter a command similar to this to change the 802.1X authentication password:

```
configure Network.Eapol.Password y6ttr4ub
```

Related information

[Configuring the time server](#) on page 73

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

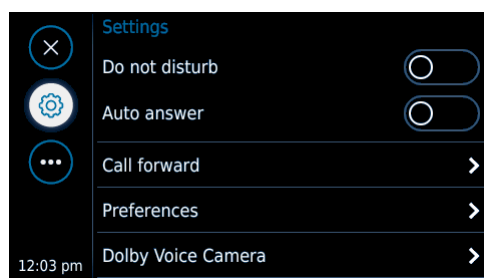
[Changing the administrative password](#) on page 77

6.10 Running Dolby Conference Phone diagnostics

The phone user interface includes a menu that enables you to test the various components of the Dolby Conference Phone, such as the microphones and speakers.

Procedure

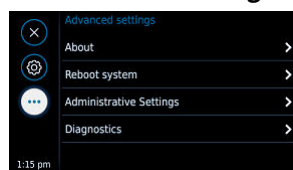
1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap

The **Advanced settings** screen displays.



3. Tap **Diagnostics**, then **Test hardware**.
4. Select one of these menu items to perform a hardware diagnostic test:

- **Touch screen:** Test the touch screen response area.

Tap the screen anywhere in the grid and determine whether the crosshair that appears is exactly where you touched. Try again in a different location. If there is a discrepancy, see the instructions below for calibration.

- **LED:** Test the LED halo on the phone.
- **Keys:** Press the three keys, one by one.

- **Microphones:** Test the phone microphones. If satellite microphones are connected, they can also be tested.
 1. Select a microphone (1, 2, or 3) and tap **Start test**.
 2. Speak into the phone and wait for the phone to play back the recording.
 3. Repeat with the other microphones.
- **Speakers:** Test the speakers.

Results

If the touch screen needs calibration:

1. Open the diagnostic shell and log in.
2. Type this command to connect directly to the Dolby Conference Phone:

```
ssh-to-dcp
```

3. Type `y` to bypass the trusted host message, and then enter the administrative password.
4. Enter this command to clear the calibration data:

```
system rmcalibration
```

5. Type this command to reboot the phone:

```
system reboot
```

6. When the calibration screen displays during the reboot process, follow the instructions to calibrate the touch screen.

Related information

[Opening the diagnostic shell](#) on page 79

[Rebooting Dolby Voice Room](#) on page 92

6.11 Bluetooth LE initialization and advertisement

The Bluetooth LE module is a hardware component of Dolby Conference Phone. If this component is faulty or does not initialize, then the phone cannot transmit a Bluetooth LE advertisement.

You can confirm whether the phone is advertising a Bluetooth LE signal. However, be aware that the phone only advertises Bluetooth LE under certain circumstances. Lack of Bluetooth LE advertisement is a problem only if your conferencing application allows the phone to advertise Bluetooth LE.

In order for the phone to advertise a Bluetooth LE signal, these criteria must be met:

- The Bluetooth LE hardware module on the phone is working and initialized.
- You have not disabled Bluetooth LE advertisement.
- The conferencing application on the phone allows Bluetooth and has turned on Bluetooth LE advertisement.

The `Phone.EnableBluetoothAdvertisement` parameter controls whether the phone can advertise Bluetooth LE. However, the conferencing application controls whether the phone does or does not advertise Bluetooth LE.

Related information

[Disabling Bluetooth advertisement](#) on page 75

6.12 Confirming Bluetooth LE initialization and advertisement

From the phone user interface, you can confirm that the Bluetooth LE hardware is initialized and advertisement is active.

About this task

If the Bluetooth LE module fails to initialize, a red status alarm  displays on the phone user interface and Dolby Voice Room web interface.

When the phone boots, it attempts to initialize the Bluetooth LE module (up to ten times). If initialization fails, each attempt is recorded in the `syslog` log file with this warning:

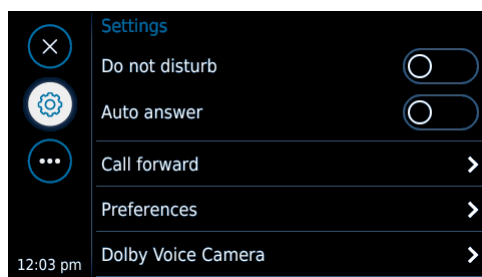
Bluetooth module initialization failed. Retrying...

If all initialization attempts fail, the phone records this warning in the `syslog` log file and quits Bluetooth LE module initialization:

Unable to initialize module. Quit application.

Procedure

1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap this sequence:



3. Scroll down to view **Bluetooth** status.

A green  or red  status icon indicates the status of this feature.

4. Tap **Bluetooth** to check the initialization and advertisement statuses.

6.13 Enabling Privacy block

When privacy block is disabled (the default), the Dolby Voice Room system can share usage data with your conferencing service provider. You can enable or disable privacy block at any time from the phone UI, web interface, or command-line interface (CLI).

About this task

Privacy block is not a provisionable feature. You must enable or disabled it one system at a time. The name of the configuration parameter that controls this feature is `Provisioning.Server.PrivacyBlock`.

From the Dolby Voice Room web interface:

1. From the **Settings** tab, expand the **Provisioning** section.
2. Expand the **Server** section.
3. Turn on **Disable Privacy-related Uploads**.

From the phone user interface:

1. Access the **Settings** screen.
2. Tap  > **Administrative Settings**.
3. Enter the administrative password.

4. Tap **Network configuration > Provisioning Server**.

5. Scroll down and enable **Privacy block**.

The Provisioning.Server.PrivacyBlock configuration parameter is now true. The system cannot share data with your conferencing service provider's provisioning service.

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

6.14 Call and audio statistics

Through the Dolby Voice Room web interface, you can view call statistics for active audio calls.

To view call statistics:

- Access the web interface and select **Status > Call Statistics**.

The call statistics page shows:

Call information:

- Calling party first and last name
- Calling party address (extension)
- Call duration in minutes and seconds
- Type of call (IP PBX or conferencing service provider)
- Audio codec type

Network statistics

- Packet count
- Average jitter
- Packet loss rate
- Packet loss burstiness (rate of audio packet losses occurring in a burst)
- Concealed packet count
- Jitter histogram

Audio diagnostics

- Average automatic gain control gain
- Voice capture level
- Voice rendering level
- Signal-to-noise level
- Signal-to-echo level
- Average round-trip time (RTT)
- Signal-to-noise level too low
- Signal-to-echo level too low
- Low input power detected
- Microphone input clipping

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

6.15 Enabling WebRTC tracing

Enable Web Real-Time Communication (WebRTC) tracing when instructed to do so by your conferencing service provider or Dolby Customer Support.

About this task

When enabled, WebRTC traces can be uploaded to the provisioning server. With Dolby Conferencing Console, traces are uploaded automatically. Using the Dolby Conferencing Console web interface, you can download the trace files from the **Files** tab for each system.

Procedure

1. Access the Dolby Voice Room web interface.
2. In the **Settings** tab, expand **Logging > WebRTC**.
3. Turn on **Debug Trace**.

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

6.16 Rebooting Dolby Voice Room

You might need to reboot Dolby Voice Room if the system becomes unresponsive, if you have made certain network configuration changes, or if requested to do so by Dolby Customer Support or your conferencing service provider customer support.

About this task

There are several methods for rebooting:

- With the Dolby Voice Hub power button
- With the phone UI
- With the web UI
- With a diagnostic shell command (if connection over the network is possible)
- With a diagnostic shell command through an external keyboard (if network connection is not possible)

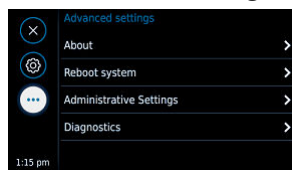
Rebooting with the Dolby Voice Hub power button

Hold down the button for five to ten seconds until you see the Dolby Voice logo on the display.

Rebooting with the phone UI

1. Access the Dolby Voice Room **Settings** screen. Ask your conferencing service provider for instructions.
2. Tap .

The **Advanced settings** screen displays.



3. Tap **Reboot system** and then tap the **Reboot** icon to confirm.

Rebooting with the web interface

1. Access the Dolby Voice Room web interface.
2. Go to **Tools** and click the **Reboot** button.
3. Click **Confirm** to proceed with the reboot.

Rebooting with the diagnostic shell

1. Open the Dolby Voice Room diagnostic shell.
2. Log in with administrator credentials. (Default credentials are *admin* and *1739*.)

3. Type this command and press Enter:

```
system reboot
```

Rebooting with the diagnostic shell through an external keyboard

Use this method when the system is unresponsive.

1. Connect a computer keyboard to one of the USB ports on the Dolby Voice Hub.
2. Type Ctrl+Alt+F3 to open a diagnostic shell on the display. Log in with the administrator credentials. (Default credentials are *admin* and *1739*.)
3. Type this command, and press Enter:

```
system reboot
```

 **Note:** If you decide not to proceed with reboot after opening the diagnostic shell, type Ctrl+Alt+F2 to return to the normal image on the display.

Results

Reboot proceeds even if a call is active.

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

6.17 Resetting to factory defaults

Resetting Dolby Voice Room to factory defaults clears the user data, such as the recent calls list and contact directory, resets user preferences, removes log files and cached data, and restarts the out-of-the-box (OOB) provisioning process.

About this task

You can also reset to factory defaults to recover from a situation in which the system becomes unresponsive.

There are a few methods for resetting to factory defaults:

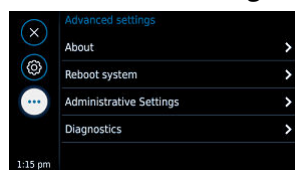
- With the phone UI
- With the web interface
- With the multiple key combination
- With a diagnostic shell command (if connection over the network is possible)
- With a diagnostic shell command through an external keyboard (if network connection is not possible)

There is also a fail-safe method that you can use if the system is in an unrecoverable state and the above methods are unavailable.

Resetting with the phone UI

1. Access the Dolby Voice Room **Settings** screen. Ask your conferencing service provider for instructions.
2. Tap .

The **Advanced settings** screen displays.



3. Tap **Administrative Settings** and enter the administrator password (default is *1739*).

4. Tap **Factory reset**.

Resetting with the web interface

1. Access the Dolby Voice Room web interface.
2. Go to **Tools** and click the **Factory Reset** button.

Resetting with the multiple key combination

1. Press the mute and volume-down buttons simultaneously for 10 seconds.
2. When prompted, confirm the reset.

 **Note:** You can also access the **Settings** screen for advanced settings and system diagnostics.

Resetting with the diagnostic shell

1. Open the Dolby Voice Room diagnostic shell.
2. Log in with administrator credentials. (Default credentials are *admin* and *1739*.)
3. Type this command, and press Enter:

```
system factoryReset
```

4. When prompted for confirmation, press Enter.

Resetting with the diagnostic shell through an external keyboard

Use this method when the system is unresponsive.

1. Connect a computer keyboard to one of the USB ports on the Dolby Voice Hub.
2. Type Ctrl+Alt+F3 to open a diagnostic shell on the display. Log in with the administrator credentials. (Default credentials are *admin* and *1739*.)
3. Type this command, and press Enter:

```
system factoryReset
```

4. When prompted for confirmation, press Enter.

 **Note:** If you decide not to proceed with factory reset after opening the diagnostic shell, type Ctrl+Alt+F2 to return to the normal image on the display.

Resetting with the fail-safe method

Use this method if the system is in an unrecoverable state and the above methods are unavailable.

1. Connect a computer keyboard to one of the USB ports on Dolby Voice Hub.
2. Hold down the power button on Dolby Voice Hub to reboot the system, and when the display goes blank, hold down the R key until you see a blue screen displaying choices to perform a factory reset or cancel.
3. Select **Factory reset device** to reset to factory defaults or **Cancel** to continue with a normal reboot.

Results

The system reboots to complete the reset process even if a call is active.

Related information

[Accessing the Dolby Voice Room web interface](#) on page 77

[Opening the diagnostic shell](#) on page 79

[Rebooting Dolby Voice Room](#) on page 92

6.18 Updating the Dolby Voice Room software

This section describes how to update the Dolby Voice Room software.

In a plug and play provisioning environment, your conferencing service provider manages all Dolby Voice Room software updates.

In an automatic provisioning environment, you obtain the latest software image from your conferencing service provider, upload it to Dolby Conferencing Console, and then install it by either pushing the software image from Dolby Conferencing Console, or by initiating reprovisioning from the Dolby Voice Room system.

After the initial (out-of-the-box) provisioning, reprovisioning can occur at regular intervals or on demand. If a software update is available on Dolby Conferencing Console, reprovisioning causes the update to be installed.

During a software update, the software is first installed on Dolby Voice Hub, and Dolby Voice Hub then updates the Dolby Conference Phone software and Dolby Voice Camera firmware.

For large-scale installations with automatic provisioning and Dolby Conferencing Console, we recommend pushing the software update from Dolby Conferencing Console.

See the *Dolby Conferencing Console Administrator's Guide* for details on uploading software images and on pushing software updates to devices.

Related information

[Initiating Reprovisioning](#) on page 85

6.19 Configuring HDMI input audio

Dolby Voice Room allows you to stream HDMI input audio with a conference uplink stream for webRTC calls or SIP video calls. By default, HDMI input audio is enabled.

To stream HDMI input audio into conference uplink audio for SIP video calls, the telephony application handles the streaming of HDMI input audio when you select the **HDMI share** option during the call.

To stream HDMI input audio into conference uplink audio for webRTC calls, the webRTC application must select the audio device exposed as **Dolby HDMI Input Audio**.

The WebRTC application can select **Dolby HDMI Input Audio** in-call or out-of-call scenario. In an out-of-call scenario, HDMI input audio is rendered locally. In an in-call scenario, HDMI input audio is additionally streamed into conference uplink audio.

You can see the status of the HDMI Input Audio streams and respective pulseaudio device status through the Dolby Voice Room UI.

1. Log into the Dolby Voice Room UI.
2. Navigate to **Status > Dolby Voice Hub > Device > Alarms and Status**.

6.19.1 Disabling HDMI input audio

Dolby Voice Room allows you to stream HDMI input audio with a conference uplink stream for webRTC calls or SIP video calls. By default, HDMI input audio is enabled.

About this task

You can disable HDMI input audio through the Dolby Voice Room web UI.

Procedure

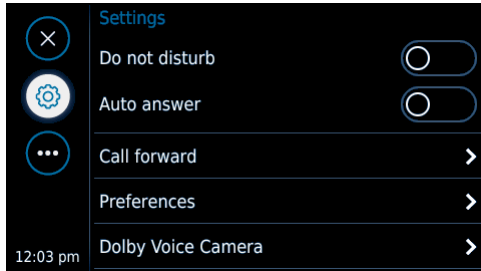
1. Log into the Dolby Voice Room web UI.
2. Navigate to **Settings > Features**.
3. Turn off **Stream HDMI Input Audio To Remote End**

6.20 Finding the system serial number and software version

You need the system serial number and software version when reporting a problem.

Procedure

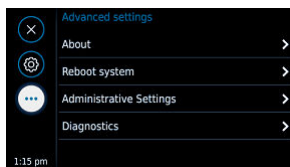
1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap

The **Advanced settings** screen displays.



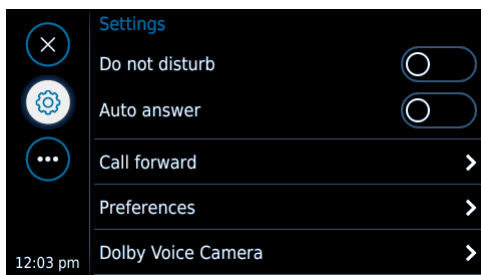
3. Tap **About > System**.
4. The software version and serial number are displayed.

6.21 Finding the Dolby Voice Camera serial number and software version

You need the Dolby Voice Camera serial number and software version when reporting a problem.

Procedure

1. Access the Dolby Voice Room **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap > **About > Status > Camera**.

The software version and serial number are displayed.

Advanced network configurations

You can perform advanced configurations, such as setting up a detached configuration on an 802.1x environment and converting a detached mode system to a tagged VLAN configuration.

- [Setting up a detached configuration on an 802.1x environment](#)
- [Setting up a detached configuration with a tagged VLAN](#)

7.1 Setting up a detached configuration on an 802.1x environment

You can setup a Dolby Voice Room detached configuration on an 802.1x environment.

Procedure

1. Load the 802.1x certificate into Dolby Voice Room and Dolby Conference Phone.
2. Access Dolby Conference Phone through the Dolby Voice Room user-interface:
`https://<URL>:30443`
3. Navigate to **Network > 802.1x**.
4. Enter the Dolby Conference Phone 802.1x credentials.
Dolby Conference Phone will reboot.
5. Connect Dolby Conference Phone to an 802.1x network port.
6. Dolby Conference Phone will power up and automatically authenticate.

7.2 Setting up a detached configuration with a tagged VLAN

You can setup Dolby Voice Room in detached mode with a tagged Dolby Conference Phone VLAN configuration.

Prerequisites

Verify the following prior to performing the configuration:

- Dolby Conference Phone must be in attached mode.
- The detached Dolby Conference Phone network port must be configured with an untagged VLAN.

Procedure

1. Access Dolby Conference Phone through the Dolby Voice Room user-interface:
`https://<URL>:30443`
2. Navigate to **Network > VLAN**.
3. Change the VLAN value.
Dolby Conference Phone will reboot.
4. When the user-interface is a blank screen, disconnect Dolby Conference Phone and convert it to detached mode.

Troubleshooting

If you encounter an issue with Dolby Voice Room, we recommend that you attempt basic troubleshooting before you contact Support. The information in this chapter describes likely causes of issues and corrective actions you can take to resolve them. Follow the available troubleshooting procedures to review logs and error messages, run diagnostics, and possibly upgrade the system software.

- [Report to Dolby](#)
- [Service QR code](#)
- [Web administration portal access over Wi-Fi](#)
- [Enabling screenshots](#)
- [Dolby Voice Room system issues and solutions](#)
- [Intelligent scene framing diagnostics](#)
- [Dolby Voice Camera issues and solutions](#)
- [Dolby Voice Camera Pro issues and solutions](#)
- [Display issues and solutions](#)
- [HDMI input issues](#)
- [CEC issues](#)
- [Dolby Conference Phone issues and solutions](#)

8.1 Report to Dolby

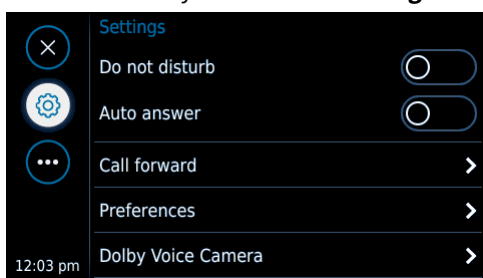
Report to Dolby allows you to easily send system logs to Dolby customer support.

About this task

You can send log files to *diagnostics.dolby.com*. Your service provider must work with Dolby customer support to retrieve this data.

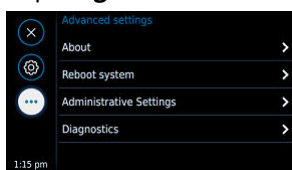
Procedure

1. Access the Dolby Voice Room **Settings** screen.

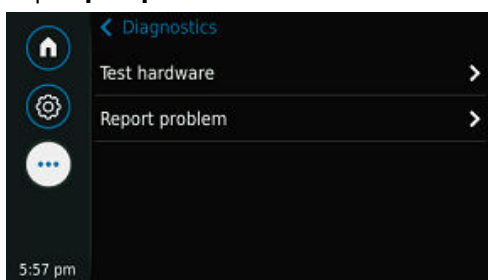


2. Tap  to call up the **Advanced Settings** screen.

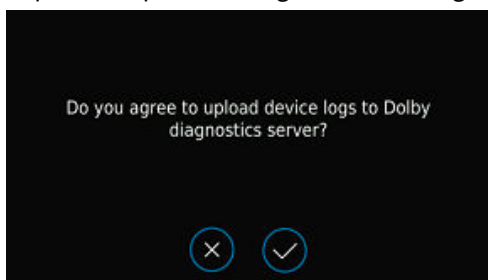
3. Tap **Diagnostics**.



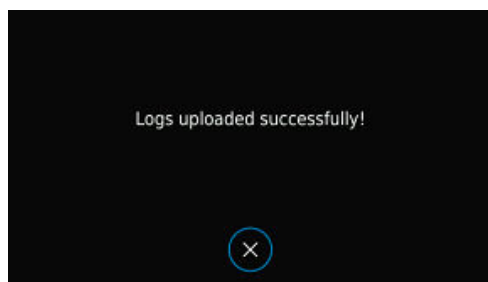
4. Tap **Report problem**.



5. Tap **Yes** to upload the log files to the diagnostics server.



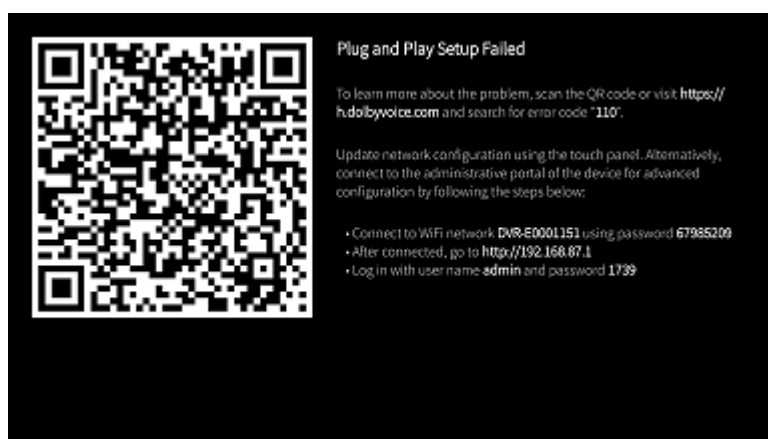
If the logs were uploaded successfully, a dialogue box displays.



8.2 Service QR code

The service QR code provides Dolby Voice Room system information and status during out-of-the-box configuration.

If plug-and-play capability fails during the out-of-the-box configuration, a service QR code displays on your TV screen.



You scan the service QR code using your mobile device, and a URL to a troubleshooting page is provided. Dolby may automatically direct you to your service provider's support website, based on the serial number of the Dolby Voice Room.

The troubleshooting page provides the following information:

- Serial number and version
- Problem description and resolution
- Request for additional support
- Advanced system information

8.3 Web administration portal access over Wi-Fi

Dolby Voice Room provides access to the administrative portal over Wi-Fi during out-of-the-box configuration.

If plug-and-play capability fails during the out-of-the-box configuration, the Wi-Fi access point is automatically enabled, allowing you to connect to Dolby Voice Room.



You can connect to Wi-Fi through a mobile device or any Wi-Fi enabled device using the Wi-Fi network name and password displayed on the TV, along with the service QR code.

Once connected, you can launch a web browser and log into the administrative portal. The administration web portal IP address and login credentials are displayed on the TV, along with the service QR code.

You can download log files through the Dolby Voice Room web administration portal.

In addition, you can modify the network configuration, as well use the web administration portal.

Wi-Fi is disabled when out-of-the-box configuration is complete.

8.4 Enabling screenshots

You can enable screenshots in Dolby Voice Room.

Prerequisites

Contact Dolby Customer Support to obtain the power user key to enable screenshots in Dolby Voice Room.

Procedure

1. Apply the power user key through SSH command-line.

```
DolbyCli>system poweruser key A123:4567:BC8D:910E
Your device serial ID is 100
Power user key is set to: A123:4567:BC8D:910E
Validating ... Key is valid.
Current time is: Mon Jun 03 04:21:56 2019
Key expires: Wed Jul 03 14:14:07 2019
```

2. Enable the system poweruser devtool.

```
system poweruser devtool enable
```

3. Log into the Dolby Voice Room web UI.
4. Navigate to **Status > Dolby Voice Hub > Device > Power User** and verify that **Screenshot Enabled** is set to true.
5. Navigate to **Tools > TV Screenshot** through the web-UI and you should be able to see Dolby Voice Room display image.

8.5 Dolby Voice Room system issues and solutions

Use this information for likely causes and corrective actions for the Dolby Voice Room system.

8.5.1 Power and start-up issues

For power and start-up issues, see these possible solutions.

Issue	Solution
The Dolby Voice Room system appears to have no power.	Determine whether the problem is caused by the system or the AC outlet. Do one of the following: • Check the power light indicator is on. It lights up when the unit is receiving power. • Ensure that the display (TV or monitor) is powered on and connected to the HDMI port. • Ensure that the power adapter is attached securely. • Ensure that the A/C outlet is functioning. • Check for a faulty power adapter. Replace the adapter and try again.
The Dolby Voice Hub has power but the phone has no power.	Determine if the problem is caused by the PoE connection. Do one of the following: • Ensure that the CAT6 cable is plugged into the PoE port on Dolby Voice Hub. This port has a Dolby Conference Phone icon. • Check the cable connection at the bottom of the phone. • Check for a faulty cable. Replace the cable and try again.

8.5.2 Dolby Voice Room power saving issues and solutions

Use this information for likely causes and corrective actions for Dolby Voice Room power saving issues.

Issue	Solution
Verify if the TV supports power saving control	Through the Dolby Conference Phone diagnostic page, navigate to: Settings > Advanced > Diagnostics > Test hardware > TV The TV status can be: • is connected • is disconnected • is powered on • is powered off If the status is powered on or is powered off, then power saving should work. The Wake and Sleep buttons force the TV to wake and sleep, as well as indicate if power saving works with the attached TV.
Disable power saving	1. Access the Dolby Voice Room web interface. 2. Expand Settings > User Preferences > Power Saving. 3. Disable power saving parameters. 4. Click Save.
TCL TV goes to the home screen when the system wakes	TCL TV has a feature that launches the home screen or last used TV input or specific input. By default home screen is enabled. This setting can be changed.

8.5.3 Network access issues

Here are possible solutions to network access issues.

Issue	Solution
The web interface Status tab indicates an alarm, and an alarm icon appears for the network status Status > Dolby Voice Hub > Device > Alarms and status . Position your mouse pointer over the alarm icon to view the network status code.	If one of these status codes appears, check whether the switch is correctly set up and if the Ethernet cable is acceptable: - Link is dormant: Network is pending authentication. - Link is down. - Unknown link error: There is an unknown network error. If this status code appears, check the network status: FAIL_DHCP: Failed. If this status code appears, check whether another device on the network is using the same IPv4 address: mac-address: Another device on the network is using the same IPv4 address. If one of these status codes appears, check whether the default gateway is up: - Default gateway is unreachable. - Default gateway is unconfigured. If one of these status codes displays, check whether the DNS servers are up: - DNS servers are unreachable. - DNS servers are unconfigured. - Only secondary DNS server is reachable: The primary DNS server is unreachable, whereas the secondary DNS server is reachable. If one of these status codes displays, check that the dynamic DNS is correctly configured on the DNS servers and the DHCP server: - DNS hostname unresolvable: The host name VPU9000-xxxxxxx is not resolvable. - DNS servers are unconfigured.

8.5.4 Authentication issues

Here are possible solutions to authentication issues.

Issue	Solution
You forgot the administrative password.	Perform a factory reset. The default administrative password (1739) is restored. After factory reset, you must reenter user preferences, such as language selection, date format, brightness, auto-answer, and so on.  Note: If you are not already logged in, you must use the fail-safe method to perform a factory reset.

Related information

[Resetting to factory defaults](#) on page 93

8.5.5 HDMI input issues

For HDMI input issues, see these possible solutions.

Table 7: HDMI input troubleshooting issues

Issue	Solution
HDMI input detection issues	- Check the HDMI Input Detected parameter. It should be set to <i>TRUE</i>. - Check the detection errors using the HDMI Input Error parameter - Validate the device type and name using the HDMI Input Device Type parameter.
Audio streaming issues	Verify that the status of these parameters are green HDMI Input Audio Device Status HDMI Input Audio Stream Status If the status for both parameters is not green, check the audio output settings on the HDMI input device and confirm the audio is routed to HDMI. Further isolate the issue by connecting the HDMI input source directly to the TV. If HDMI input audio is not transmitted in the meeting but renders fine without the meeting, consult your conference service provider.
Video streaming issues	Check the video stream status using HDMI Input Stream Active, it should be <i>TRUE</i> when active HDMI Input Stream Rate is reported.
Log collection for detection issues	1. Log-in to web UI. 2. Navigate to Logging > HDMI Input. 3. Set Detection Logs to <i>On</i>. 4. Set Logging Interval (seconds) to <i>1</i>. 5. Reproduce issue, download and share logs with support team member.
Video capture issues	1. Log-in to web UI. 2. Navigate to Logging > HDMI Input. 3. Set Detection Logs to <i>On</i>. 4. Set Logging Interval (seconds) to <i>1</i>. 5. Reproduce issue, download and share logs with support team member. Reproduce the issue, download and share the logs with a support team member.

You can check the HDMI status through the UI, by navigating to **Status > Device > Alarms and status**.

You can check the HDMI status through the UI, by navigating to **Status > Dolby Voice Room > Device > Alarms and status**.

You can check the HDMI logs through the UI, by navigating to **Settings > Logging > HDMI Input**

8.5.6 Software update issues

Here are solutions for Dolby Voice Room software update issues.

Issue	Solution
The message Invalid signature on software image file appears on the phone.	The image is signed by Dolby and the system performs a validation upon software upgrade. If you see this message, you are using an unauthorized image. Contact Dolby Customer Support if you received the image from Dolby. Otherwise, contact your conferencing service provider.
The message Invalid software image file appears on the phone.	After downloading the image, the system performs a checksum validation. This message appears if the validation failed. Reasons for this include transmission errors or a corrupt image file. If the image file came from Dolby, contact Dolby Customer Support. Otherwise, contact your conferencing service provider. If the issue persists, check with your organization's IT department to investigate any network transmission errors.
The message Failed to complete software update for an unknown reason appears on the phone.	During the software upgrade, a possible flash error was detected. Retry the software update. If the message appears again, contact Dolby Customer Support.

8.5.7 Video quality issues

Use this information for likely causes and corrective actions for video quality issues.

Issue	Solution
No image out of the camera in the preview (picture-in-picture (PiP)) or Camera not connected message.	Refer to the section on Dolby Voice Camera issues and solutions.
Remote participants report no video signal, but video preview (PiP) looks good.	Likely problem with the conference service provider application. Reboot the system and report the problem to the service provider.
Remote participants say video looks grainy, noisy, or out of focus.	Adjust lighting so that the room is well lit. If that does not remedy the issue, then adjust image settings as follows: 1. End the call and access the Settings screen. 2. Select Dolby Voice Camera to get a full-screen preview of the outgoing video signal. 3. Tap Change mode and select Room mode to get a full view of the room. 4. Tap Adjust image quality and then try these adjustments: • Adjust brightness, color intensity, and contrast. Avoid over-saturated skin tones. • If the room has no outside-facing windows, turn HDR off. Otherwise, if the image is washed out or faces are dark due to bright daylight coming in through a window, turn HDR on. In a well-lit room with these adjustments, the image should be sharp at a distance of one to three meters. If the preview still shows an out-of-focus image, return the camera for a replacement.
After making above adjustments, video preview looks good but remote participants say video is still poor quality.	There may not be enough up-link bandwidth, so the Dolby Voice Room system has automatically reduced transmission resolution to cope with the network. Consult your conferencing service provider for more information regarding minimum network requirements.

Issue	Solution
Poor video quality coming from the remote room.	Ask the remote conference participants to adjust the lighting in the room and adjust the image settings on their camera. If this does not resolve the issue, there may not be enough down-link bandwidth, so the remote Dolby Voice Room system has automatically reduced transmission resolution to cope with the network. Consult your conferencing service provider for more information regarding minimum network requirements.
Artifacts on the screen, such as flickering colored dots.	Replace the high-definition video cable to the display. We recommend using the cable included in the packaging. If you use your own cable, ensure that it is a high-speed cable.

Related information

[Dolby Voice Camera issues and solutions](#) on page 109

8.5.8 Spatial audio issues

Here are solutions to spatial audio issues.

Issue	Solution
Dolby Conference Phone speaker output does not reflect the speaking position (left or right side of room) of remote participants	Spatial audio functions only with headphones. Connect USB headphones to a computer, or connect headphones to a mobile device.
Audio in headphones does not align with speaking position (left side or right side) of participant in the remote room.	Ask remote participants to orient the Dolby Conference Phone to match the setting of the <code>Dvms.Audio.SpatialCaptureOrigin</code> configuration parameter, or to change the parameter. See Setting up individual Dolby Voice Room systems on page 38.

8.5.9 Corporate directory issues

Here are solutions for corporate directory issues.

Issue	Solution
Cannot connect to the LDAP directory. The system status shows there is a directory error.	The possible reasons for the failure are: • An incorrect LDAP server IP address • Incorrect LDAP server access credentials • An incorrect or missing LDAP server root CA certificate (necessary when accessing the LDAP server using Secure Sockets Layer (SSL)) In the Dolby Voice Room web interface, go to Status > Dolby Voice Hub > Device > Alarms and status. Find LDAP connected. A red circle indicates an LDAP error. Position your mouse pointer over the red circle to see the specific error that occurred.

8.5.10 Fatal error issues

Here are possible solutions to fatal error issues.

Issue	Solution
This message appears on the display: The services application has stopped responding. Press Debug to stop rebooting.	You can choose to debug the problem. This stops the system from performing the automatic reboot. If you do not tap Debug, the system automatically reboots after five seconds. If the system is configured to upload log files, the system uploads the core dump (if there is any) to the provisioning server during the next reboot. The core dump file name is <i>SN-timestamp-core.appname</i>. For example, <i>D0025416-1402090840-core.services_app</i> is a core dump generated from the <i>services_app</i> application on the device numbered D0025416. Attach the uploaded log files as well as the core dump file, and create a bug report for Dolby Customer Support to perform further analysis.  Note: Uploading a core dump during booting prolongs boot time.
The phone user interface continues to display the Dolby logo and does not display the home screen, even after 90 seconds.	A combination of configuration and software issues have left the system in a state from which it cannot recover. Try rebooting again by holding down the power button for five to ten seconds until you see the Dolby Voice logo on the display. If the reboot fails, then perform a factory reset. If the reboot still stalls at the Dolby logo, contact Dolby Customer Support.

Related information

[Hardware issues](#) on page 113

[Rebooting Dolby Voice Room](#) on page 92

[Resetting to factory defaults](#) on page 93

8.6 Intelligent scene framing diagnostics

Dolby Voice Room provides a diagnostics tool to help troubleshoot intelligent scene framing issues.

If the intelligent scene framing has these issues, you can turn on the intelligent scene framing diagnostics to perform a recording to help Dolby and the service provider support engineer understand why it is occurring:

- Zooms in and out even though people are not moving around.
- Cuts people off.
- Leaves people out of the frame.

The recording helps you understand how to adjust the camera for it to work optimally.

If the camera leaves people out or cuts people off in the frame, look at the recording and see if the people are in the source feed. If they are not in the source feed, then they will not be in the frame.

In addition, ensure the people around at the edge of the source feed are fully captured. If they are not fully captured, the camera may zoom in and out to exclude or include the partial view of the people at the edge.

Make sure to organize the room, or move the camera so that it is harder for people to be outside of the source feed including no partial view of the people.

By default, intelligent scene framing diagnostics is disabled. You can enable it through the Dolby Voice Room UI. Navigate to **Settings > Dolby Voice Camera > Diagnostics > Enable**.

When you enable intelligent scene framing diagnostics, the system starts recording whenever the intelligent scene framing feature moves the camera. The recorded file is saved into the system diagnostics package, which is available through the web UI download, “report a problem to Dolby”, or “report a problem to service provider,” if the service provider application implemented a reporting function.

Once they receive the recording, they can play it using a video player. The video is recorded in .mkv (Matroska Multimedia Container) format.

Each frame in the recording contains these elements:

- Source feed

What the camera sees in the raw form, the intelligent scene framing feature uses this feed to determine the best way to frame.

- Overlay feed

What the intelligent scene framing feature sees in the source feed - people detected are outlined with greenboxes, and the target frame is outlined with a blue rectangle.

- Target feed

What the user sees from the camera view.

- Metadata feed

Contains the local time of each frame, as well as additional information useful for Dolby to debug.

See the Dolby Voice Camera section for horizontal field of view of each camera type. For the Dolby Voice Camera Pro, see the panoramic camera field of view.

8.6.1 Setting up intelligent scene-framing diagnostics

You can setup intelligent scene-framing diagnostic for Dolby Voice Camera and Dolby Voice Camera Pro.

Procedure

1. Set **DVCamera.OperationMode** to **PEOPLE**.

2. Set **DVCamera.DebugMode** to **true**.

The system reboots automatically.

3. Make a video call, or enter camera preview mode.

4. Walk in front of the camera.

The camera follows the person who moves in front of it.

5. Download the syslog package.

6. Extract the files and check the autoframe directory.

The text files can be viewed in any editor. The .mkv files can be viewed on any video player which supports MPEG-4.

8.7 Dolby Voice Camera issues and solutions

Use this information for likely causes and corrective actions for Dolby Voice Camera issues.

Issue	Solution
No image out of the camera in the preview (PiP) or Camera not connected message.	Check the USB cable connections at the back of the camera. Remove and reconnect the cable at both ends. Wait 15 seconds for the Dolby Voice Camera connected message. - Try a different USB port on Dolby Voice Hub. - Try a different USB 3.0 cable To test the camera without joining a conference: 1. Access the Dolby Conference Phone Settings screen on the. 2. Tap Dolby Voice Camera. The indicator light on the camera should illuminate and the image from the camera should show on the display. (Ensure that the display is properly connected and powered on.)  Note: If the system notices that the camera is not connected, when you attempt to go to the Settings screen, the phone might bypass that screen and immediately display the Status screen, indicating an alarm next to the Camera entry. In this case, the display also shows Check system status. If the issue persists after reconnecting or replacing the cable, reboot the system. If the problem continues, contact Dolby Customer Support.
This message shows on the display: Cannot complete the Dolby Voice Camera software update	The Dolby Conference Phone user interface displays the reason why the software update failed. Check for an alarm on the Status screen on the phone user interface. Disconnect and reconnect the camera with Dolby Voice Hub powered on. The system retries the software update. You can check the software update status of the camera in the web interface by going to Status > Dolby Voice Camera > Status
Intelligent scene framing (people mode) not tracking people properly	- Maximum recommended conference room size for optimal intelligent scene framing performance is 15 ft × 15 ft (4.5 m × 4.5 m). - Check the image quality of the faces in the camera preview (PiP). Access the Settings screen on the phone UI, and then tap Dolby Voice Camera > Adjust image quality. If the skin tones are over-saturated, adjust the color intensity. If the faces in the preview are too bright or too dark, adjust the brightness, and if the faces are washed out, adjust the brightness and contrast, and turn on HDR. If the faces are dark or washed out due to a bright background, turn on HDR.
The camera image occasionally resets for a brief second.	This could be a cable issue. Ensure that you are using the USB 3.0 cable supplied with the system. If the problem persists, try replacing the cable.

Related information

[Video quality issues](#) on page 106

8.8 Dolby Voice Camera Pro issues and solutions

Use this information for likely causes and corrective actions for Dolby Voice Camera Pro issues.

Issue	Solution
Intelligent Scene Framing (or “People” and “Auto” mode) does not work	- Make sure the Intelligent Scene Framing feature is enabled under Settings > Dolby Voice Camera > Intelligent Scene Framing. - If the feature is turned on, and Intelligent Scene Framing is still not working, make sure both USB ports on the camera are connected to the Dolby Voice Hub. - Check the camera status under Settings > ... > About > Status. If there is an alarm stating “Panoramic camera not found,” that means the Dolby Voice Hub did not detect the connection from the Panoramic camera. Check the camera USB cable connections from the port with label USB-2, and use the Settings > Status menu to ensure the panoramic camera is detected.
Camera is not detected by the Dolby Voice Hub	- Make sure the camera power supply is connected, and the power indicator LED is on (should be green color) - Make sure both USB ports are connected to the Dolby Voice Hub.

8.9 Display issues and solutions

Use this information for likely causes and corrective actions for display (TV or monitor) issues.

Issue	Solution
No picture	- Ensure that the display is powered on. - Ensure that Dolby Voice Hub is powered on. (The power indicator light should be lit up.) - Ensure that the display is connected to one of the high-definition video out ports, labeled 1 and 2, not the high-definition video in port. - Disconnect and reconnect the high-definition video cable - Reboot - Try another high-speed (CAT2) high-definition video cable - Try another display
In a dual display configuration, images are out of order (swapped).	Swap the images. (On the web interface, go to Settings > Displays > Swap monitor position).

Related information

[Video quality issues](#) on page 106

8.10 HDMI input issues

For HDMI input issues, see these possible solutions.

Table 8: HDMI input troubleshooting issues

Issue	Solution
HDMI input detection issues	- Check the HDMI Input Detected parameter. It should be set to <i>TRUE</i>. - Check the detection errors using the HDMI Input Error parameter - Validate the device type and name using the HDMI Input Device Type parameter.
Audio streaming issues	Verify that the status of these parameters are green HDMI Input Audio Device Status HDMI Input Audio Stream Status If the status for both parameters is not green, check the audio output settings on the HDMI input device and confirm the audio is routed to HDMI. Further isolate the issue by connecting the HDMI input source directly to the TV. If HDMI input audio is not transmitted in the meeting but renders fine without the meeting, consult your conference service provider.
Video streaming issues	Check the video stream status using HDMI Input Stream Active, it should be <i>TRUE</i> when active HDMI Input Stream Rate is reported.
Log collection for detection issues	- Log-in to web UI. - Navigate to Logging > HDMI Input. - Set Detection Logs to <i>On</i>. - Set Logging Interval (seconds) to <i>1</i>. - Reproduce issue, download and share logs with support team member.
Video capture issues	- Log-in to web UI. - Navigate to Logging > HDMI Input. - Set Detection Logs to <i>On</i>. - Set Logging Interval (seconds) to <i>1</i>. - Reproduce issue, download and share logs with support team member. Reproduce the issue, download and share the logs with a support team member.

You can check the HDMI status through the UI, by navigating to **Status > Device > Alarms and status**.

You can check the HDMI status through the UI, by navigating to **Status > Dolby Voice Room > Device > Alarms and status**.

You can check the HDMI logs through the UI, by navigating to **Settings > Logging > HDMI Input**

8.11 CEC issues

For CEC connection issues, see these possible solutions.

Issue	Solution
If the Dolby Voice Hub is connected to the TV through a High-Definition Multimedia Interface switch, the CEC control feature may not work.	Perform a diagnostic test from the Dolby Conference Phone to verify if the setup is working.
The CEC controller on the Dolby Voice Hub may run into a bad state and stop sending or receiving commands.	Full power cycle of the Dolby Voice Hub restarts controller and it starts working as normal.
Some TVs may behave unexpectedly for CEC controls.	For Self-Monitoring, Analysis, and Reporting Technology (SMART) TVs, update the TV firmware to the latest version before conducting the test. This should resolve the CEC control problems.

Use these commands if dual display is being used and the CEC control is not working as expected:

Command	Description
<code>system hdmi-cec list-firmware</code>	Shows the current firmware version, as well as firmware available for updating.
<code>system hdmi-cec update-firmware</code>	Automatically updates firmware to latest version. If the firmware is up-to-date, an error will be returned.
<code>system hdmi-cec update-firmware <fw_file_to_flash></code>	Writes firmware from the specified file and disregards the version information. This can be used when downgrading or manually updating to a specific version.

8.12 Dolby Conference Phone issues and solutions

Use this information for likely causes and corrective actions for the Dolby Conference Phone.

8.12.1 Hardware issues

These are possible solutions to hardware issues.

Issue	Solution
Buttons do not work. For example, when the phone is idle, pressing the volume button does not bring up the ringer widget, and you cannot hear the ringer tone.	Tap the phone icon on the home screen. Tap the keypad digits on the dialer screen: - If you can hear the DTMF tone, then there might be a problem with the hardware. To confirm whether the keys are working, access the Settings menu, then tap  > Diagnostics > Test hardware > Keys and follow the instructions on the screen. - If you cannot hear the DTMF tone, wait until the fatal error screen displays (see Fatal error issues on page 107) and the phone reboots. Log files from the error are uploaded to the provisioning server and will assist you in finding the problem.
Touch screen is not accurate. For example, when you touch the screen, the coordinates in the phone software do not compare to where you touched the screen.	It is possible that the touch screen is not accurately calibrated. Access the Settings menu, then tap  > Diagnostics > Test hardware > Touch Screen. Tap the screen anywhere in the grid and determine whether the crosshair that appears is exactly where you touched. Try again in a different location. If it appears that the phone touch screen is not accurately calibrated, then recalibrate. See Running Dolby Conference Phone diagnostics.
One or more speakers are not working.	To verify, access the Settings menu, then tap  > Diagnostics > Test hardware > Speakers and test each speaker.

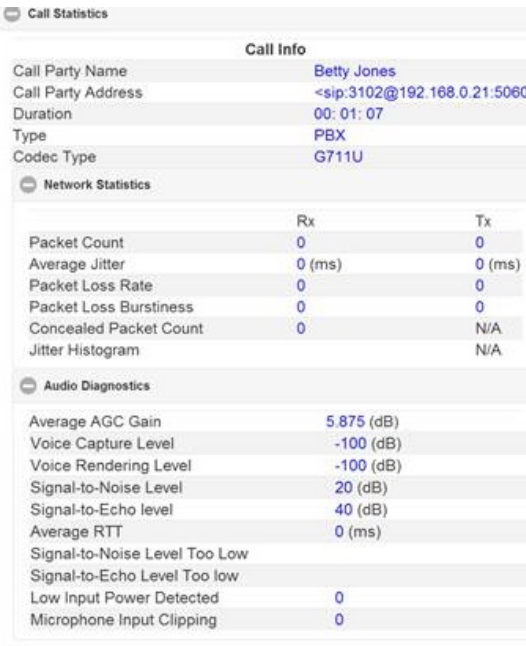
Issue	Solution
One or more microphones are not working.	To verify, access the Settings menu, then tap  > Diagnostics > Test hardware > Microphones and test each microphone one at a time. When testing a microphone, speak a short phrase and wait for the phone to play back the recording.
The LED halo flashes red and the phone unexpectedly mutes and unmutes.	Check for and remove any HDMI devices, other than the satellite microphones, connected to the phone microphone ports.

8.12.2 IP PBX calling issues

Here are possible solutions for IP PBX calling issues.

Issue	Solution
The user cannot make a call because the phone button is grayed out, and there are no alarms (red circle with exclamation point) relating to the call server.	The system is not configured for IP PBX calling. Ensure that the system is registered with the call server and check these configuration parameters: • <code>Sip.Account.DisplayName</code> • <code>Sip.Account.UserName</code> • <code>Sip.Account.DisplayNumber</code> • <code>Sip.Pbx.OutboundProxy1</code> • <code>Sip.Pbx.OutboundProxy1Port</code> You can view these parameters on the web interface here: • Settings > IP PBX Settings > Account • Settings > IP PBX Settings > Server
The user cannot make a call because the phone button is grayed out and the Status screen shows an alarm.	Access the Status screen to determine whether there is an alarm (red circle with exclamation point) adjacent to Call Server. If there is, then the system is not registered with the call server or cannot connect to the call server. Through the Dolby Voice Room web interface, go to Status > Dolby Voice Hub > Device > Alarms and status. Find SIP Registered, and position the mouse pointer over the alarm icon to obtain the failure details. The reason could be one of the following: • The system cannot connect to the call server due to network errors. • The system cannot connect to the call server due to authentication failures (incorrect credentials). • The system cannot connect to the call server due to other configuration errors. You can review all configuration parameters listed under SIP on the web interface Help page.

Issue	Solution
The user always gets a busy tone after dialing a number.	The possible reasons for the failure are: • The dial plan on the server side is set incorrectly. • The phone the user is calling is no longer registered to the server. • The phone is using an incompatible codec configuration. Through the Dolby Voice Room web interface, go to Settings > IP PBX Settings > Server and scroll down to the PBX Codec List. It is possible that the phone that the user is trying to call does not support the same codec.
The user cannot receive a call.	The possible reasons for the failure are: • The do not disturb feature is enabled. To disable, access the Settings screen, then change the do not disturb Off / On option to Off. If do not disturb is enabled, a scrolling message displays on the home screen. • The call forwarding feature is enabled. To disable, access the Settings screen, tap Call Forward, and then delete the appropriate forwarded number. If call forwarding is enabled, a scrolling message displays on the home screen. • The phone is using an incompatible codec configuration. To see a list of supported codecs, in the web interface, navigate to Settings > IP PBX Settings > Server and scroll down to the PBX Codec List. It is possible that the phone the user is trying to call does not support the same codec.

Issue	Solution
During a call, the user cannot hear or be heard by the other end of the call, but the call timer shows an increasing call length.	The possible reasons for the failure are: • Network connectivity is lost between the phone and the other end of the call. • The system encountered a serious error. Go to Fatal error issues on page 107. • The signal processing on the phone does not get packets from the network or send packets to the network. Through the Dolby Voice Room web interface, navigate to Status > Call Statistics to view the run-time call statistics as shown below. If the receiver (Rx) or transmitter (Tx) counts grow, this indicates that the signal processing is sending or receiving packets to or from the network. If the receiver counts do not grow, this indicates that the phone is not receiving any packets from the other end. 

Issue	Solution
The IP-PBX registration is successful but calls cannot be made and are not received.	Check the logs to see if the system is continuously trying to register with IP-PBX. If this occurs, you must set the Allow Contact Header Rewrite parameter to OFF. 1. Using the admin web page, navigate to Settings > IP PBX > Account. 2. Set Allow Contact Header Rewrite to OFF. 3. Click Save.
SIP registration is successful, but intermittent calls fail.	Check if the outbound proxy server hostname resolves to more than eight IP addresses. If this occurs, you must set the Set Active Server to Preferred Server's Order. 1. Using the admin web page, navigate to Settings > IP PBX > Server > Call Failover Mode. 2. Set Set Active Server to Preferred Server's Order. 3. Click Save.

8.12.3 Dolby Satellite Microphone issues

Here are possible solutions for Dolby Satellite Microphone issues.

Issue	Solution
The user plugs in a Dolby Satellite Microphone, but no tone plays.	• Try the other microphone port. If the same microphone works in one port but fails in the other, replace the Dolby Conference Phone. • Check the microphone on another phone that is known to work with satellite microphones. If the failure continues, replace the microphone.

8.12.4 Dolby Conference Phone pairing issues

For Dolby Conference Phone connection issues, see these possible solutions.

Issue	Solution
The Dolby Conference Phone cannot connect to the Dolby Voice Hub and the Dolby Voice Hub Unreachable message appears.	Tap  for more information. The screen shows the phone and Dolby Voice Hub IP addresses. You can use the phone IP address to access the Web interface to perform diagnostic tasks. If the hub IP address is 192.168.67.1, it means the phone was unable to discover the hub IP address. Verify that Dolby Conference Phone and Dolby Voice Hub are powered on and connected to the network. Perform these tasks to correct this issue: 1. Go to the Dolby Conference Phone Web interface. 2. Under Status, locate Bluetooth Module Initialized and ensure there is a green bubble next to it, indicating it is working. If the bubble is red, reboot the Dolby Conference Phone. If rebooting does not work, you can: • Continue to use the attached mode configuration. • Manually configure the Dolby Voice Hub IP address. You must use a fixed IP address. • Return the device to the manufacturer. If the screen shows a non 192.168.67.1 IP address and the phone still cannot connect to the IP address, the phone may not be able to reach the discovered hub IP address. You must ensure the phone is on the same network as the hub.

Error Messages

These tables list error messages, possible causes, and suggested actions.

When an error occurs, users will see:

- A warning icon on the phone home screen
- An informative message when the warning is first detected
- [General error messages](#)
- [Dolby Voice Console error messages](#)
- [Provisioning service error messages](#)
- [Dolby Conference Phone setup app error messages](#)
- [Dolby Voice Hub application error messages](#)

9.1 General error messages

Most of the errors listed here are logged to the system log file. However, if you are having trouble connecting to the provisioning server, the system may not be able to upload every error.

- [Error messages that appear on the Dolby Conference Phone](#) on page 120
- [Error messages that appear on the display \(TV or monitor\)](#) on page 123

Error messages that appear on the Dolby Conference Phone

Error message	What the administrator should do
Parameter does not exist.	You have added a configuration parameter that is not supported by the system in the configuration file. Remove it, and try again.
Invalid value.	Enter a valid value.
Invalid value range.	Enter a value within the valid range.
Invalid value type.	You entered a value of a different type (for example, using an integer instead of a Boolean value). Enter a value of the correct type.
Invalid value length.	You entered a string that is too short or long. Enter a valid string.
Write access denied.	You are trying to set a configuration parameter in the configuration file that cannot be changed in that way. For information on parameters, see Configuration parameters on page 41.
Invalid numeric value.	You entered a numeric value that does not match the parameter range rule. (For example, it is larger than the maximum value or smaller than the minimum value.) Enter a valid numeric value.
Invalid alphabetic value.	You entered a value that contains invalid characters for an alphabetical parameter. Enter a valid alphabetical value.
Invalid alphanumeric value.	You entered a value that contains invalid characters for an alphanumeric parameter. Enter a valid alphanumeric value.
IPPBX service is not available. Try again later.	The IP PBX is not available, or the system is not registered to the IP PBX. Contact the IP PBX administrator.
Invalid input.	You entered an invalid value. Verify and try again.
Invalid server address. Try again.	You entered an invalid provisioning server address. Verify and try again.
Reboot is required to apply new configuration, would you like to reboot now? System will reboot automatically in x seconds.	A change in the system configuration parameter required the system to reboot. If you want to delay rebooting the system, tap No immediately.
The services application has stopped responding, press Debug to stop rebooting.	See Fatal error issues on page 107.
Unable to download software image file.	See Software update issues on page 105.
Invalid signature on software image file.	See Software update issues on page 105.
Unable to download meta file.	See Software update issues on page 105.

Error message	What the administrator should do
Invalid software image file.	See Software update issues on page 105.
Invalid meta file.	See Software update issues on page 105.
Software update failed. Contact system administrator.	See Software update issues on page 105.
Invalid IP.	You entered an IP address that does not conform to the IP address format requirement. Enter a valid IP address.
Invalid IP address. Try again.	You entered an invalid IP address. Verify the address, and enter a valid IP address.
Invalid domain value.	You entered a domain name that does not conform to the domain-name format requirement. Enter a valid domain name.
Invalid URL value.	You entered a URL that does not conform to the URL format requirement. Enter a valid URL.
Failed to connect to server.	The system cannot connect to the provisioning server. Check network connections.
Failed to connect to the server due to SSL error.	The system cannot connect to the HTTPS provisioning server. Check network connections, verify that the provisioning server CA certificate was downloaded to the system, and confirm that the system time is current.
Invalid credentials. Try again.	The system cannot connect to the provisioning server. For an HTTPS provisioning server, verify that a CA certificate was downloaded to the system from the provisioning server. Check provisioning server credentials. Check that system credentials match the device pool credentials setup on Dolby Conferencing Console.
Could not find master configurations.	The master configuration files are not on the provisioning server.
Invalid configurations. Press the "More Options" button for more information.	See Provisioning issues .
Provisioning failed. Try again.	There was an error during the first-time provisioning of the system.
Server address is required for static provisioning.	If you want to use static provisioning, you must enter a provisioning server IP address.
Write access not allowed for provisioning.	A <i>.json</i> file on the provisioning server contains a configuration parameter that is not writable to the current <i>.json</i> file on the system. The parameter is not allowed to be configured from the provisioning server. See Provisioning issues .
Failed to get or set configuration.	The configuration files or certificates listed in the master configuration file are not on the provisioning server.
Not allowed to update provisioned configuration.	There was an attempt to update a configuration parameter that is presently in the configuration file on the provisioning server. Remove these parameters from the configuration files on the provisioning server, and try again.

Error message	What the administrator should do
Write not allowed for provisioning.	You specified a parameter in the configuration file on the provisioning server that cannot be modified by provisioned interface. Remove the parameter from the configuration file, and try again.
IEEE 802.1X is unconfigured but challenged.	You have to configure the IEEE 802.1X user identity and password to be able to use the system on the current network.
Unsupported IEEE 802.1X authentication method required by RADIUS server.	The current network requires an authentication method that is not supported by the system. To use the system, you must move it to another network or get the network administrator to change the current authentication method to one that the system supports.
IEEE 802.1X authentication failed due to wrong credentials.	Verify that the correct credential has been provided for 802.1X authentication.
IEEE 802.1X authentication failed due to certificate validation.	Verify that the correct CA certificate is installed on the system prior to 802.1X authentication if you are using EAP-PEAP authentication. Verify the installed certificate on the system through the web interface. Select Status > System > Device and scroll down to CA Certificates Details. If the certificate is successfully installed, you should see the CA certificate for the 802.1X TLS authentication. If the certificate is not present, you must set up the configuration file for the system to download the CA certificate. See Adding certificate authority certificates on page 43.
ERROR: Certificate <certificate fingerprint> is expired: <ddd mmm dd hh:mm:ssyyy>	This message displays on the CA certificate status screen only. This certificate has expired and needs to be replaced on the system. Contact the issuer (for example, Dolby or your conferencing service provider) for a new certificate.
WARNING: Certificate <certificate fingerprint> will expire soon: ddd mmmdd hh:mm:ss yyyy>	This message displays on the CA certificate status screen only. This certificate will expire within the next seven days. Contact the issuer (for example, Dolby or your conferencing service provider) for a new certificate.
Unsupported authentication method.	An unsupported 802.1X authentication method was used in the configuration files. Change it to a supported method, and reboot the system.
Failed to download PAC file.	The PAC file did not download. The administrator needs to check and, if necessary, correct the URL for downloading the PAC file; confirm that the system can reach that URL; and then download the PAC file.
Invalid PAC file.	The downloaded PAC file is invalid.
PAC syntax error.	The PAC file has syntax errors and cannot be processed. Correct the errors and try again.
PAC file too large.	The PAC file size is greater than 64 kB. This is the maximum supported size.
Unresolvable hostname error	The proxy server host name is not resolvable. Verify the server name in the PAC file.
Unreachable error	The proxy server is unreachable. Verify the server name in the PAC file. Verify that the proxy server is online.

Error message	What the administrator should do
Authentication error	Proxy authentication failed. Verify the user name and/or password for the proxy server.
Higher logging level is enabled. Performance might be impacted due to excessive logging. Contact administrator to reset.	Reset logging level. See Log levels on page 82.
Not allowed to update provisioned configuration.	You attempted to make a local configuration change to a parameter that is provisioned, where local means either on the web interface or through a shell command. The phone user interface disables entry of the provisioned parameters, so you can view the value but not make a change.
Invalid PBX codec list.	You entered an invalid IP PBX codec list as the last entry in the table.
Invalid domain.	You entered an invalid SIP domain name.
Invalid URL.	You entered an invalid URL.
Duplicate parameter.	The provisioned configuration file contains multiple instances of the same parameter. Remove the duplication, and try again.

Error messages that appear on the display (TV or monitor)

Error message	What the administrator should do
Invalid conference service software hub configuration.	Hub application URL is empty. Check the hub application URL on the Dolby Conferencing Console. Or, on the web interface, check Settings > Dolby Voice Conferencing Service > Hub Application > Application URL . Or check configuration parameter <code>Dvms.HubApp.Url</code> .
Unable to load the conference service software on the hub.	Unable to download the hub application. Check the hub application URL on the Dolby Conferencing Console. If correct, your conferencing service provider's server might be unavailable or you may have a network issue preventing Dolby Voice Room from accessing the server. Check proxy settings and ensure that your firewall policy permits access to the server. Obtain the server address from your conferencing service provider. You can also check the web interface at Settings > Dolby Voice Conferencing Service > Hub Application > Application URL, or check configuration parameter <code>Dvms.HubApp.Url</code>.  Note: The system will continuously attempt to reload the conference service software.
Invalid conference service software phone configuration.	Dolby Conference Phone application URL is empty. Check the Dolby Conference Phone application URL on the Dolby Conferencing Console. Or, on the web interface, check Settings > Dolby Conference Phone > Application > Application URL . Or check configuration parameter <code>Phone.App.Url</code> .

Error message	What the administrator should do
Unable to load the conference service software on the phone.	Unable to download the application that runs in the phone. Check the Dolby Conference Phone application URL on the Dolby Conferencing Console. If correct, your conferencing service provider's server might be unavailable or you may have a network issue preventing Dolby Voice Room from accessing the server. Check proxy settings and ensure that your firewall policy permits access to the server. Obtain the server address from your conferencing service provider. You can also check the web interface at Settings > Dolby Conference Phone > Application > Application URL, or check configuration parameter <code>Phone.App.Url</code>.  Note: The system will continuously attempt to reload the conference service software.
Unable to load the conference service setup software on the hub.	Check the OOB application URL on the Dolby Conferencing Console. If correct, your conferencing service provider's server might be unavailable or you may have a network issue preventing Dolby Voice Room from accessing the server. Check proxy settings and ensure that your firewall policy permits access to the server. Obtain the server address from your conferencing service provider. You can also check the web interface at Settings > Dolby Voice Conferencing Service > Hub Application > OOB Application URL, or check configuration parameter <code>Dvms.HubApp.OobUrl</code>.
Cannot complete the Dolby Voice Hub software update. Check the phone for more information.	The Dolby Conference Phone user interface displays the reason why the software update failed. Check for an alarm on the Status screen on the phone user interface by closing this message.
Cannot complete the Dolby Conference Phone software update. Check the phone for more information.	The Dolby Conference Phone user interface displays the reason why the software update failed. Check for an alarm on the Status screen on the phone user interface by closing this message.
Cannot complete the Dolby Voice Camera software update. Check the system settings for more information.	The Dolby Conference Phone user interface displays the reason why the software update failed. Check for an alarm on the Status screen on the phone user interface by closing this message. Disconnecting and reconnecting the camera retries the software update.
Cannot complete the Dolby Conference Phone and the Dolby Voice Camera software update. Check the phone for more information.	The Dolby Conference Phone user interface displays the reason why the software update failed. Check for an alarm on the Status screen on the phone user interface by closing this message.
The connected display's resolution is not supported. Connect a display with a resolution of at least 1920 x 1080.	See Display on page 18 for Dolby Voice Room display requirements.

9.2 Dolby Voice Console error messages

These are possible solutions to Dolby Voice Console errors when deploying Dolby Voice Room with the plug and play provisioning method.

These messages appear on the Dolby Conference Phone screen.

Error code	Message displayed	Solution
101	Plug and play setup failed because the system has failed 802.1X authentication, which is required by your network. You can modify 802.1X settings in the system according to your network security policy. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system has failed 802.1X authentication, which is required by your network. You can modify 802.1X settings in the system according to your network security policy. 802.1X, also known as port guard, configuration on the system often requires the security team in your IT organization to provide help and guidance. If you are not sure about the 802.1X policy, consult your security team for more information. You can modify basic 802.1X settings by tapping the Next button on the Dolby Conference Phone to navigate to the network settings screen. You can toggle between 802.1X authentication methods, turn off 802.1X authentication, enter 802.1X authentication credentials, or accept server certificates to get the device authenticated against your network. You may also need to import a server certificate into the system through its web user interface. For more information on 802.1X deployment, see Configuring 802.1X during out-of-the-box setup on page 31. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
102	Plug and play setup failed because the system was unable to obtain a valid IP address. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to obtain a valid IP address for communicating with the Dolby Voice Console . The system needs an IP address that can reach Internet in order to function properly. If there are multiple network ports in the room, you can try to plug the main network interface to a different port. If your network uses VLAN, the system may need to be configured for a specific VLAN. The system supports manual VLAN configuration. You can modify the VLAN settings by tapping the Next button on the Dolby Conference Phone. You may need to consult your IT help desk for assistance. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
103	Plug and play setup failed because a DNS server was not configured for the system. Contact your IT help desk for assistance.	The plug and play provisioning method failed because a DNS server was not configured for the system. The system requires a valid DNS server to resolve the host name to the IP address on the Internet. If your DHCP server did not supply a DNS configuration to the system, you can configure the DNS server manually on the Dolby Conference Phone. Follow the instructions on the Dolby Conference Phone to provide the DNS server settings; for example you can enter the public DNS server 8.8.8.8. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
104	Plug and play setup failed because the system cannot complete proxy auto detection. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the PAC file to complete the setup. The system can automatically detect proxy server on your network. It has detected a PAC file, but was unable to download the file to complete the auto discovery process. Follow the instructions on the Dolby Conference Phone to disable WPAD. You can either leave the PAC URL field blank, or enter a valid URL for the system to download the PAC file. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
105	Plug and play setup failed because the proxy on your network requires authentication. Provide the proxy authentication credentials on the next screen.	The plug and play provisioning method failed because the proxy on your network requires authentication. If your proxy server requires authentication, the system must be configured with authentication credentials. If you do not know what credentials to use, contact your IT organization's security team. If you already know the authentication credentials, tap the Next button and move to the next screen to supply proxy authentication credentials and try again. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
106	Plug and play setup failed because of an unknown proxy error. Contact your service provider for assistance.	The plug and play provisioning method failed because of an unknown proxy error. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
107	Plug and play setup failed because the Dolby Voice Console (https://console.dolbyvoice.com) host name is not resolvable. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console host name is not resolvable. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is required for the plug and play provisioning method. The DNS server configured on system must be able to resolve the Dolby Voice Console host name. Tap the Next button to review the network configuration. You can manually override the DNS server settings on the next screen. Configuring the public DNS server 8.8.8.8 may help resolve this issue. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
108	Plug and play setup failed because the Dolby Voice Console (https://console.dolbyvoice.com) is unreachable from this network. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console is unreachable from this network. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the system to connect to the Dolby Voice Console. Make sure that your firewall whitelists <i>*.dolbyvoice.com</i> to allow the plug and play provisioning method and service data traffic from the system to pass through. Consult your conferencing service provider's network preparation guidelines for all addresses required by the service. For more information, see DHCP and DNS servers on page 20 and Firewall on page 21. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
109	Plug and play setup failed because of an unknown SSL error. Contact your service provider for assistance.	The plug and play provisioning method failed because of an unknown SSL error. The system detected an unknown SSL error and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
110	Plug and play setup failed because of a communication error with the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because of a communication error with the Dolby Voice Console. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The system experienced a communication error with the Dolby Voice Console and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
111	Plug and play setup failed because of an error in discovering provisioning information for the system. Contact your service provider for assistance.	The plug and play provisioning method failed because of an error in discovering provisioning information for the system. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The system experienced an error while trying to retrieve the provisioning service information from the Dolby Voice Console and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
112	Plug and play setup failed because the system's provisioning information could not be found on the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because of an error in discovering provisioning information for the system. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The provisioning information for the system was not found on the Dolby Voice Console and the system was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
113	Plug and play setup failed because the system cannot reach the proxy server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system cannot reach the proxy server. The system has detected a proxy and is trying to connect to the Dolby Voice Console using that proxy. However, the proxy server is unreachable. Follow the instructions on the Dolby Conference Phone to disable WPAD and clear the setting for PAC configuration. If your network allows Internet access bypassing the proxy server, this may resolve the issue. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
114	Plug and play setup failed because the system failed authentication with the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because the system failed authentication with the Dolby Voice Console. The Dolby Voice Console authenticates the devices during the plug and play provisioning method. The system failed authentication due to security reasons. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
115	Plug and play setup failed because the system encountered an SSL certificate issue connecting to Dolby Voice Console (https://console.dolbyvoice.com). Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system encountered an SSL certificate issue connecting to Dolby Voice Console. The system is unable to verify the identity of the Dolby Voice Console. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface. See Accessing the Dolby Voice Room web interface on page 77 and Adding certificate authority certificates on page 43 for information on connecting to the Dolby Voice Room web interface using a web browser and importing the CA certificate into the device from the web interface. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
116	Plug and play setup failed because the Dolby Voice Console is blocked by a firewall. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console is blocked by a firewall. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the system to connect to the Dolby Voice Console. Make sure that your firewall whitelists <i>*.dolbyvoice.com</i> to allow the plug and play provisioning method and service data traffic from the system to pass through. For additional help, contact your conferencing service provider. For more information, see Firewall on page 21. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
150	Plug and play setup failed because the main network interface of the system is down. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the main network interface of the system is down. Contact your IT help desk for assistance. Dolby Voice Room has two network interfaces, one for connecting to your corporate network and one for connecting to the Dolby Conference Phone. Make sure the correct interface is connected to the corporate network. If the problem persists, replace the network cable and try again. You may need to consult your IT organization to verify that the port is active on the Ethernet switch. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
151	Plug and play setup failed because the system detected another device on your network with a duplicate IP address. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system detected another device on your network with a duplicate IP address. Contact your IT help desk for assistance. You can find the conflicting device's MAC address on the Dolby Voice Room web interface, and provide the MAC address to your network administrator to locate the device. This usually occurs if the device is set to a static IP address that is in the DHCP address range. If your network uses VLAN for video conferencing equipment, you can enter the VLAN ID on the Dolby Conference Phone so that the Dolby Voice Hub can use a different IP address. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
152	Plug and play setup failed because the default gateway is unreachable. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the default gateway is unreachable. Contact your IT help desk for assistance. Verify that the DHCP assigned default gateway address on the Dolby Conference Phone is valid. This error may be due to incorrect DHCP configuration. If your network uses VLAN for video conferencing equipment, you can enter the VLAN ID on the Dolby Conference Phone so that the Dolby Voice Hub can use a different IP address and default gateway. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

9.3 Provisioning service error messages

These are possible solutions to provisioning service errors when deploying Dolby Voice Room with the plug and play provisioning method.

These messages appear on the Dolby Conference Phone screen.

Error code	Message displayed	Solution
201	Plug and play setup failed because the system could not connect to the provisioning server. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system could not connect to the provisioning server. The system needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the system to connect to the Dolby Voice Console. Log in to the system user interface to determine which provisioning server the system is configured with. Make sure that your firewall whitelists *.dolbyvoice.com to allow the plug and play provisioning method and service data traffic from the system to pass through. Consult your conferencing service provider's network preparation guidelines for all addresses required by the service. For more information, see DHCP and DNS servers on page 20, Firewall on page 21, and Handling network issues during provisioning on page 30. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
202	Plug and play setup failed because the system encountered an SSL error while connecting to the provisioning server. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system encountered an SSL error while connecting to the provisioning server. The system is unable to verify the identity of the provisioning server. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface. See Accessing the Dolby Voice Room web interface on page 77 and Adding certificate authority certificates on page 43 for information on connecting to the Dolby Voice Room web interface using a web browser and importing the CA certificate into the device from the web interface. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
203	Plug and play setup failed because the system failed authentication with the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system failed authentication with the provisioning server. The system is attempting to connect to a provisioning server that requires authentication. If you do not know the credentials, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
204	Plug and play setup failed because the system cannot find the master configuration file from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system cannot find the master configuration file from the provisioning server. The master configuration file is the first file the system downloads during the provisioning process. The system was unable to download the master configuration file. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
205	Plug and play setup failed because the system cannot find the configuration file from the provisioning server. Contact your service provider for help.	The plug and play provisioning method failed because the system cannot find the configuration file from the provisioning server. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
206	Plug and play setup failed because the system found an invalid master configuration file from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system found an invalid master configuration file from the provisioning server. The master configuration file is the first file the system downloads during the provisioning process. The system found an invalid master configuration file. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
207	Plug and play setup failed because the system cannot download the upgrade package metadata file. Contact your service provider for assistance.	The plug and play provisioning method failed because the system cannot download the upgrade package metadata file. The upgrade package metadata file contains software upgrade information for the system. The system uses this file to determine whether it needs to upgrade. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
208	Plug and play setup failed because the system found an invalid upgrade package metadata file. Contact your service provider for assistance.	The plug and play provisioning method failed because the system found an invalid metadata file. The upgrade package metadata file contains software upgrade information for the system. The system uses this file to determine whether it needs to upgrade. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
209	Plug and play setup failed because the system detected an invalid configuration from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system detected an invalid configuration from the provisioning server. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
210	Plug and play setup failed because the system detected an invalid or empty certificate URL from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the system detected an invalid or empty certificate URL from the provisioning server. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

9.4 Dolby Conference Phone setup app error messages

These are possible solutions to setup app errors when deploying Dolby Voice Room with the plug and play provisioning method.

For the plug and play provisioning method, the system must be able to connect to your conferencing service provider's setup app through the Internet. After the system connects to the setup app, screens and workflows specific to your conferencing service provider display on the phone. After out-of-the-box setup is complete, the system uses your conferencing service provider's conferencing application, which is separate from the setup app.

Error code	Message displayed	Solution
301	Plug and play setup failed because the Dolby Conference Phone setup app encountered a program failure. Contact your service provider for assistance.	The plug and play provisioning method failed because the Dolby Conference Phone setup app encountered a program failure. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system detected a program error in the Dolby Conference Phone setup app and your conferencing service provider will need to examine the error. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
302	Plug and play setup failed because the Dolby Conference Phone setup app has missing files. Contact your service provider for assistance.	The plug and play provisioning method failed because the Dolby Conference Phone setup app has missing files. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system detected missing files in the Dolby Conference Phone setup app and your conferencing service provider will need to examine the error. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
303	Plug and play setup failed because the system was unable to download the Dolby Conference Phone setup app. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Conference Phone setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Conference Phone setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
304	Plug and play setup failed because the server refused the Dolby Conference Phone setup app download request. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the server refused the Dolby Conference Phone setup app download request. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Conference Phone setup app because the server has refused the download request. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
305	Plug and play setup failed because the system failed to download the Dolby Conference Phone setup app due to a network error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Conference Phone setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Conference Phone setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
306	Plug and play setup failed because the system failed to download the setup app due to a DNS error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Conference Phone setup app due to a DNS error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Conference Phone setup app. This error can occur if the DNS server assigned to the system is unable to resolve the server name. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
307	Plug and play setup failed because the system failed to download the Dolby Conference Phone setup app due to a network timeout. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Conference Phone setup app due to a network timeout. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Conference Phone setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
308	Plug and play setup failed because the system failed to download the Dolby Conference Phone setup app due to a proxy error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Conference Phone setup app due to a proxy error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Conference Phone setup app. This error can occur if the proxy that the system has automatically discovered is unable to process the setup app download request. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
309	Plug and play setup failed because the system failed to download the Dolby Conference Phone setup app due to an SSL error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Conference Phone setup app due to an SSL error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface. See Accessing the Dolby Voice Room web interface on page 77 and Adding certificate authority certificates on page 43 for information on connecting to the Dolby Voice Room web interface using a web browser and importing the CA certificate into the device from the web interface. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
310	Plug and play setup failed because the server rejected the system from downloading the Dolby Conference Phone setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the system from downloading the Dolby Conference Phone setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
311	Plug and play setup failed because of an authentication failure when the system was downloading the Dolby Conference Phone setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because of an authentication failure when the system was downloading the Dolby Conference Phone setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
312	Plug and play setup failed because the Dolby Conference Phone setup app encountered a program failure. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app encountered a program failure. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system detected a program error in the Dolby Conference Phone setup app and your conferencing service provider will need to examine the error. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
314	Plug and play setup failed because the Dolby Conference Phone setup app contains an invalid certificate. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app contains an invalid certificate. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
315	Plug and play setup failed because the Dolby Conference Phone setup app contains an invalid signature. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app contains an invalid signature. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
316	Plug and play setup failed because the Dolby Conference Phone setup app has contains an invalid checksum. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app contains an invalid checksum. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

9.5 Dolby Voice Hub application error messages

These are possible solutions to Dolby Voice Hub application errors when deploying Dolby Voice Room with the plug and play provisioning method.

Error code	Message displayed	Solution
401	Plug and play setup failed because the system failed to download the setup app due to missing files. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to missing files. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
402	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
403	Plug and play setup failed because the server rejected the request when the system tried to download the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the request when the system attempted to download the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
404	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
405	Plug and play setup failed because the system failed to download the setup app due to a DNS error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a DNS error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app due to a DNS error. This error can occur if the DNS server assigned to the system is unable to resolve the server name. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
406	Plug and play setup failed because the system failed to download the setup app due to a network error. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the system needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
407	Plug and play setup failed because the system failed to download the setup app due to a proxy error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the setup app due to a proxy error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. This error can occur if the proxy that the system has automatically discovered is unable to process the setup app download request. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
408	Plug and play setup failed because the system failed to download the setup app due to an SSL error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the system was unable to download the Dolby Voice Hub setup app due to an SSL error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the system through its web user interface. See Accessing the Dolby Voice Room web interface on page 77 and Adding certificate authority certificates on page 43 for information on connecting to the web interface using a web browser and importing the CA certificate into the device from the web interface. For additional help, contact your conferencing service provider. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Error code	Message displayed	Solution
409	Plug and play setup failed because the server rejected the request when the system tried to download the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the system from downloading the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.
410	Plug and play setup failed because the system failed to load the setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the system was unable to load the setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The system was unable to download the Dolby Voice Hub setup app. Contact your conferencing service provider for help. You can collect diagnostic information from the system. For more information on downloading diagnostic information from the device, see Viewing log files on page 84.

Glossary

1080p

See [full high definition](#).

API

Application programming interface. A set of functions that can be used to access the functions of an operating system or other type of software.

API

Application programming interface. A set of functions that can be used to access the functions of an operating system or other type of software.

Bluetooth LE

Bluetooth low energy. A chip technology that produces radio waves to connect a device to other Bluetooth devices. The low-energy version uses less power than standard Bluetooth, but maintains a similar communication range.

CLI

Command-line interface.

CSV

Comma-separated values.

DHCP

Dynamic Host Configuration Protocol.

DNS

Domain Name System. An Internet service that translates Internet domain and host names to IP addresses and conversely. DNS automatically converts between the name entered in a web browser and the IP addresses of the web server hosting the site whose URL is entered in the web browser.

DNS SRV

Domain Name System service record. A service record (SRV record) defines the location (the host name and port number) of a server in a DNS.

DoS

Denial of service. Events that render systems on a computer network temporarily unusable. DoS can happen accidentally as the result of actions taken by network users, but is often malicious.

DSCP

Differentiated Services Code Point. An IP header field that allows a router to provide different levels of service to different types of network traffic.

DSP

Digital signal processor. A specialized microprocessor optimized for digital signal processing.

DTMF

Dual-tone multifrequency. A system that is used by touch-tone telephones to assign a specific frequency (consisting of two separate tones) to each key so that it can easily be identified by a microprocessor.

FTP

File Transfer Protocol. A network-based protocol designed for transferring data using a client-server architecture.

H.264

Also known as Advanced Video Coding (AVC), ISO/IEC MPEG-4 AVC, and ISO/IEC 14496-10:2012. An MPEG standard for video compression most commonly used for high-definition video, such as Blu-ray Disc. The standard was developed jointly by the International Telecommunication Union (ITU) and ISO/IEC MPEG.

HDMI

High-Definition Multimedia Interface. A high-speed, high-capacity format for transferring digital information and the specific hardware interface for the format.

HDR

High dynamic range.

HTTP

Hypertext Transfer Protocol. An application protocol for hypermedia information systems, and the foundation for data communication for the World Wide Web.

HTTPS

Hypertext Transfer Protocol Secure. An application protocol for secure communication over a network and the Internet that provides authentication of websites and keeps user information private.

IEEE

Institute of Electrical and Electronics Engineers.

iLBC

Internet low bit-rate codec. An open-source royalty-free narrowband speech audio coding format codec and reference implementation.

IP

Internet Protocol.

IP address

Internet Protocol address. A numerical identifier assigned to a device that is a member of a network that uses the IP for communication.

JSON

JavaScript Object Notation. A lightweight data-interchange format that is easy for humans to read and write, and for machines to parse and generate.

kbps

Kilobits per second.

LAN

Local area network.

LDAP

Lightweight Directory Access Protocol. An application protocol for querying or modifying items in corporate directories that allows sharing of information about users, devices, and applications on a network.

LLDP

Link Layer Discovery Protocol . A vendor-neutral link layer protocol for Ethernet network devices such as switches, routers, and wireless local area network (LAN) access points to announce information about themselves to other nodes on the network and store the information they discover, as defined in IEEE 802.1AB.

MAC

Multiply-accumulate. In digital signal processing, the multiply-accumulate operation is a common step that computes the product of two numbers and adds that product to an accumulator.

MAC address

Media access control address. A unique identifier assigned to a network interface for communications on a network. MAC addresses are typically assigned by the network interface manufacturer.

MPEG

Moving Picture Experts Group. An ISO/IEC working group that develops video and audio coding standards.

NAPTR record

Naming Authority Pointer record. A type of resource record in the Domain Name System (DNS) of the Internet used to map servers and user addresses in the Session Initiation Protocol (SIP). Used most frequently for Internet telephony (VoIP) services.

NAT

Network Address Translation. An Internet standard that enables a local area network (LAN) to use one set of IP addresses for internal traffic and a second set of addresses for external traffic.

NTLMv2

A Microsoft security protocol that provides authentication, integrity, and confidentiality to users. NTLMv2 is part of Windows NT LAN Manager, which is a suite of security protocols.

NTP

Network Time Protocol. A network protocol for clock synchronization on computers.

PAC

Proxy automatic configuration. A file that defines how web browsers and other user agents can automatically choose the appropriate proxy server (access method) for fetching a given URL.

PBX

Private branch exchange. A phone system that is delivered as a hosted service.

PEM

Privacy-Enhanced Electronic Mail. A file format for security certificates in email communication.

PiP

Picture-in-picture.

PKI

Public key infrastructure. A set of roles, policies, and procedures needed to create, manage, distribute, use, store, and revoke digital certificates and manage public-key encryption.

PoE

Power over Ethernet. A solution in which an electrical current is run to networking hardware over Ethernet category 5 or higher data cabling.

PSTN

Public switched telephone network. A circuit-switched telephone network operated by national, regional, or local operators.

RTCP

RTP Control Protocol. A protocol for communicating quality-of-service information about Real-time Transport Protocol (RTP) sessions.

RTP

Real-time Transport Protocol. An IP network audio and video protocol.

SBC

Session border controller. A piece of network equipment or a collection of functions that controls real-time session traffic at the signaling, call control, and packet layers as they cross a notional packet-to-packet network border between networks, or between network segments.

SIP

Session Initiation Protocol. An application-layer communications protocol used for signaling and controlling communications sessions.

SNMP

Simple Network Management Protocol. A protocol for managing IP network devices

SRTP

Secure Real-time Transport Protocol. A Real-time Transport Protocol (RTP) profile that provides encryption, message authentication, and integrity, and replay protection to the RTP data in both unicast and multicast applications.

SSH

Secure Shell protocol. An encrypted network protocol for secure data communication, remote command-line login, remote command execution, and other secure network services between two networked computers.

SSL

Secure Sockets Layer . A security protocol that works at a socket level.

TCP

Transmission Control Protocol. A communications protocol that specifies how data should be formatted, addressed, transmitted, routed, and received at the destination. Part of the Internet protocols communications suite.

TCP/IP

Transmission Control Protocol/Internet Protocol. Communications protocols that specify how data should be formatted, addressed, transmitted, routed, and received at the destination. Part of the Internet protocols communications suite.

TLS

Transport Layer Security. A cryptographic protocol designed to provide communications security over a computer network.

TLV

Type length value. Within data communication protocols, a format for encoding optional information.

UDP

User Datagram Protocol. A communications protocol that uses no handshaking dialogues to establish a connection with the remote host. UDP is a member of the IP suite.

UHD

Ultra-high definition. Ultra-high-definition television or video, with a display resolution of 3840×2160 pixels in the 16:9 aspect ratio. Also referred to as 2160p.

UI

User interface.

URI

Uniform Resource Identifier. A group of characters identifying a resource on a network (typically, the Internet).

URL

Uniform resource locator.

USB

Universal Serial Bus. A standard that defines the cables, connectors, and communications protocols used in connections between computers and electronic devices.

VLAN

Virtual LAN. Any broadcast domain that is partitioned and isolated in a computer network at the data link layer (OSI layer 2).

VoIP

Voice over Internet Protocol. A methodology and group of technologies for the delivery of voice communications and multimedia sessions over Internet Protocol (IP) networks.

WebRTC

Web Real-Time Communication. API definition drafted by the World Wide Web Consortium (W3C) that supports browser-to-browser applications for voice calling, video conferencing, and P2P file sharing without the need of either internal or external plugins.

WPAD

Web Proxy Automatic Discovery. An Internet protocol that enables a web browser to automatically connect to a cache server (or proxy server) location in a network to retrieve stored web pages more quickly by than leaving the network to request the web page from the site of the originating web server.



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